Homochiral ideas: symmetry centers

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Objectives

In the whole infinite spectre tiling:

- Find the location of global 2-fold and 3-fold symmetry centers

Around a global 2-fold or 3-fold symmetry center:

- Build an homochiral inflation construction

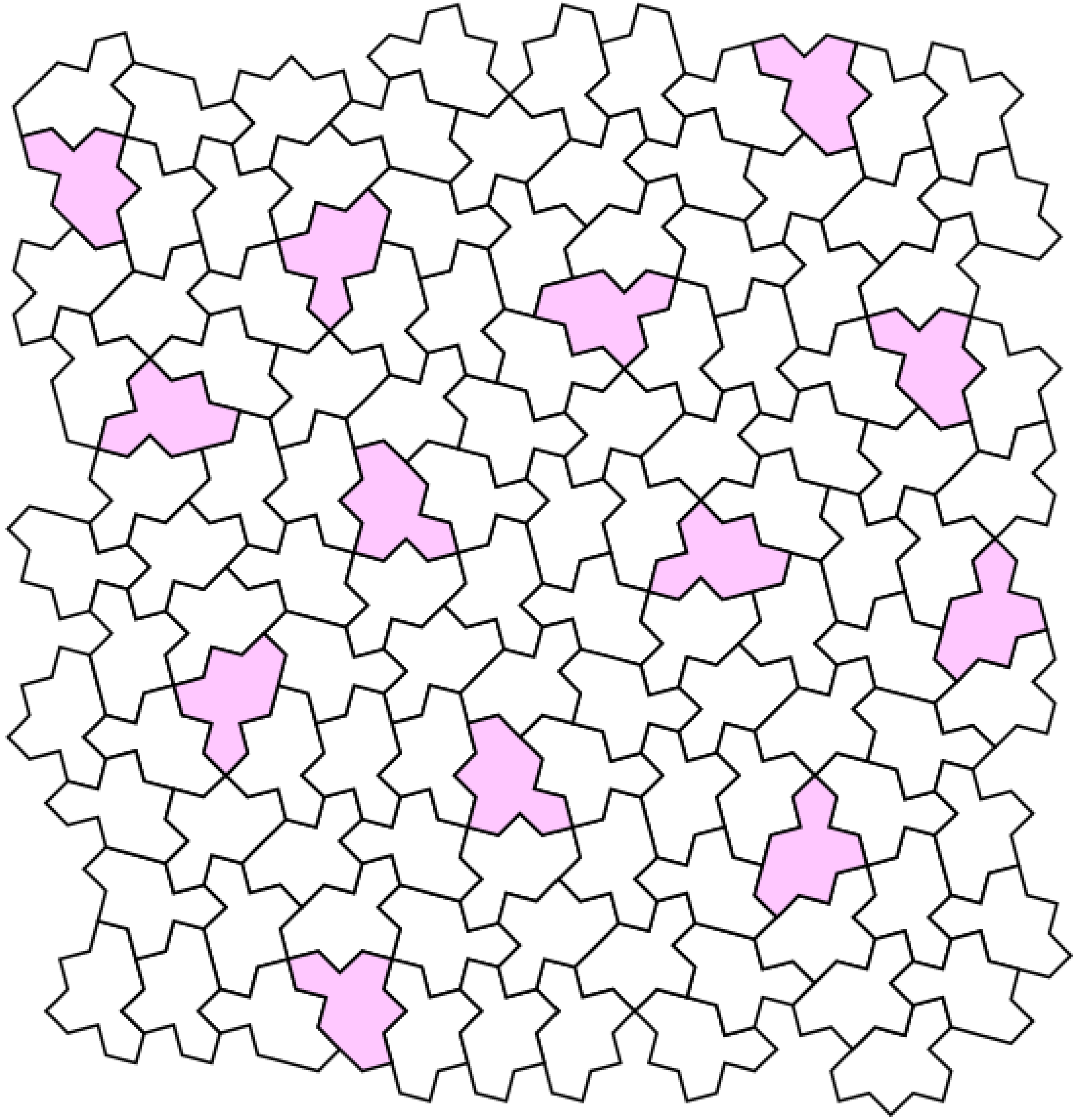
Difference between global and local symmetry

We distinguish between:

- a global symmetry center where rotation operation around the center applies to the whole infinite tiling

- And a local symmetry center, where rotation operation around the center applies locally to a finite subset of tiles

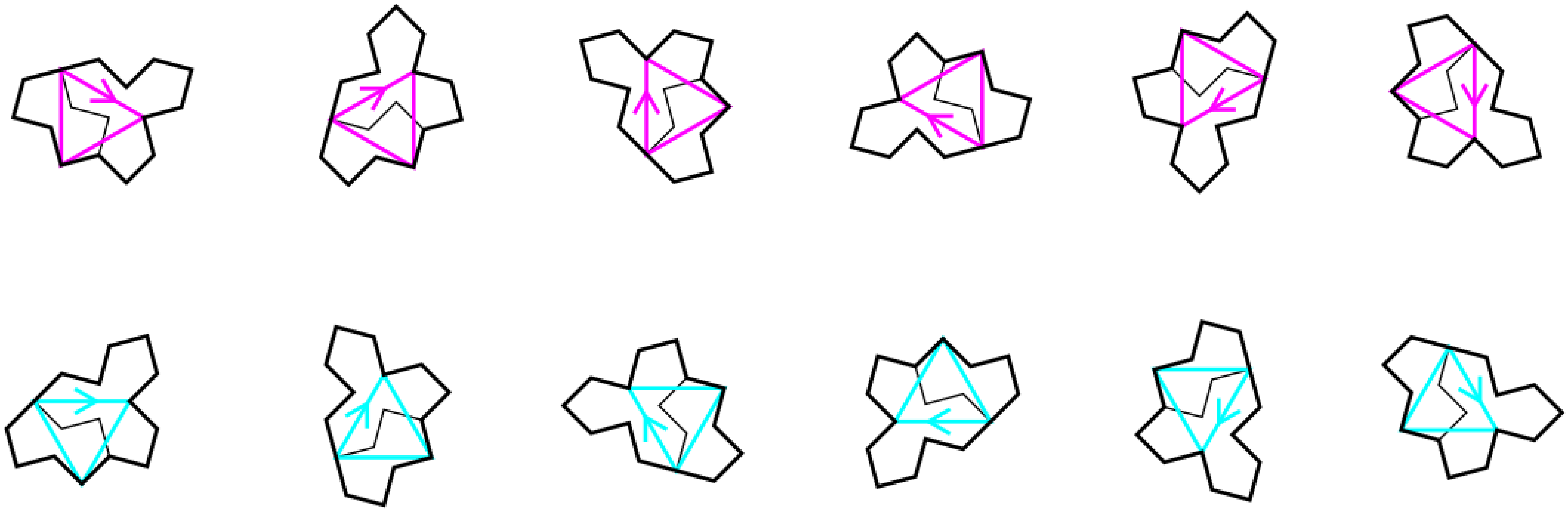
Spectre tiling (left chirality): ODD and EVEN monotiles



- ODD (pink): M1,M3,M5,M7,M9,M11
- EVEN (no color): M2,M4,M6,M8,M10,M12

Six orientations for ODD and EVEN monotiles

$M_1 = M_{O_1}$ $M_3 = M_{O_2}$ $M_5 = M_{O_3}$ $M_7 = M_{O_4}$ $M_9 = M_{O_5}$ $M_{11} = M_{O_6}$

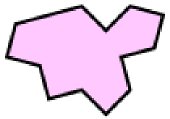


$M_2 = M_{E_1}$ $M_4 = M_{E_2}$ $M_6 = M_{E_3}$ $M_8 = M_{E_4}$ $M_{10} = M_{E_5}$ $M_{12} = M_{E_6}$

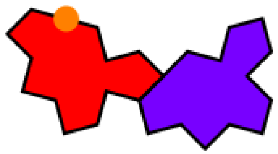
Identification of the greek letters for tiles

- ODD tiles
Pink color

$\Gamma \quad \Gamma_2$



Θ

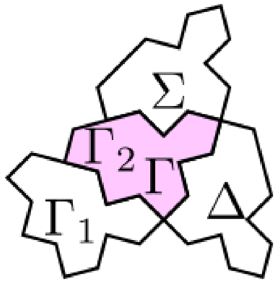


Λ

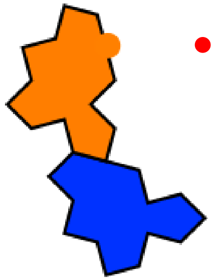
- \theta and \lambda always in couple

- Small cluster \sigma, \delta, \gamma1
white color

$\Gamma_1 \quad \Sigma \quad \Delta$

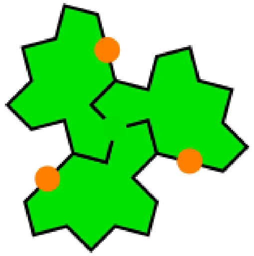


Ξ



- All \chi tiles are linked to a \pi tile and consequently to a large star.

Φ

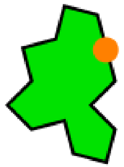


Π

Ψ

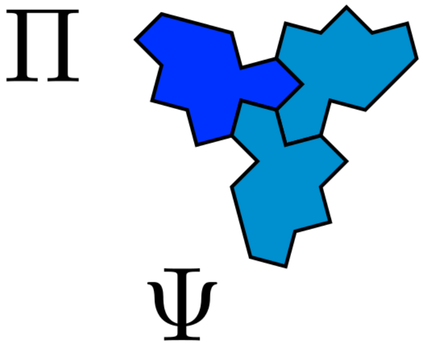
Π

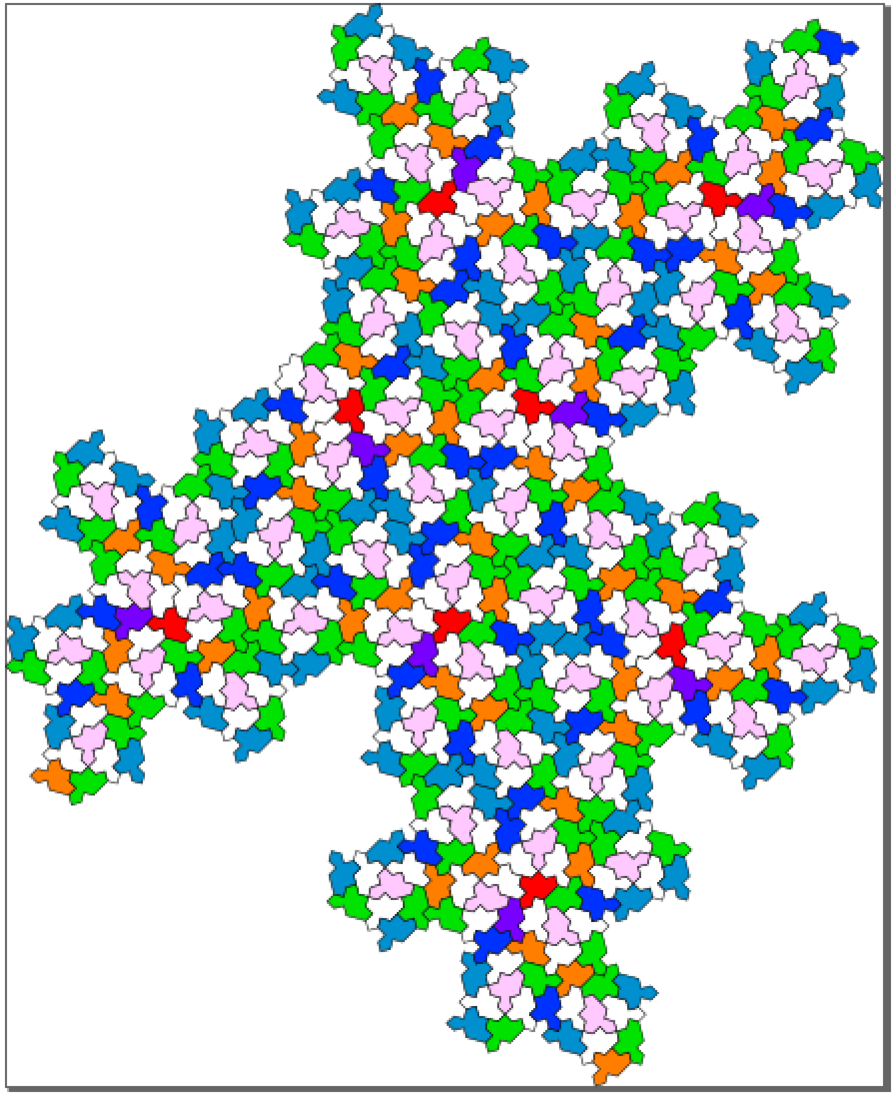
- \phi: Small green stars + some isolated EVEN tiles



- Blue large stars correspond to all \Pi and \Psi tiles with different combinations

Inflation using python code





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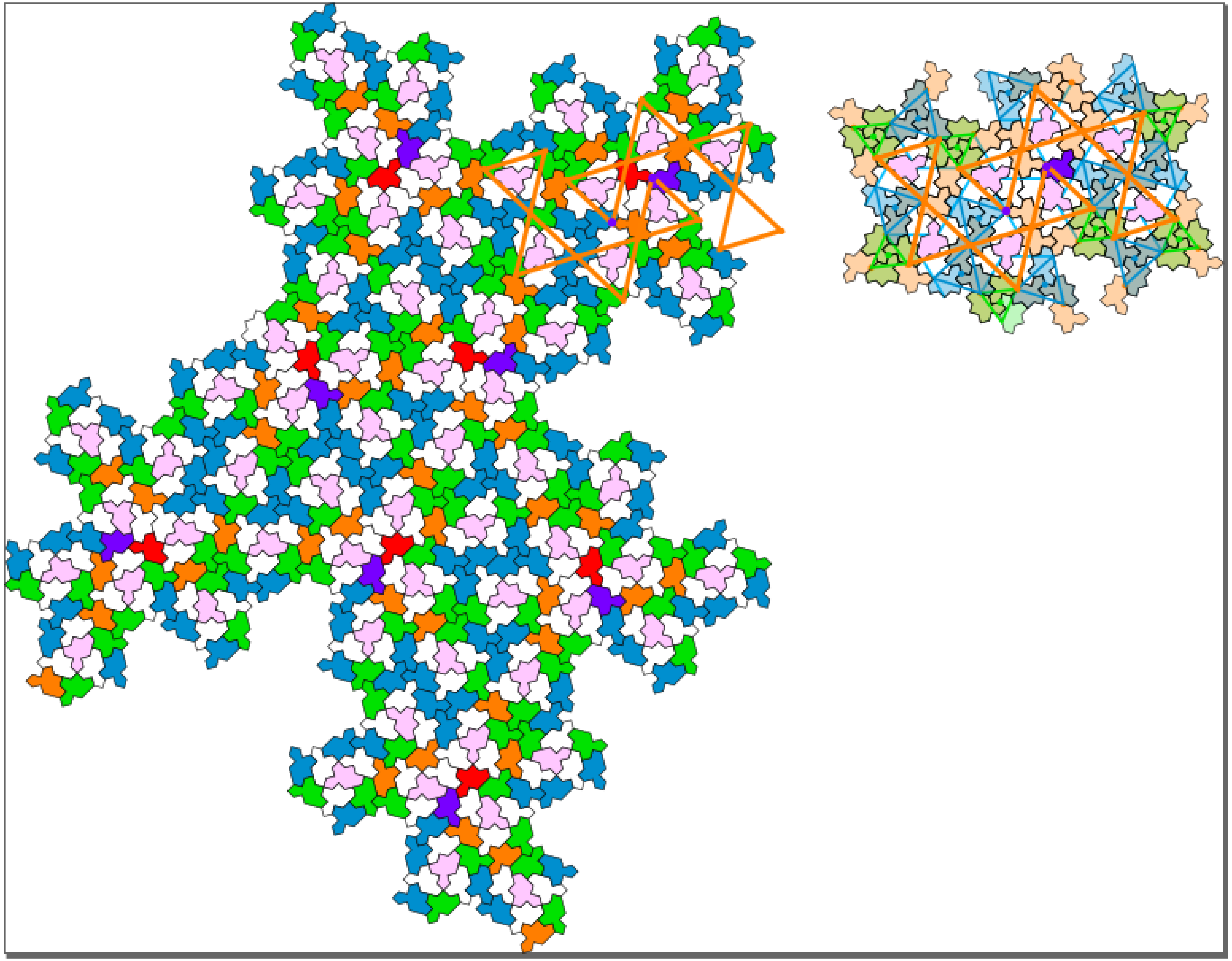
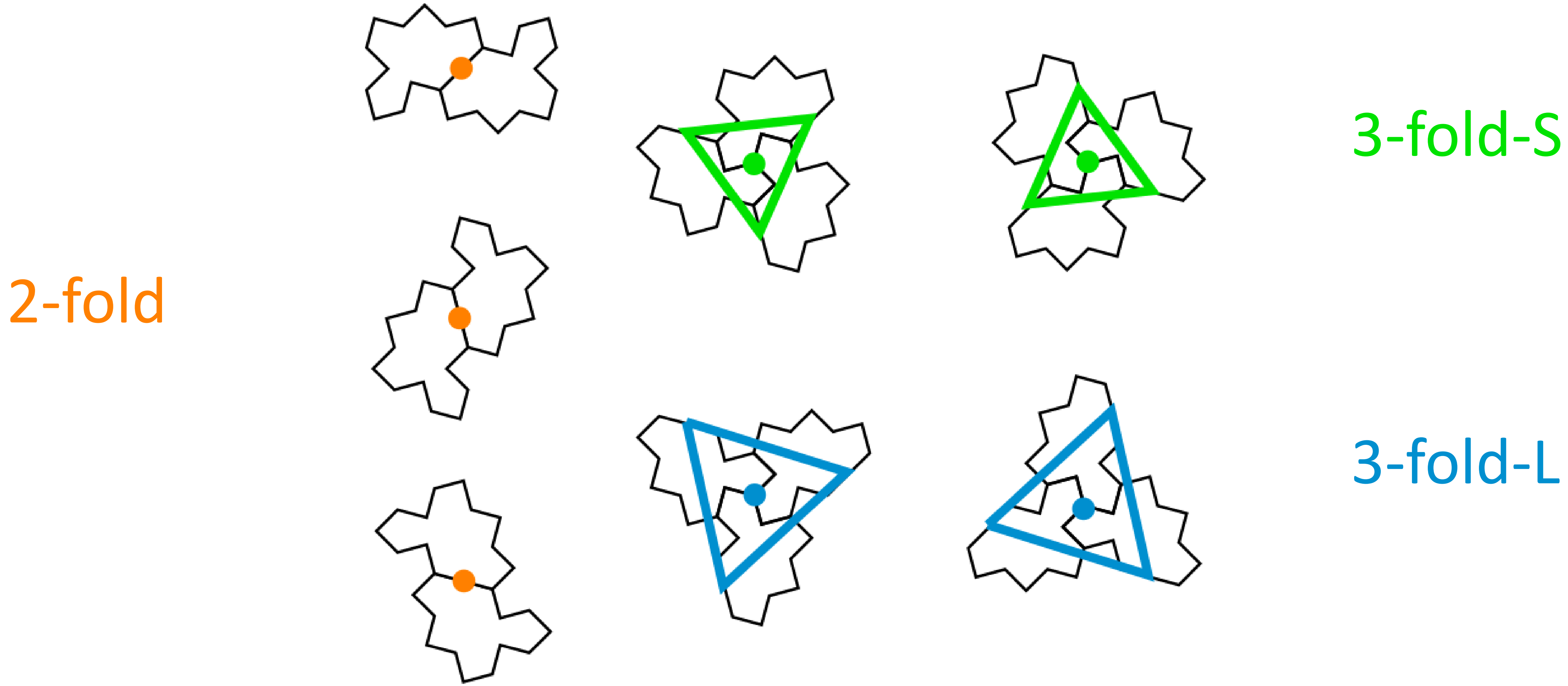
Inflation using python code

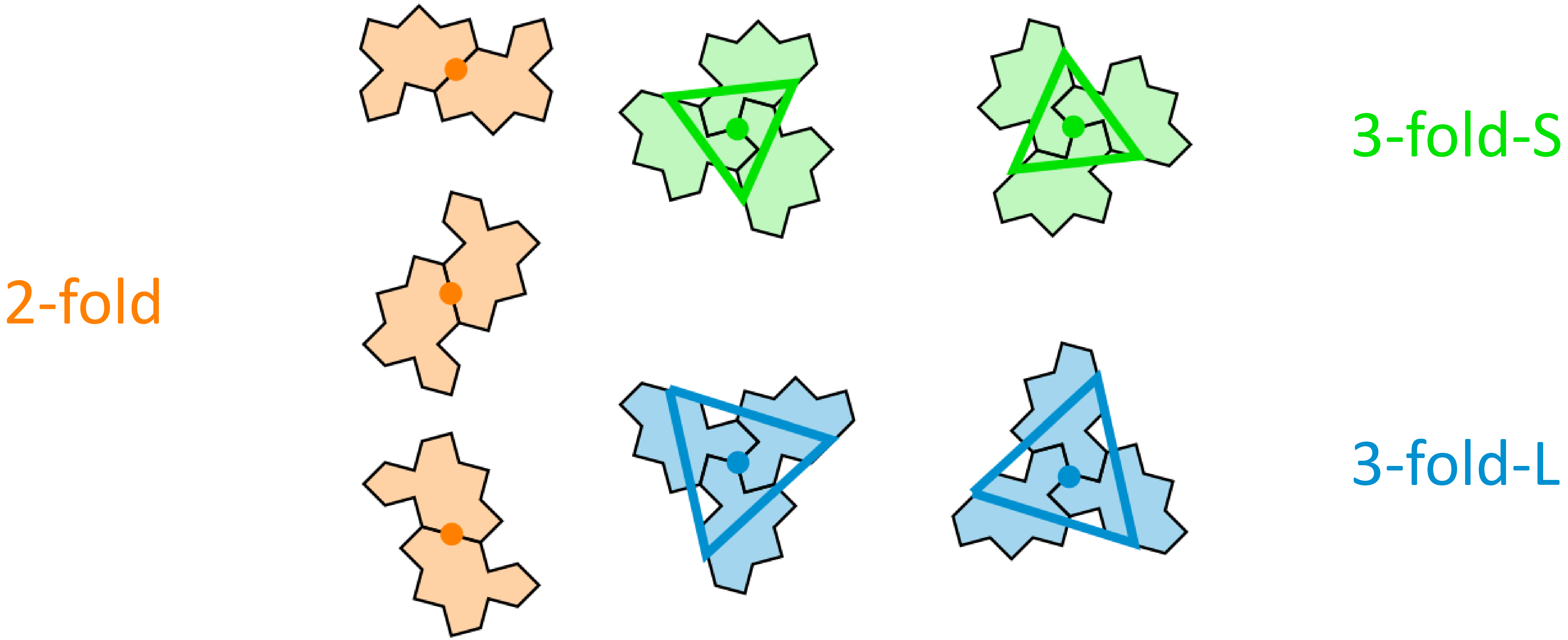


configuration of monotiles at a symmetry center



- One configuration for 2-fold (orange color) and two configuration for 3-fold (green and blue). Figures show all possible orientations in the tiling.

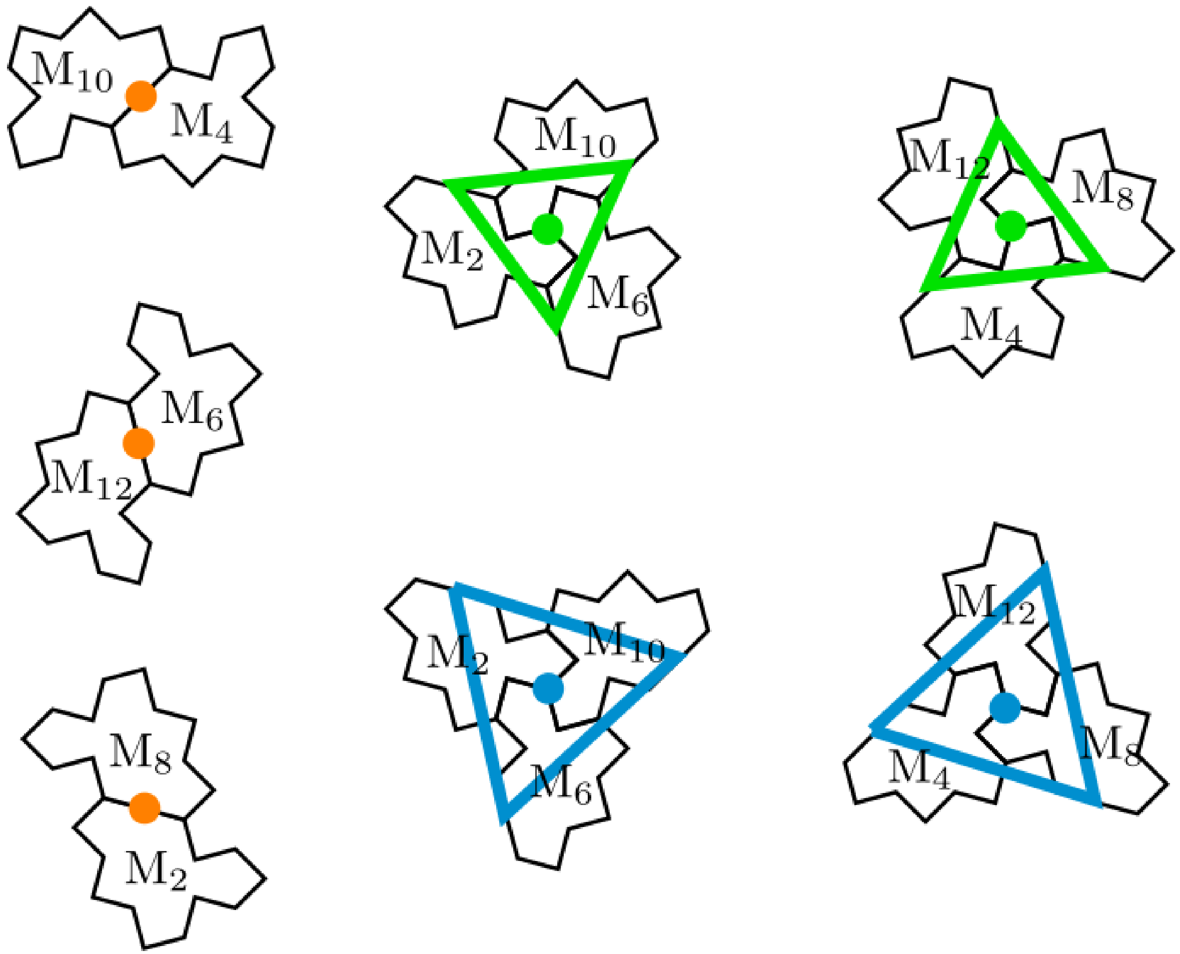
configuration of monotiles at a symmetry center



- Color is used when rotation around symmetry center is applied locally.

Symmetry centers are located on the contour of adjacent EVEN monotiles

2-fold

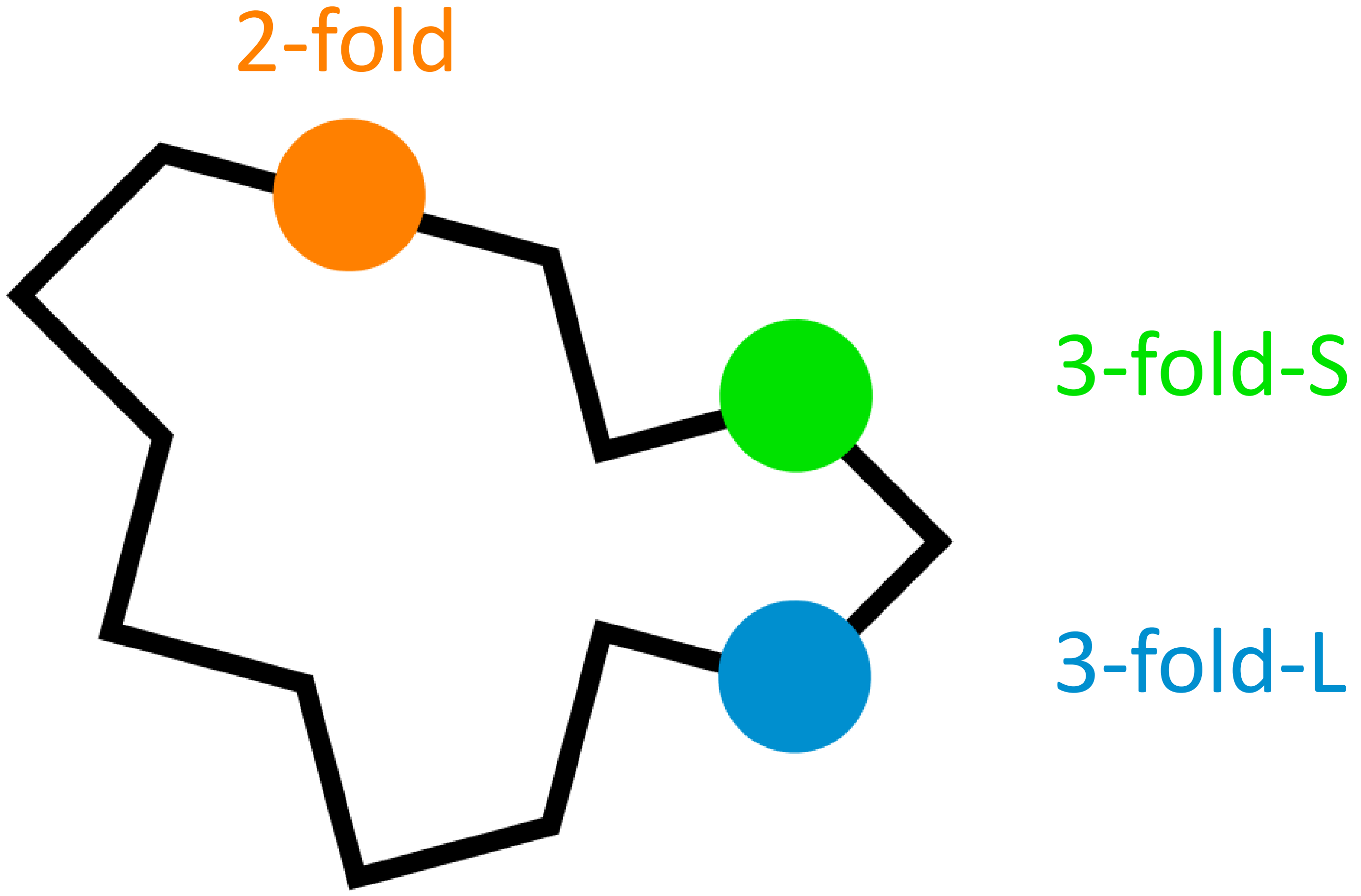


3-fold-S

3-fold-L

- EVEN monotiles are M2, M4, M6, M8, M10 and M12

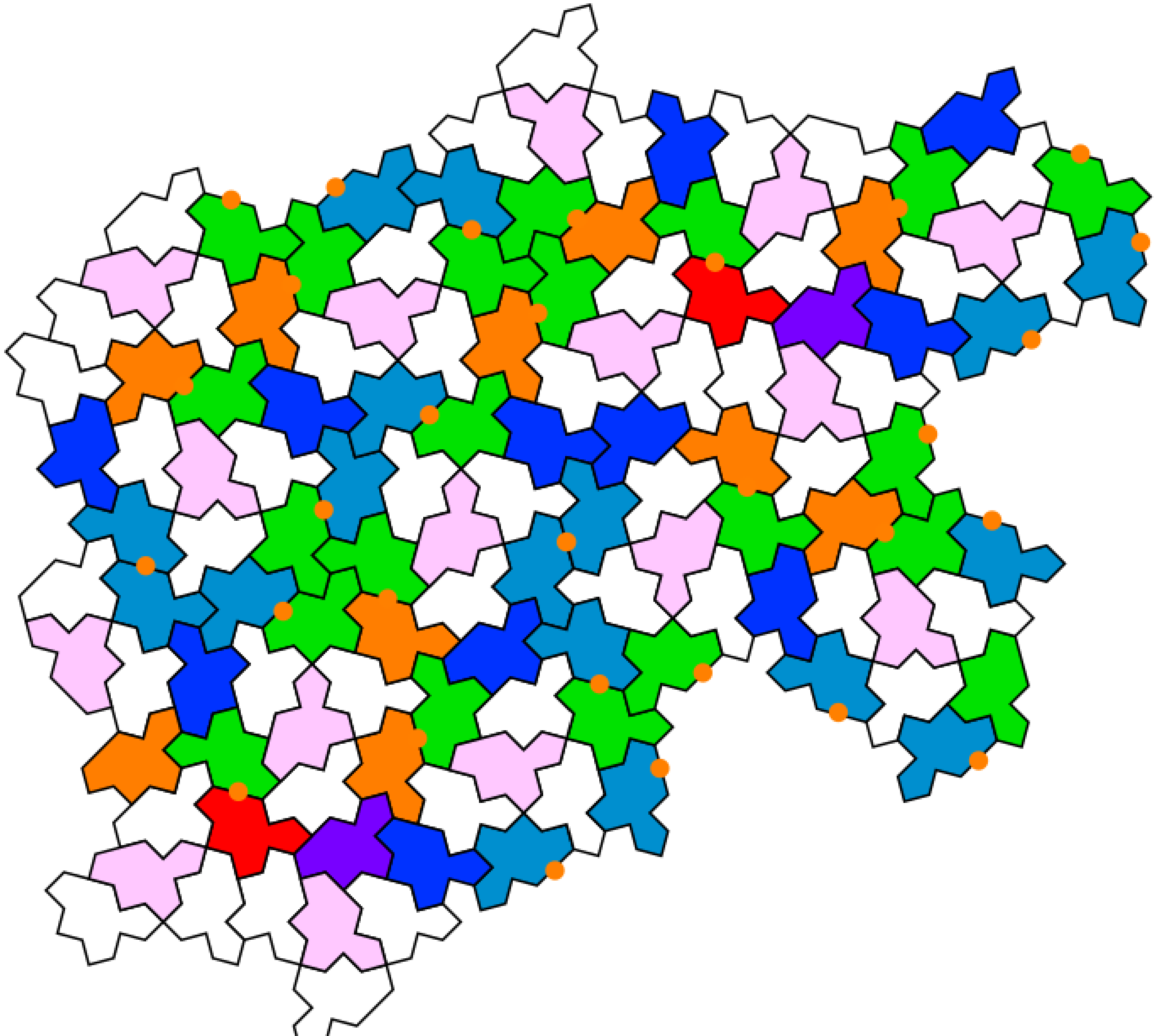
Location on the contour of one EVEN monotile



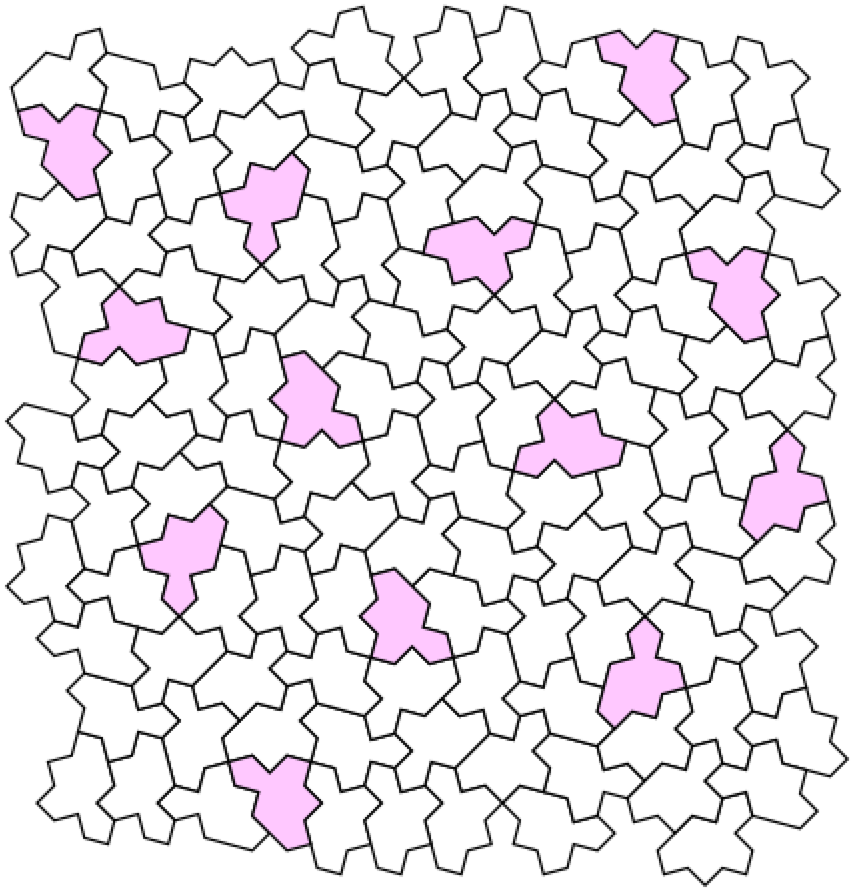
- This figure shows the origin of the two different, S and L, 3-fold centers
- It is linked with the chirality of the spectre tile (here with left handiness)

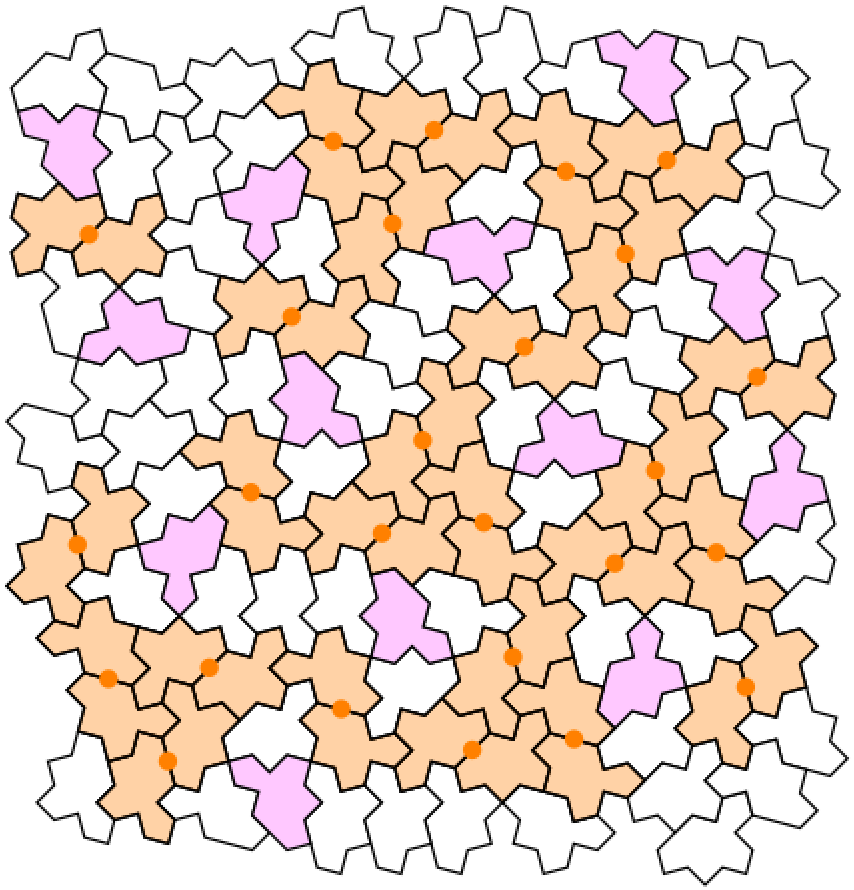
Local two fold symmetry centers

2-fold

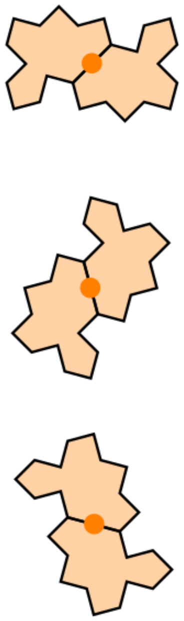


Example patch with all local 2-fold centers (orange color)

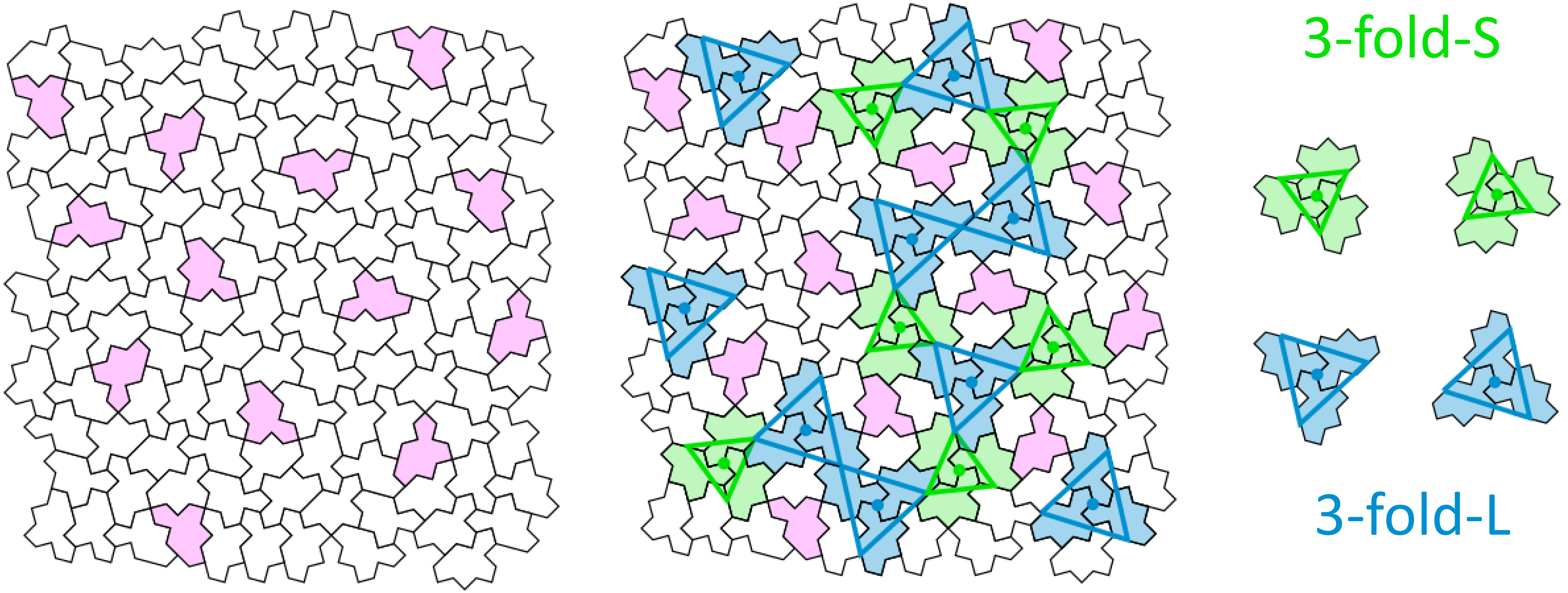




Local 2-fold

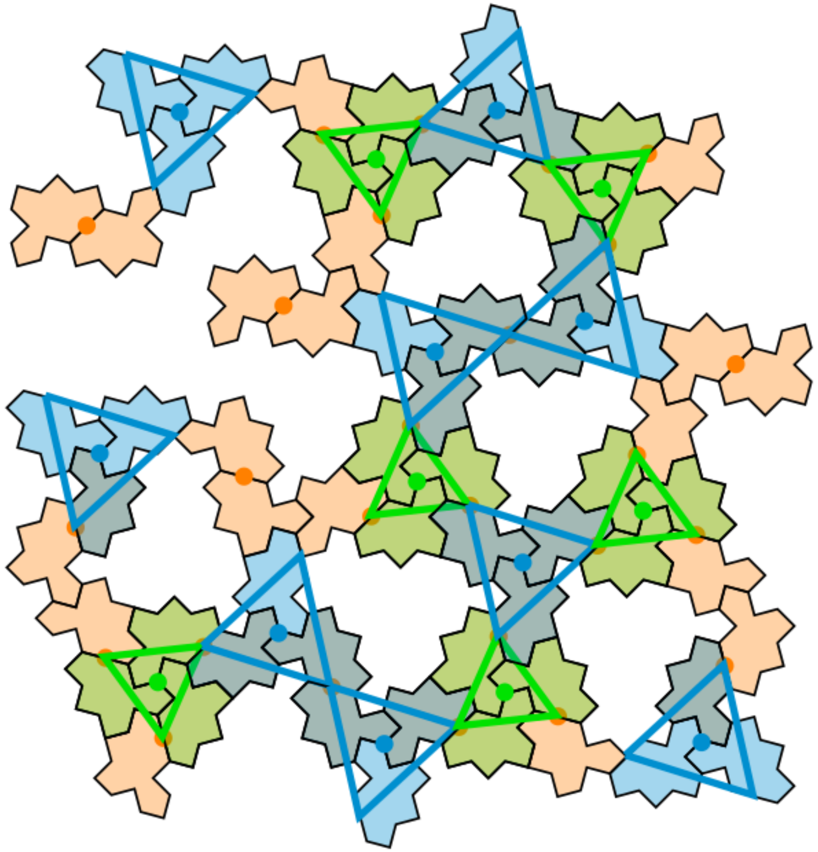


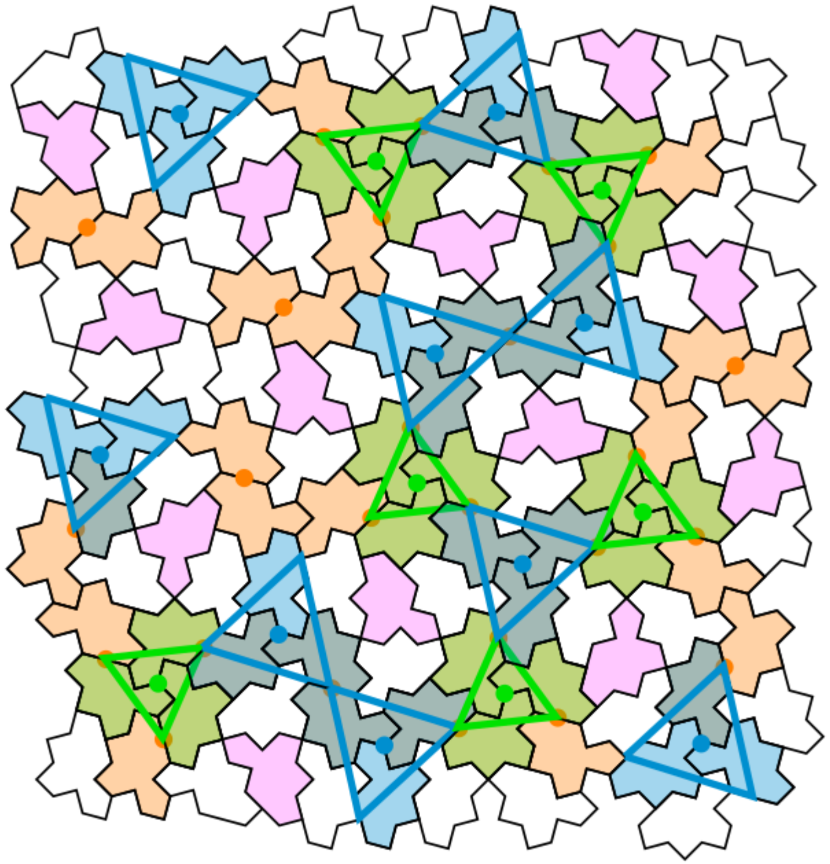
Same patch with all local 3-fold centers (green and blue colors)



Superposition of (orange/blue/green) colors

2-fold

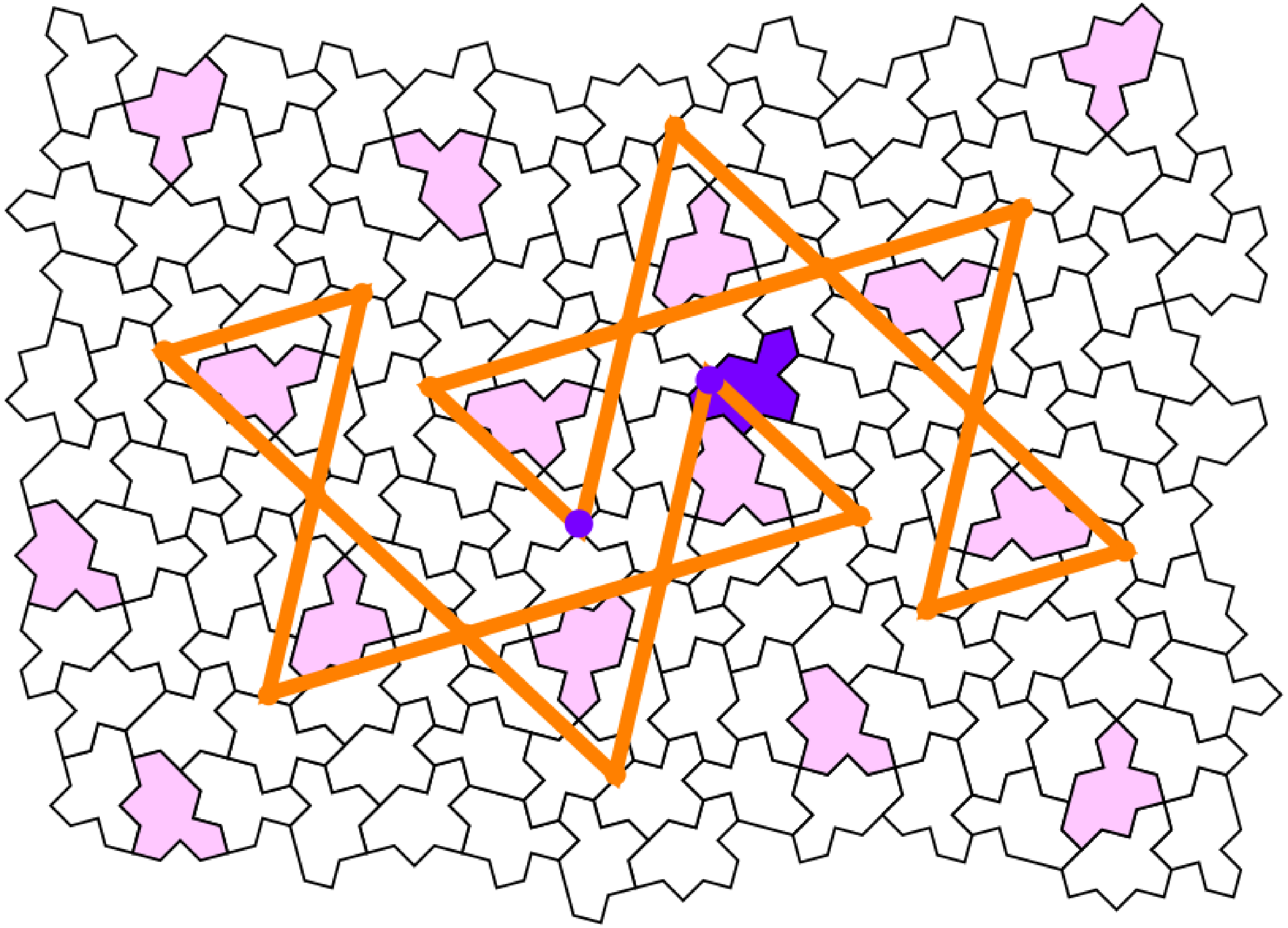




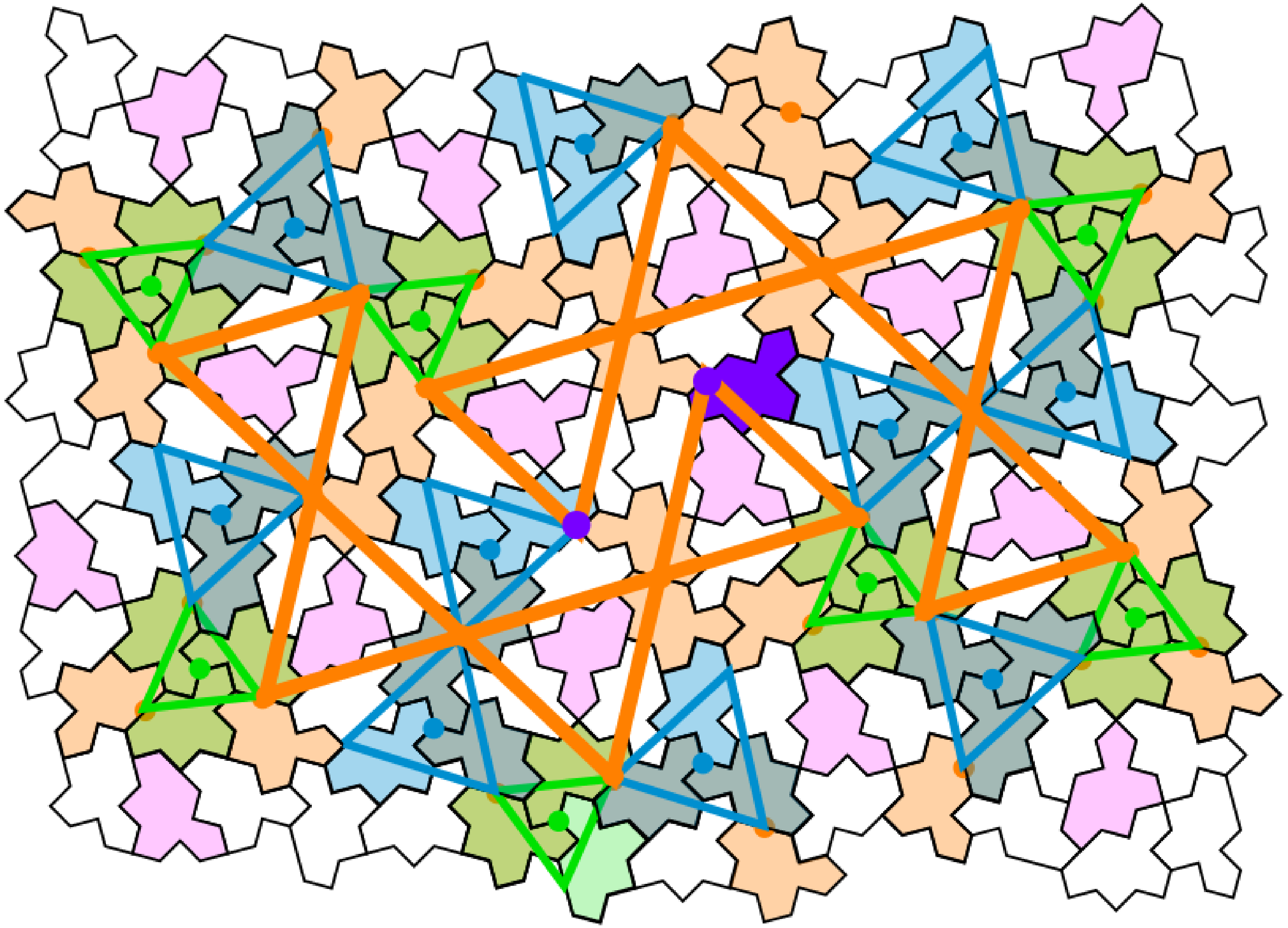
3-fold-S

3-fold-L

2-arms

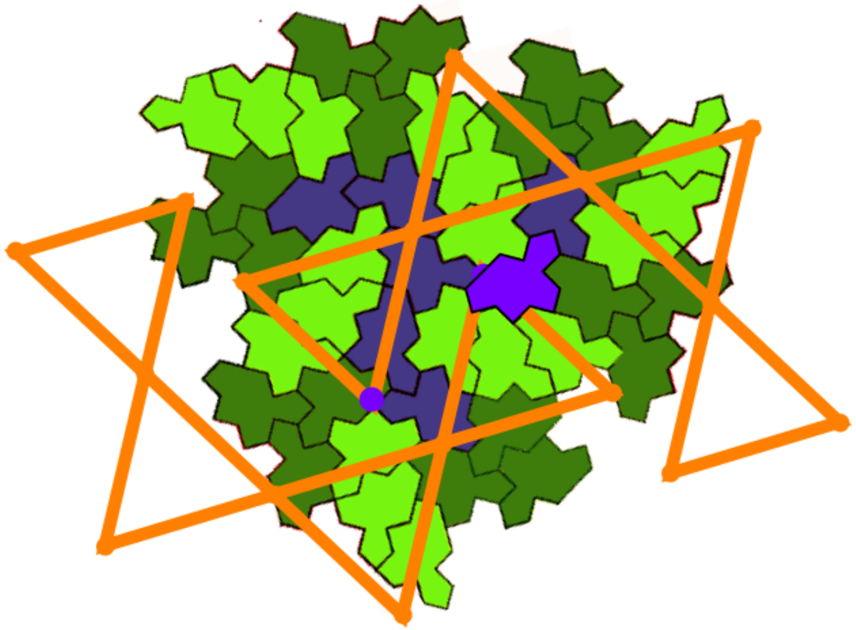


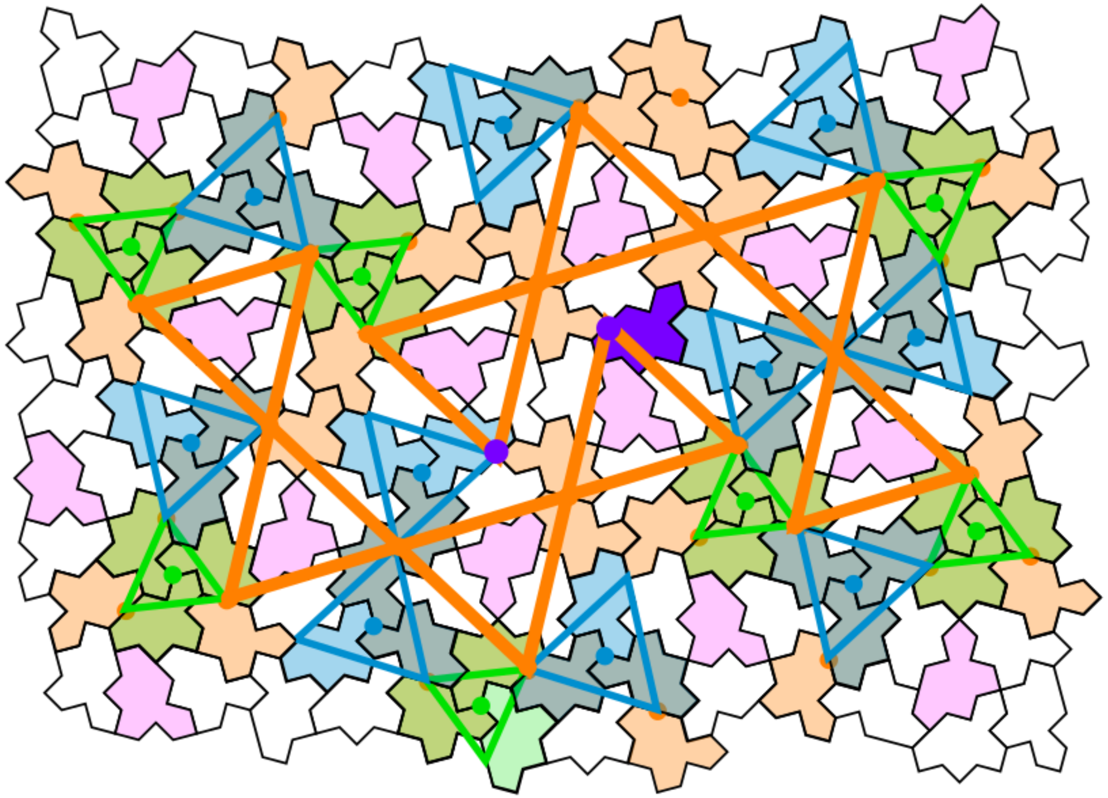
ALL EVEN tiles: orange/blue/green colors + one violet isolated tile



- 2-arms

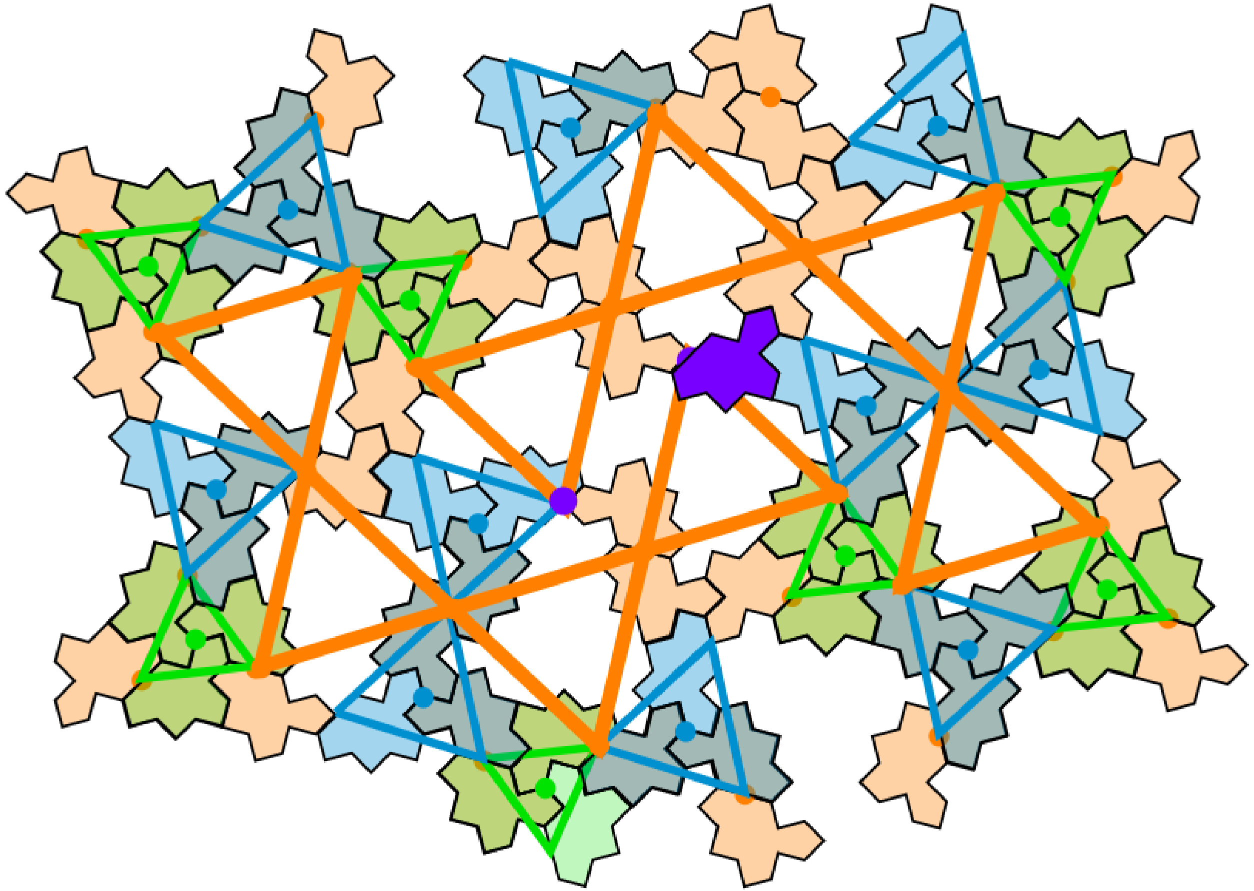
ALL EVEN tiles: orange/blue/green colors + one violet isolated tile

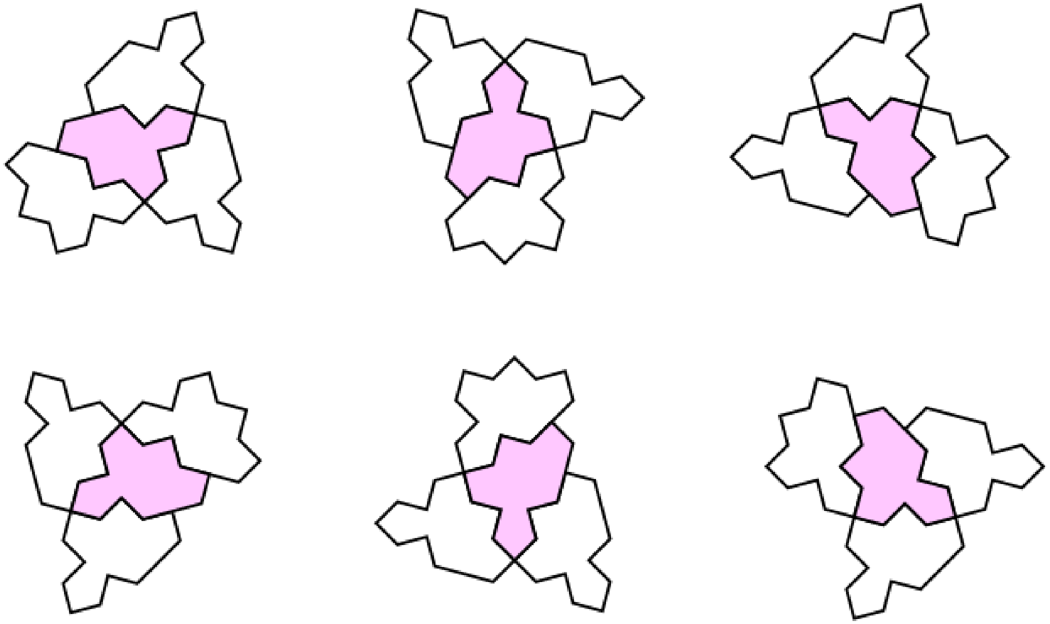




- 2-arms and superposition with chimera pattern

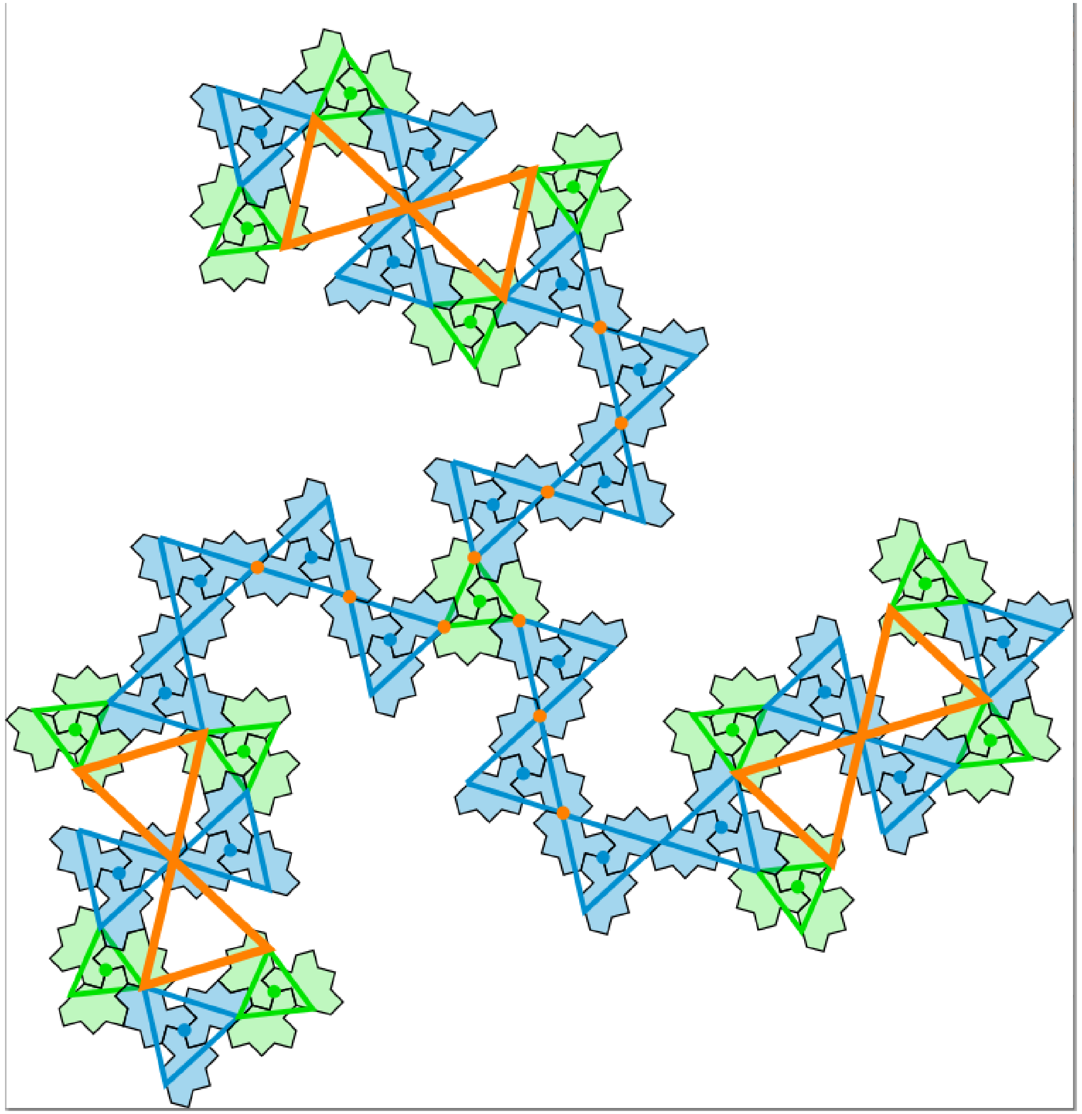
Orange/blue/green + violet: a continuous cluster of ODD tiles around small clusters.





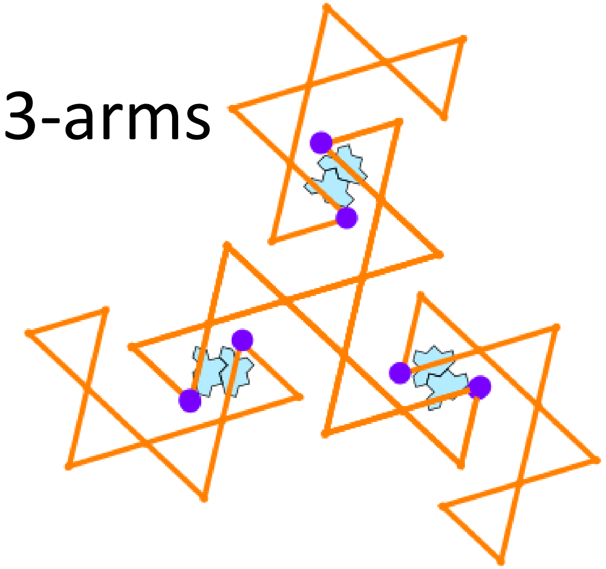
- Small clusters of 4 tiles (one ODD in pink \gamma2 + three EVEN \gamma1, \Delta and \Sigma)

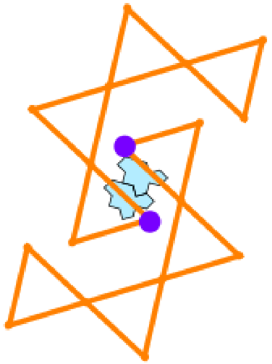
To continue ? generate star clusters by applying locally 2-fold and 3-fold symmetry between EVEN monotiles



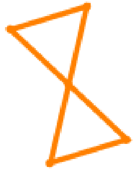
Three patterns in the whole tiling ☺

- 3-arms

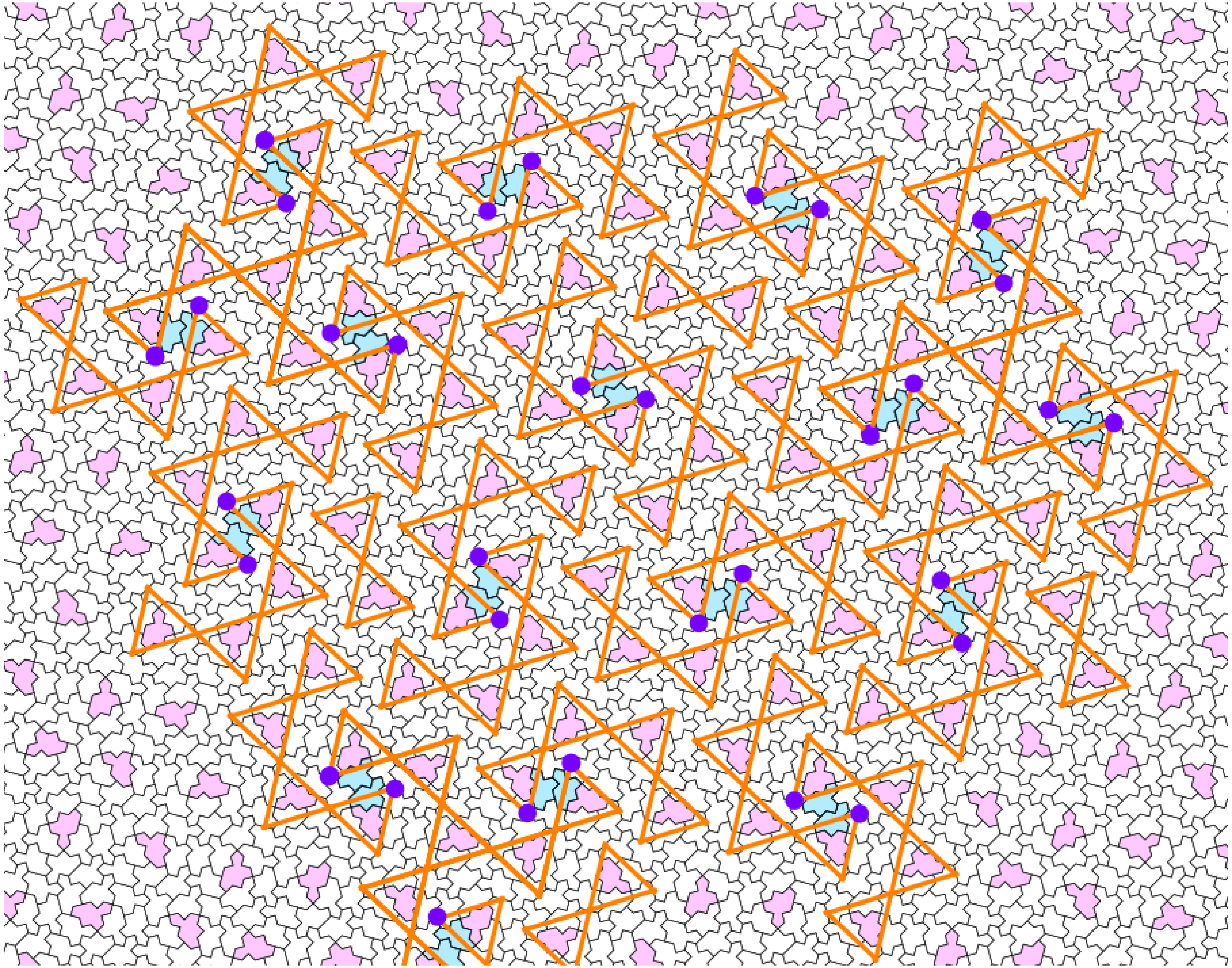




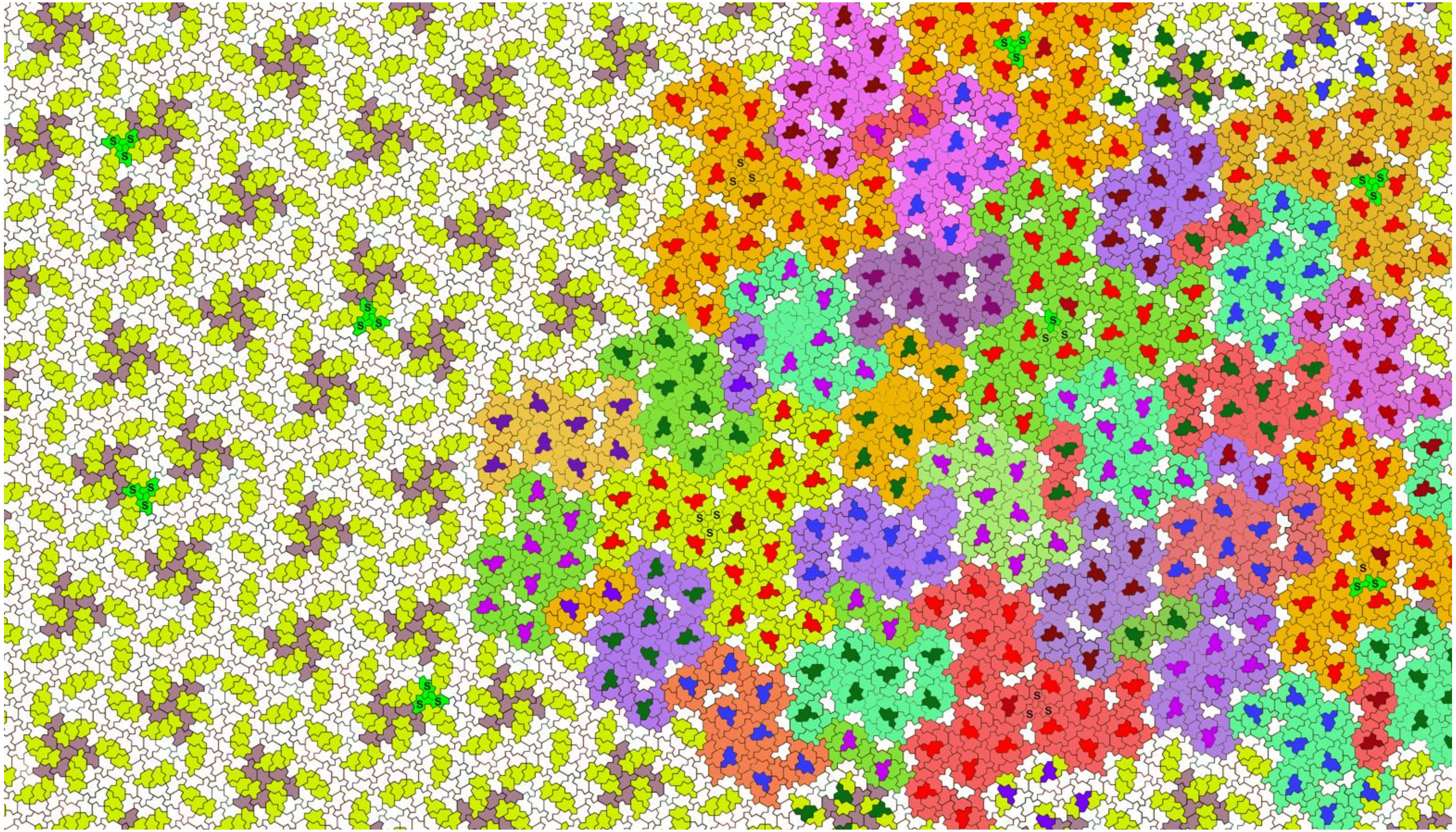
- 2-arms



- diabolo

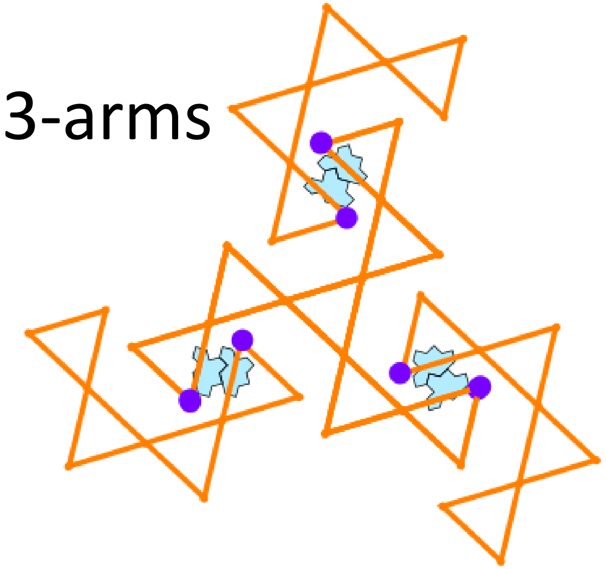


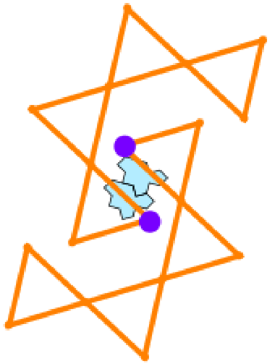
Diabolo/2-arms/3-arms at level 2 by Boris Horvat

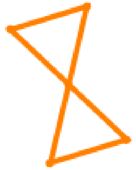


Frequency of the three patterns (Boris Horvat)

- 3-arms







- diabolo
- 2-arms

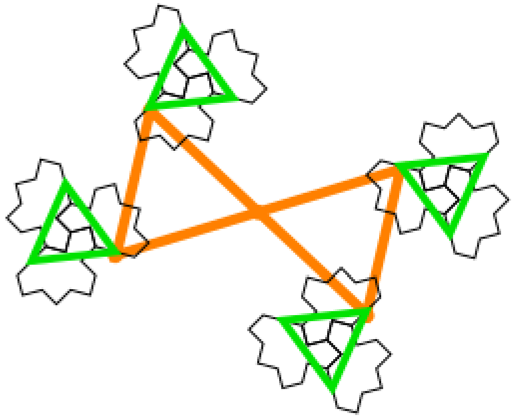
After colouring (manually) the tiling with your objects, I realized that the frequency of diabolos is exactly twice the frequency of 3-way objects. My observations confirm that **every 3-way object (Object-19) has exactly two diabolos attached to two of its arms in a consistent topological orientation**. This allows us to define a new primary super-structure: Object-23 (1×3-way+2×Diabolo), thus partitioning the tiling into two large objects only (Object-23, and Object-8). The 3-way objects (the nodes of the tiling, centred around lone stars) connect to each other via three distinct modes:

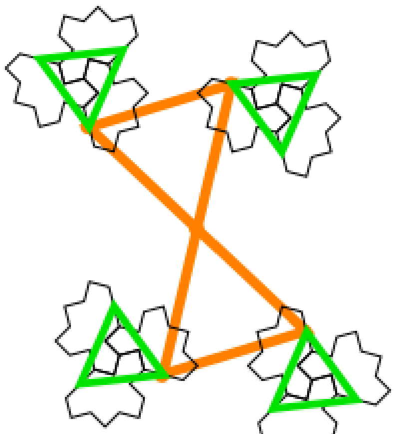
1) Direct Contact: Arm-to-arm contact between two 3-way objects (Object-19).

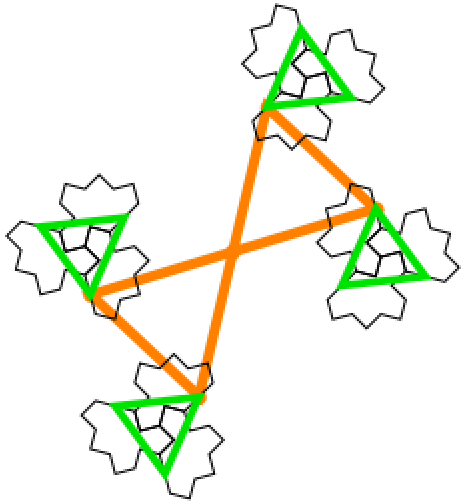
2) Diabolo Link: Connection mediated by a single diabolo (Object-2).

3) 2-way link: Connection mediated by a 2-way object (Object-8).

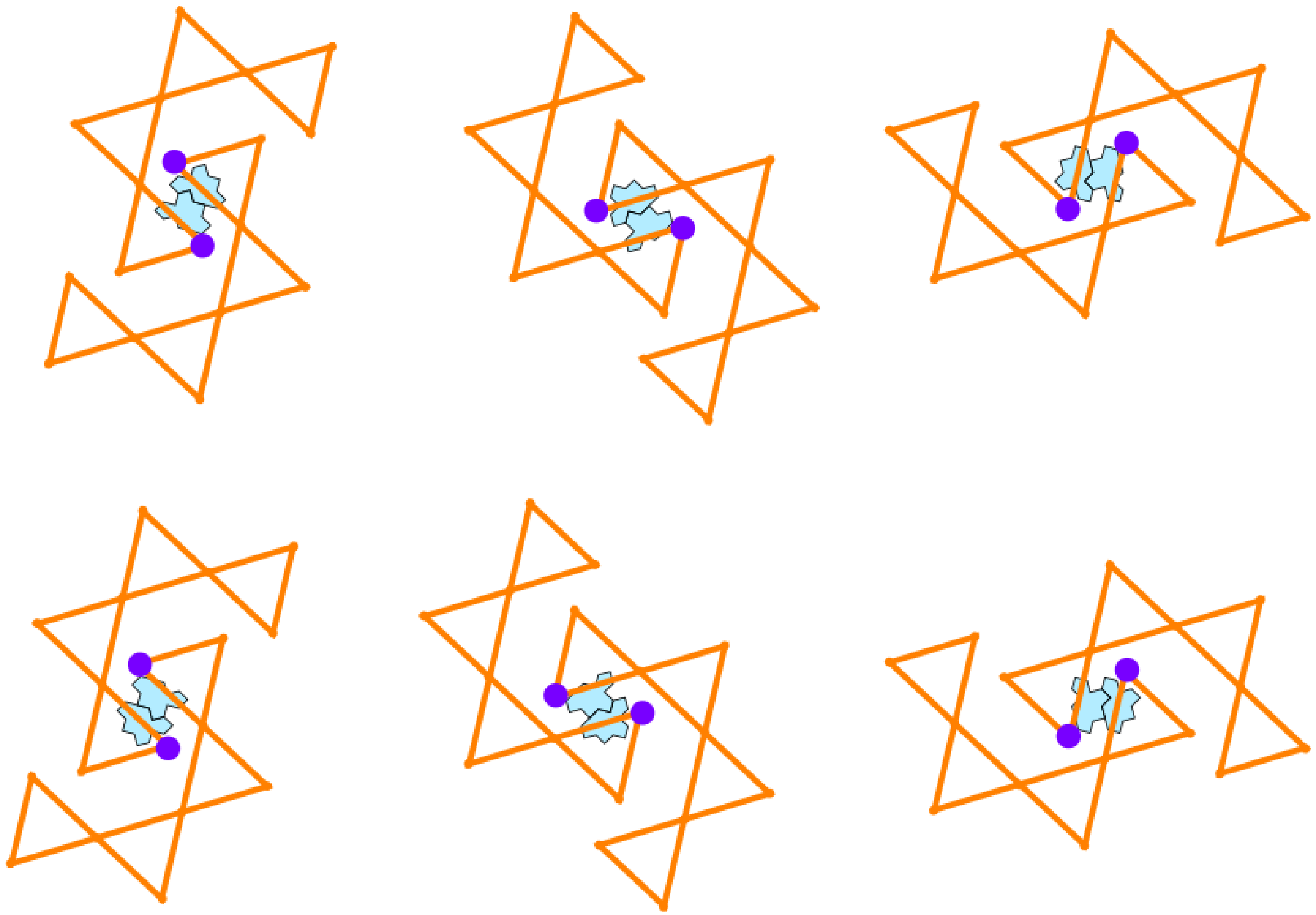
three orientations for diabolos







Six orientations of the 2-arms



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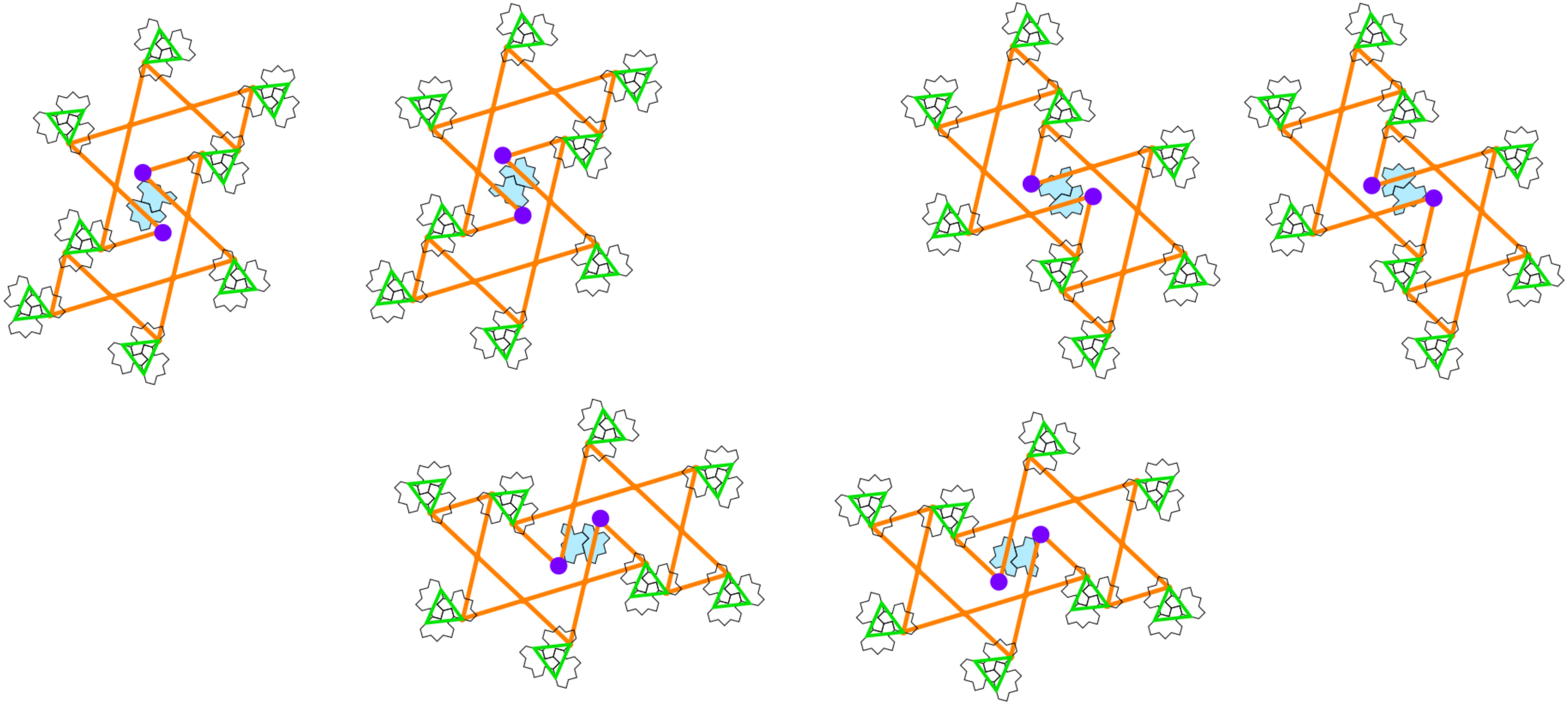
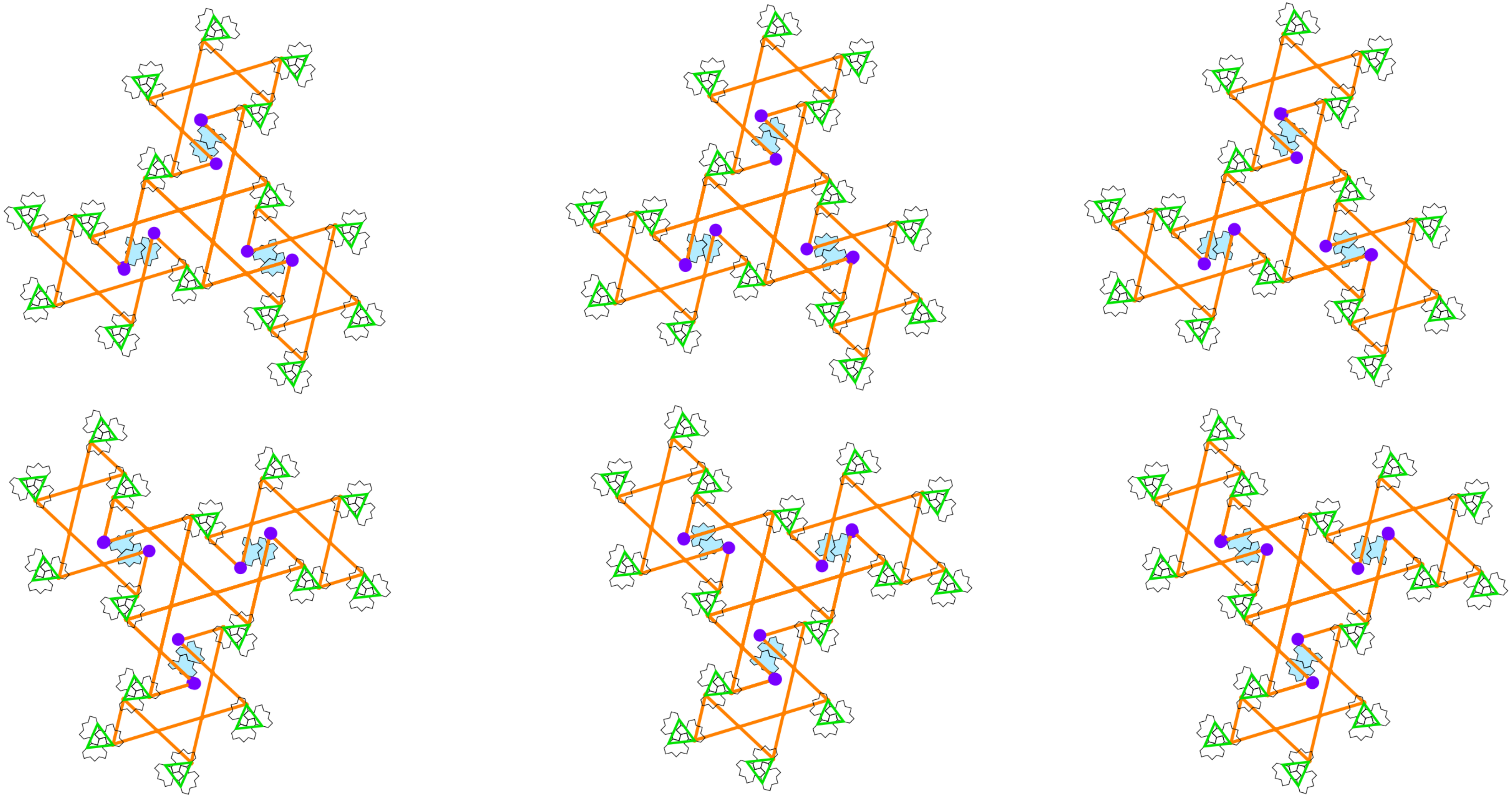
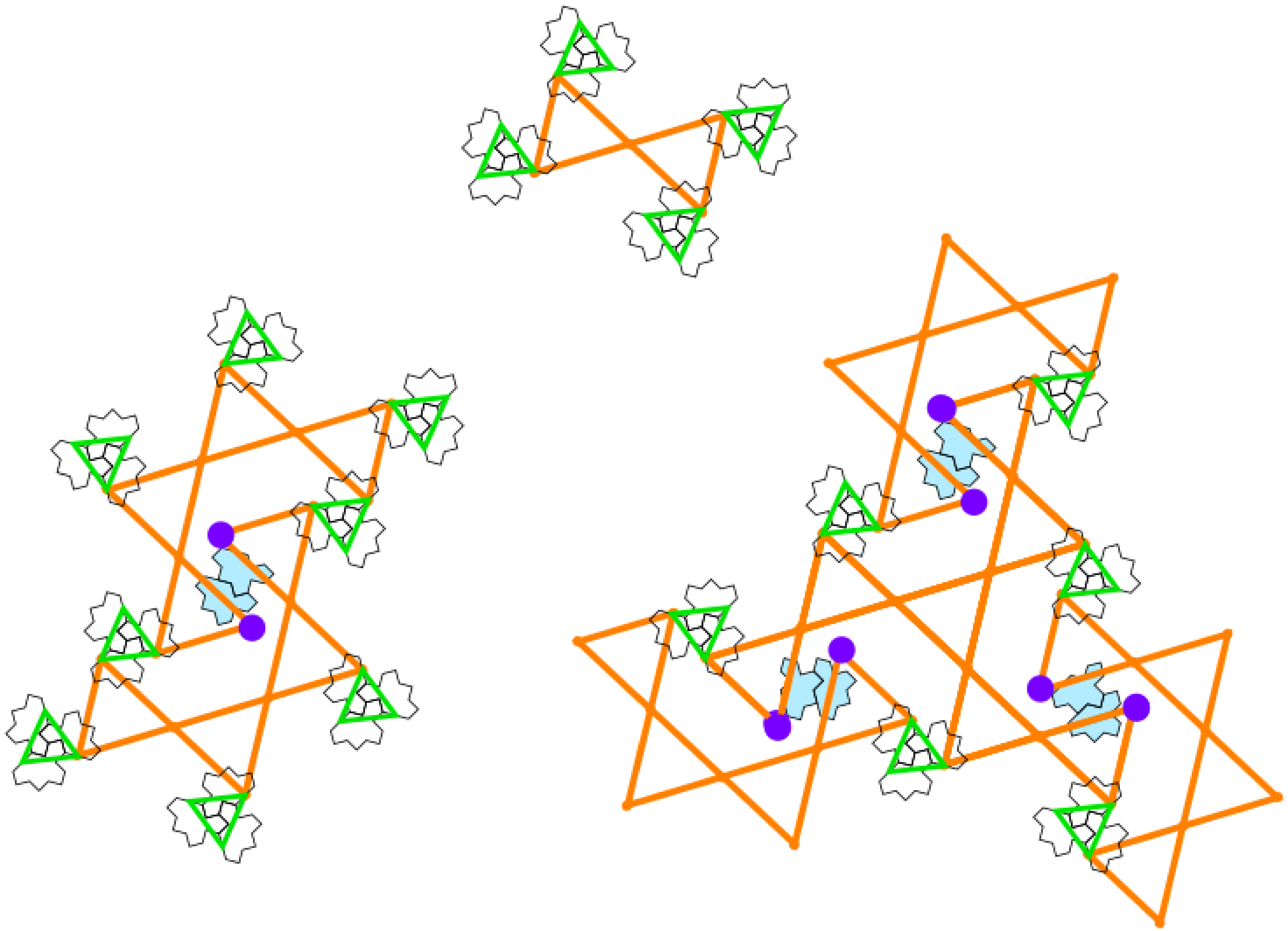


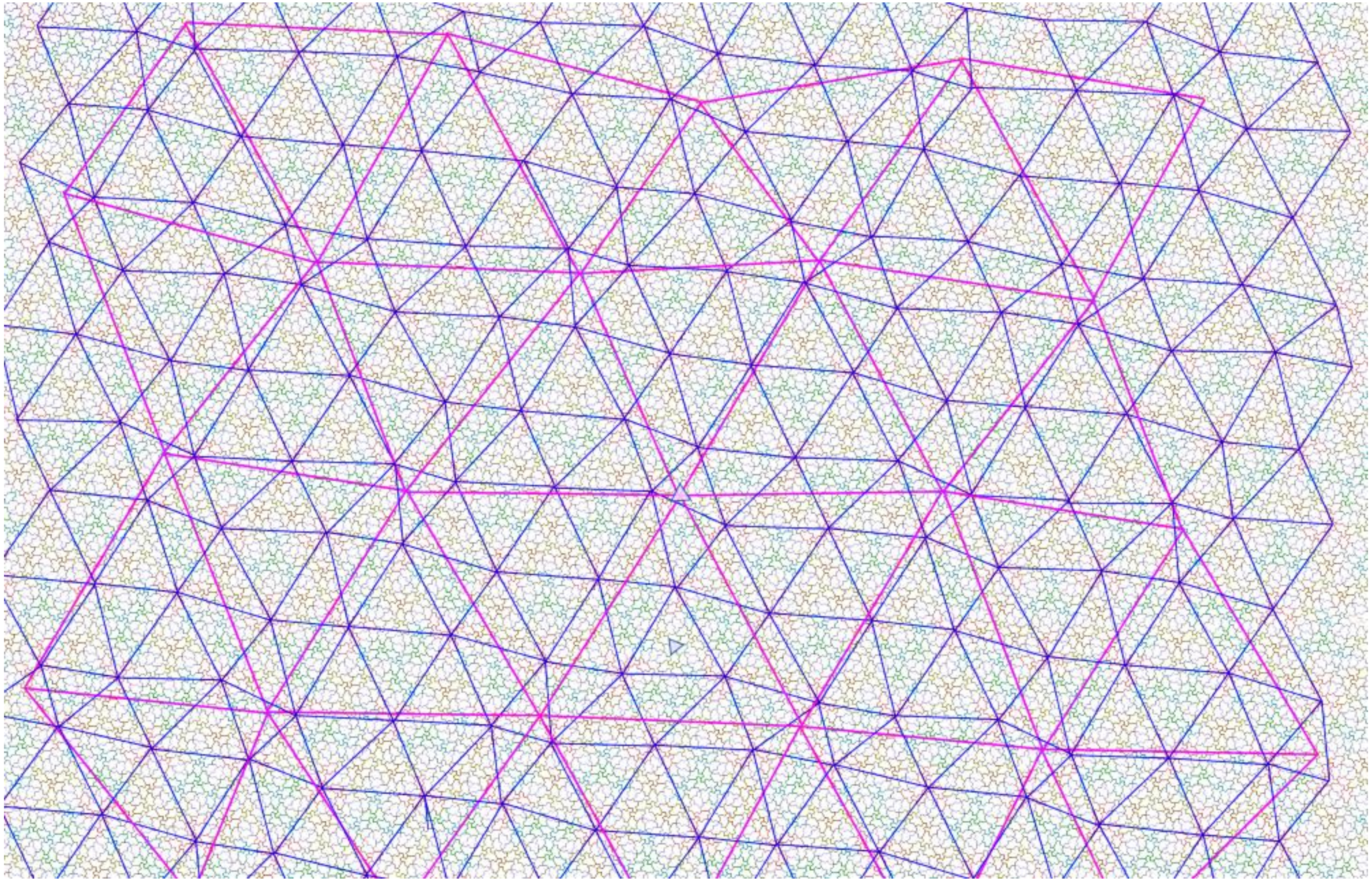
six orientations of the 3-arms



New puzzle

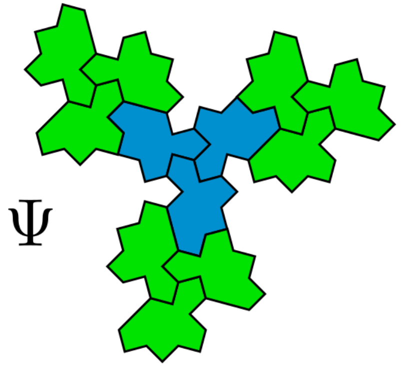


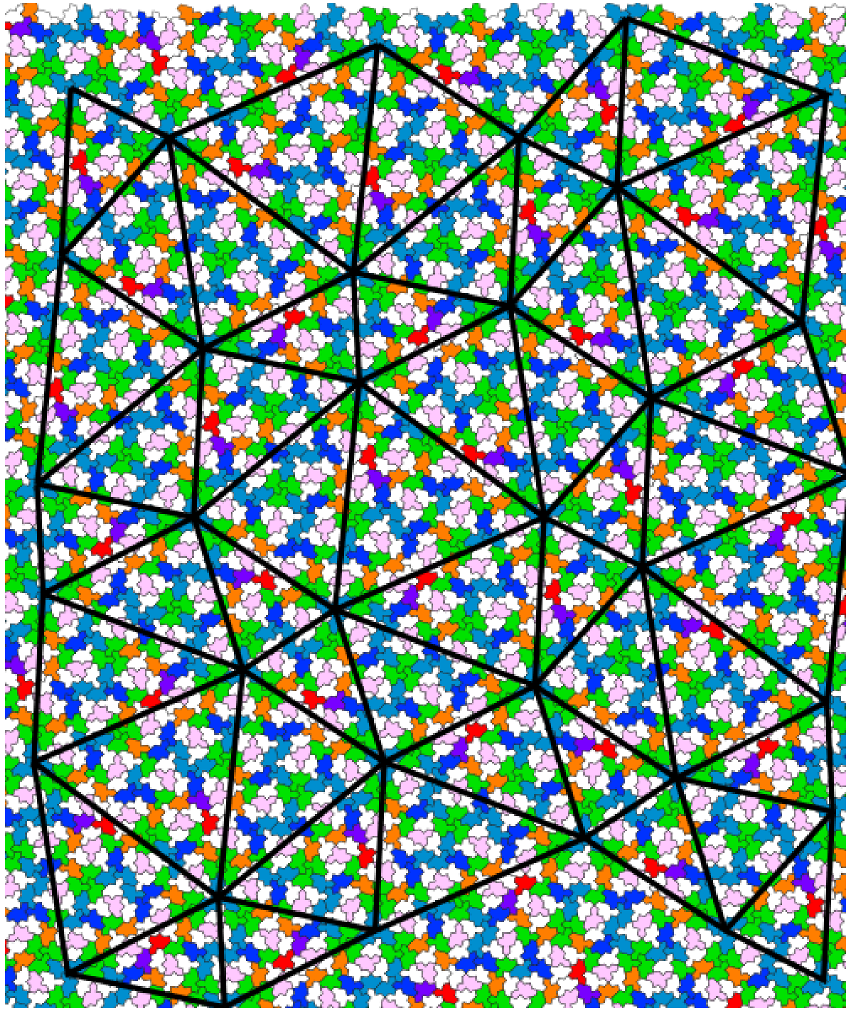
Triangular array by Boris Horvat



blue color: triangular array between the “lone stars”

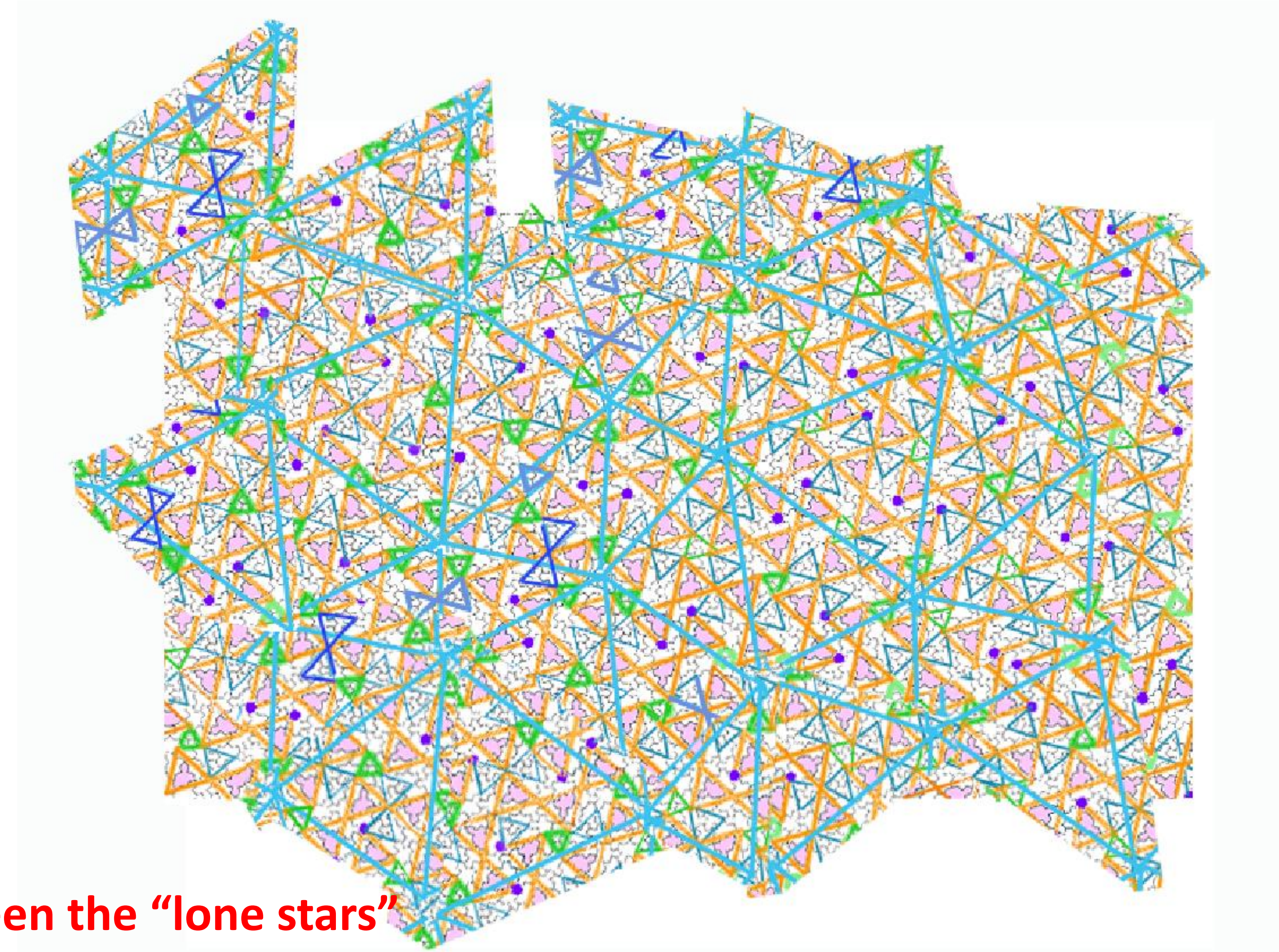
Inflation using python code





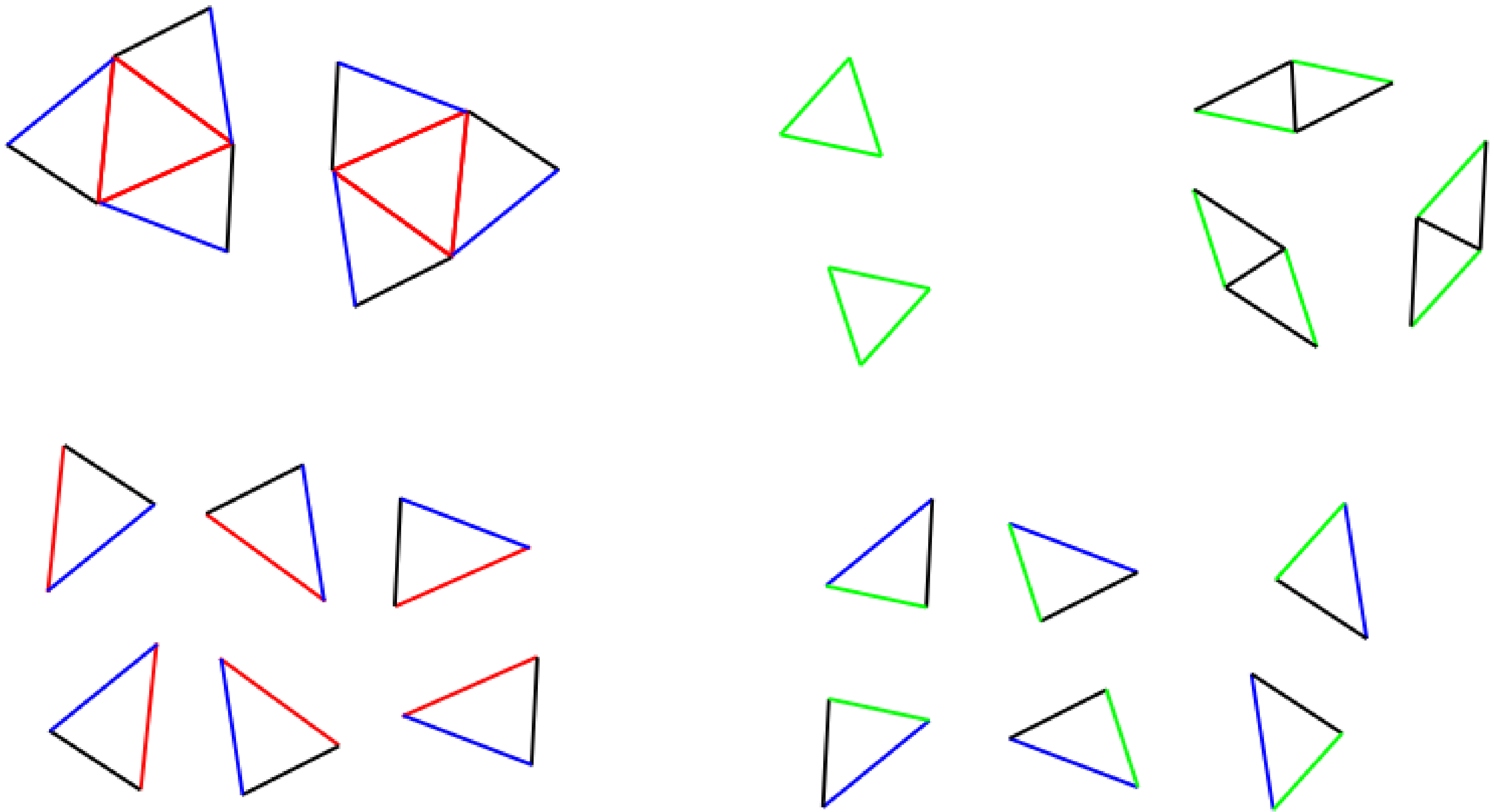
- Triangular array between large blue stars made of three \psi tiles surrounded by three S2 green stars

Triangular networks by Jean-François



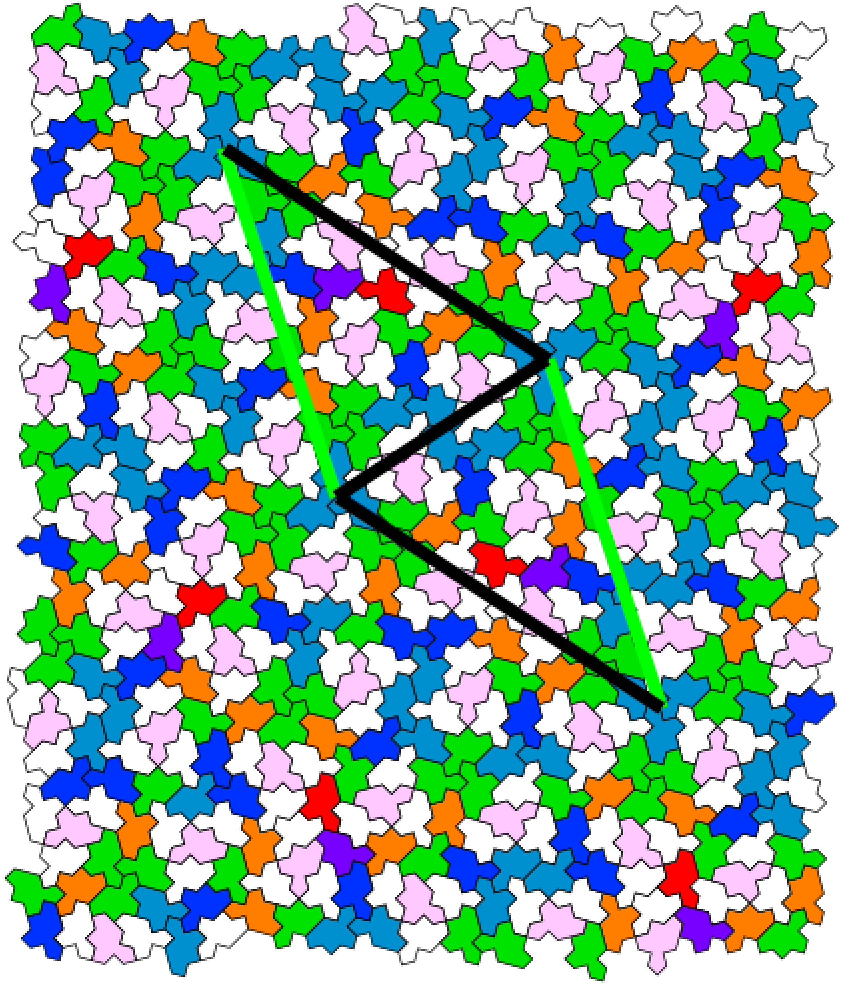
Same network between the "lone stars"

A finite set of triangles ☺

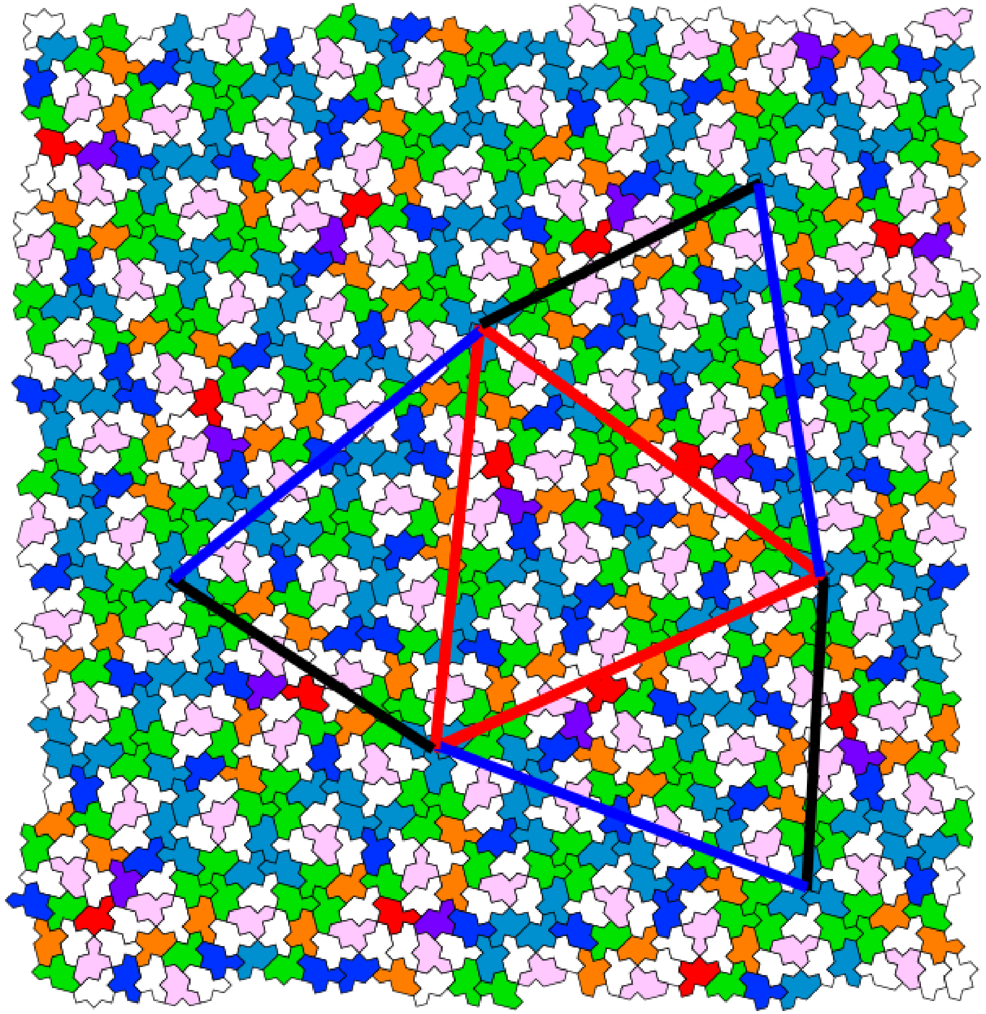


5 different triangles with two regular ones -> Calculate the edge's lengths ?

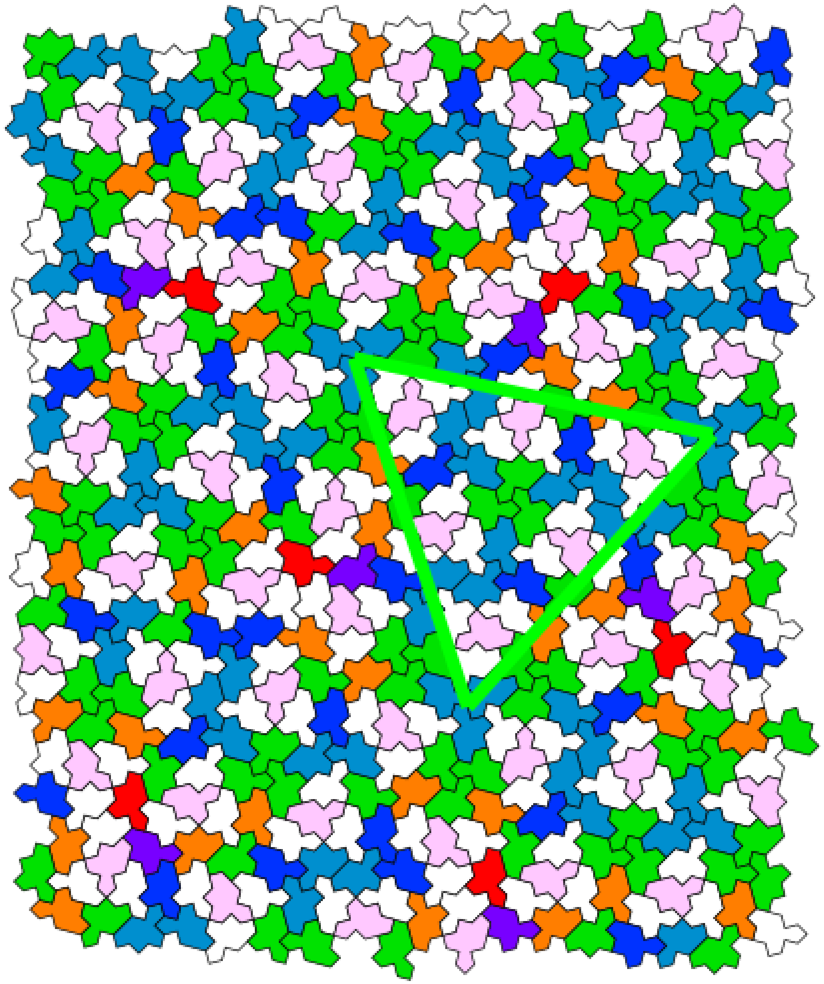
Set of triangles: diabolo



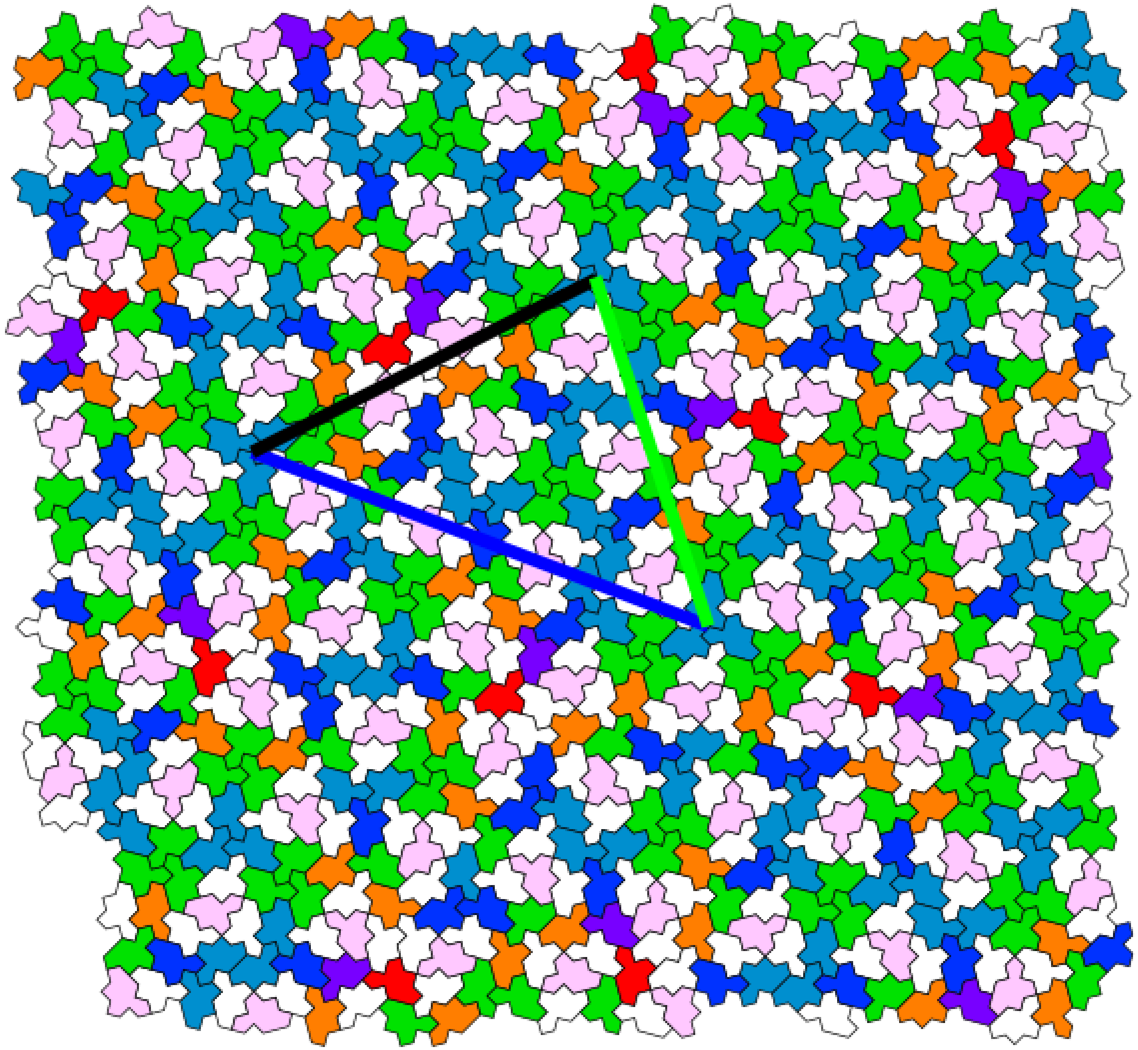
Set of triangles: three arms



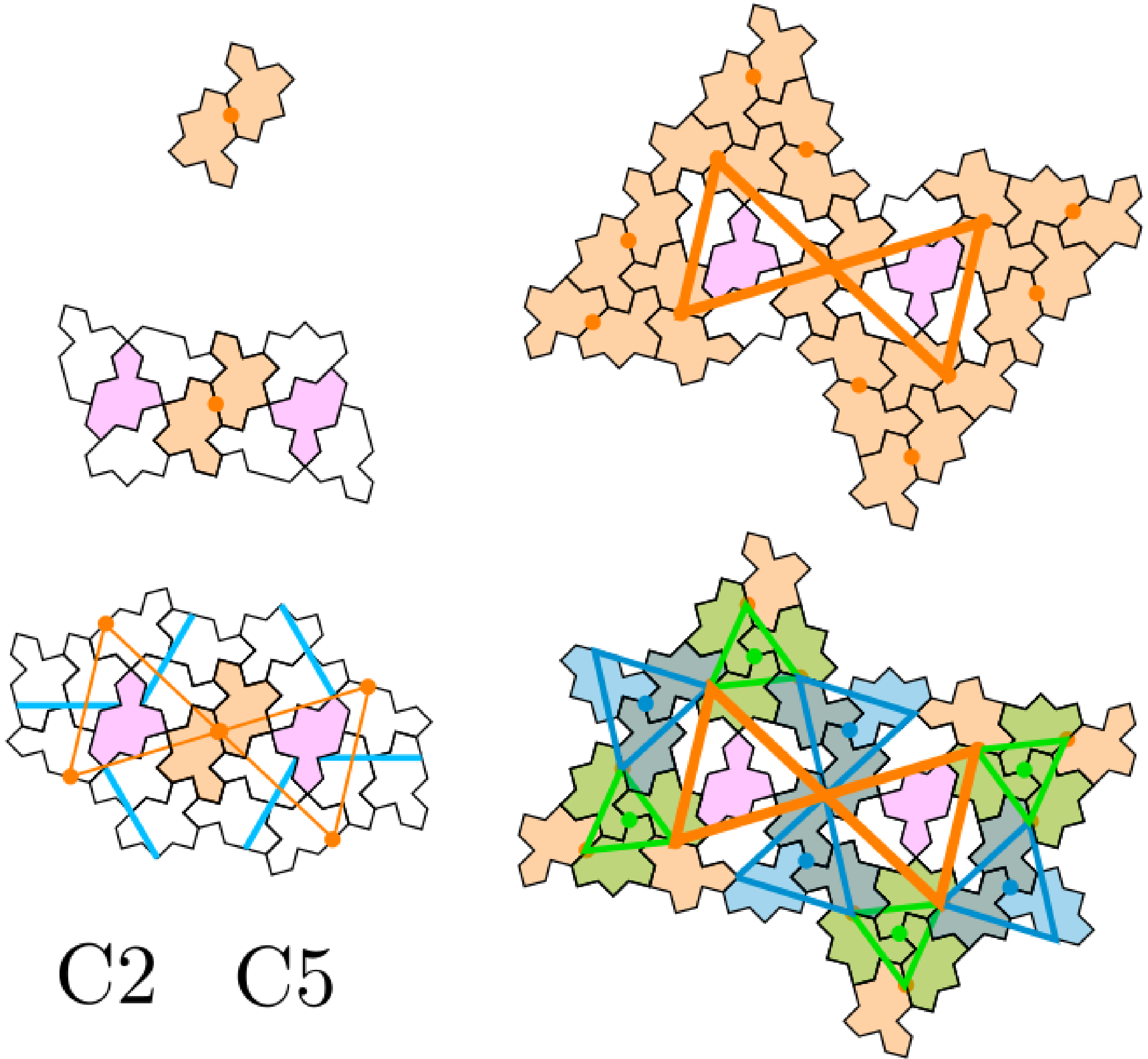
Set of triangles: green triangle



Set of triangles: last triangle

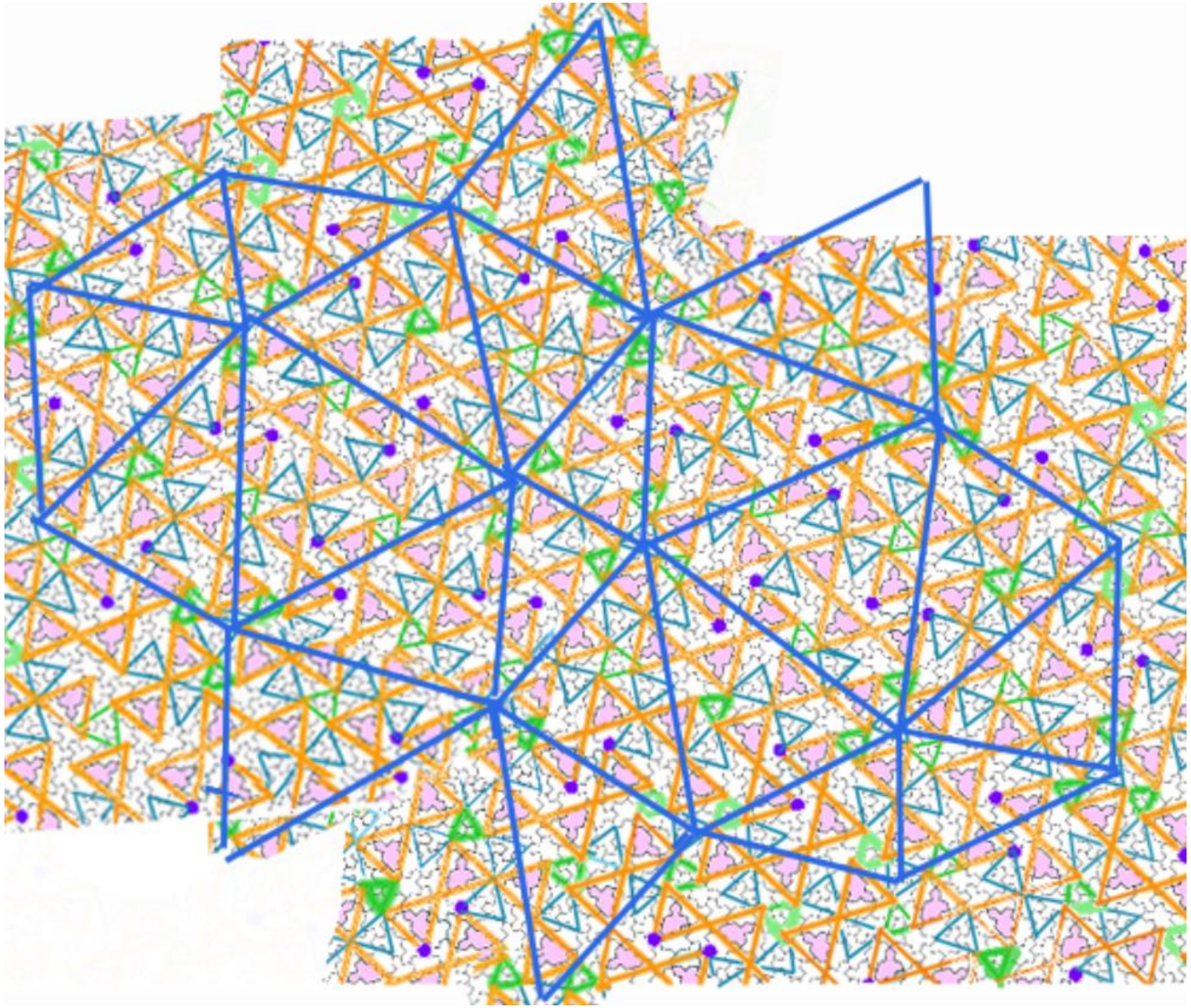


Global 2-fold symmetry: center of a diabolo pattern

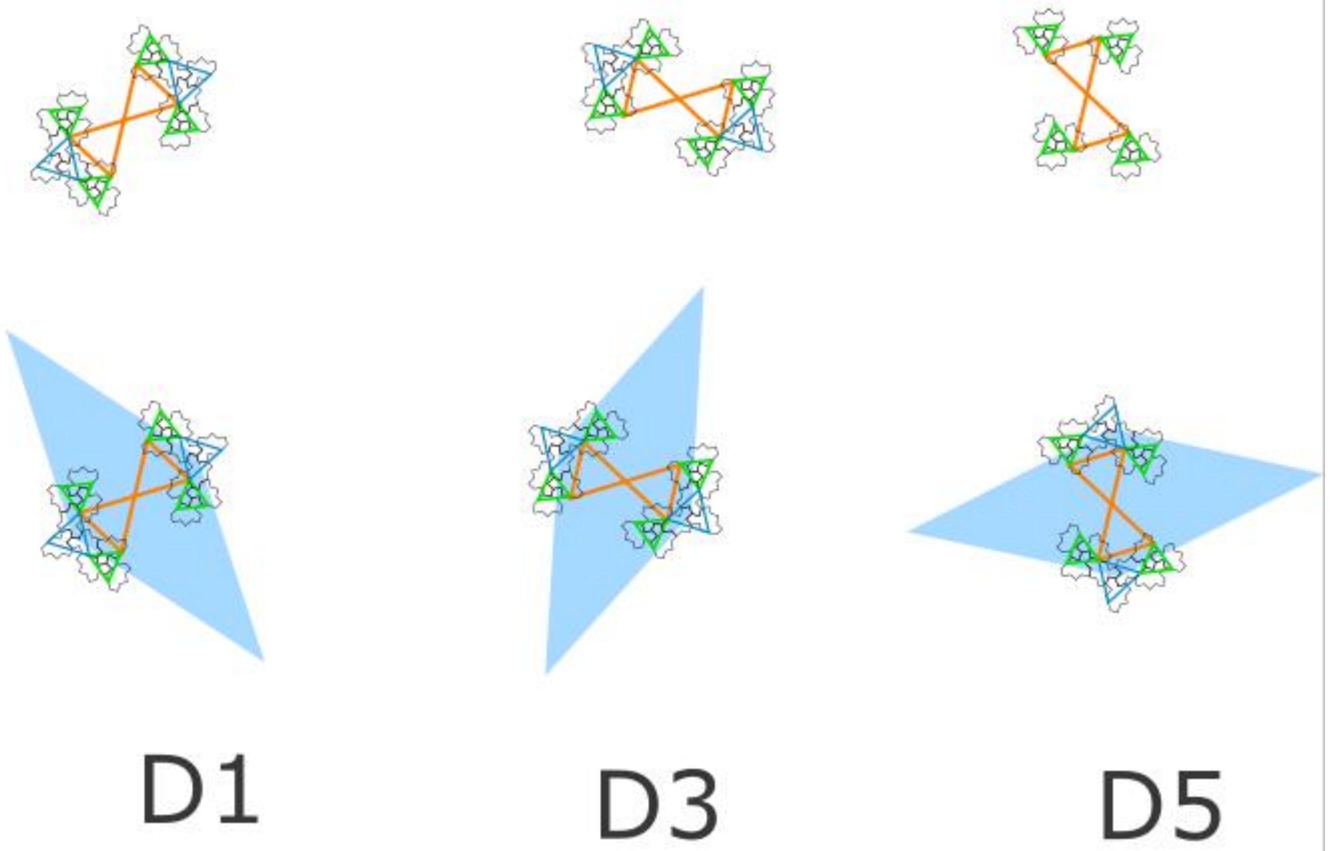


- From local to global 2-fold symmetry ???

Global 2-fold symmetry: center of a diabolo pattern

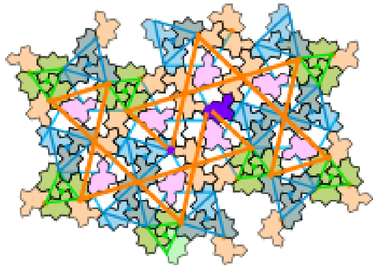


Rhombi for diabolo

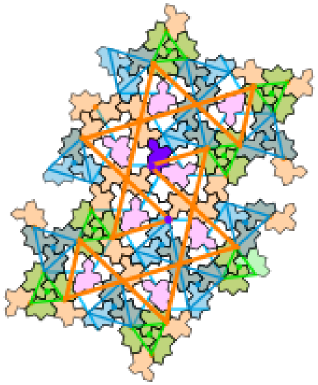


Six ORIENTATIONS OF 2-arms pattern

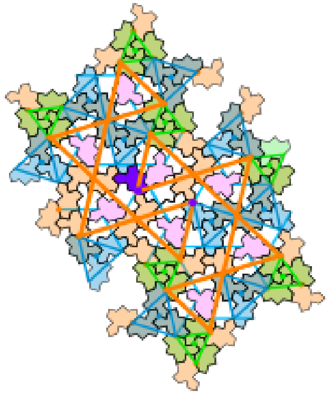
P1

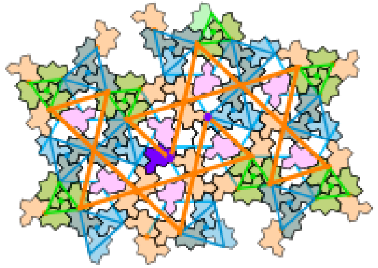


P2

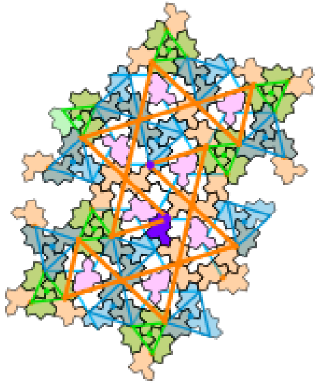


P3

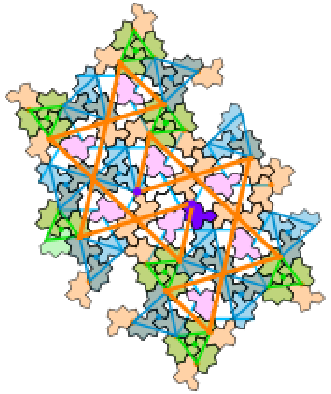




P4



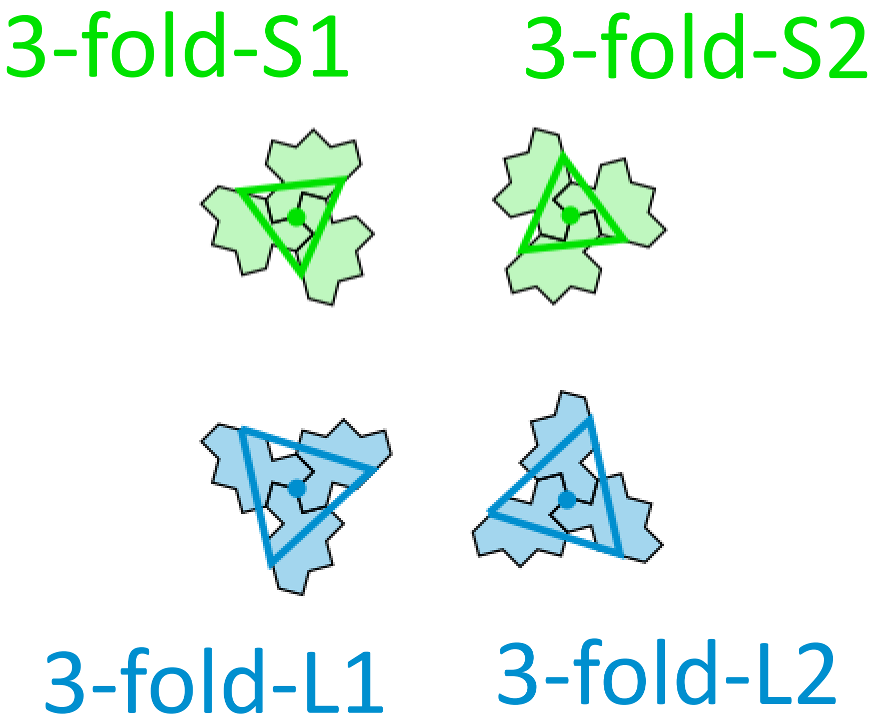
P5

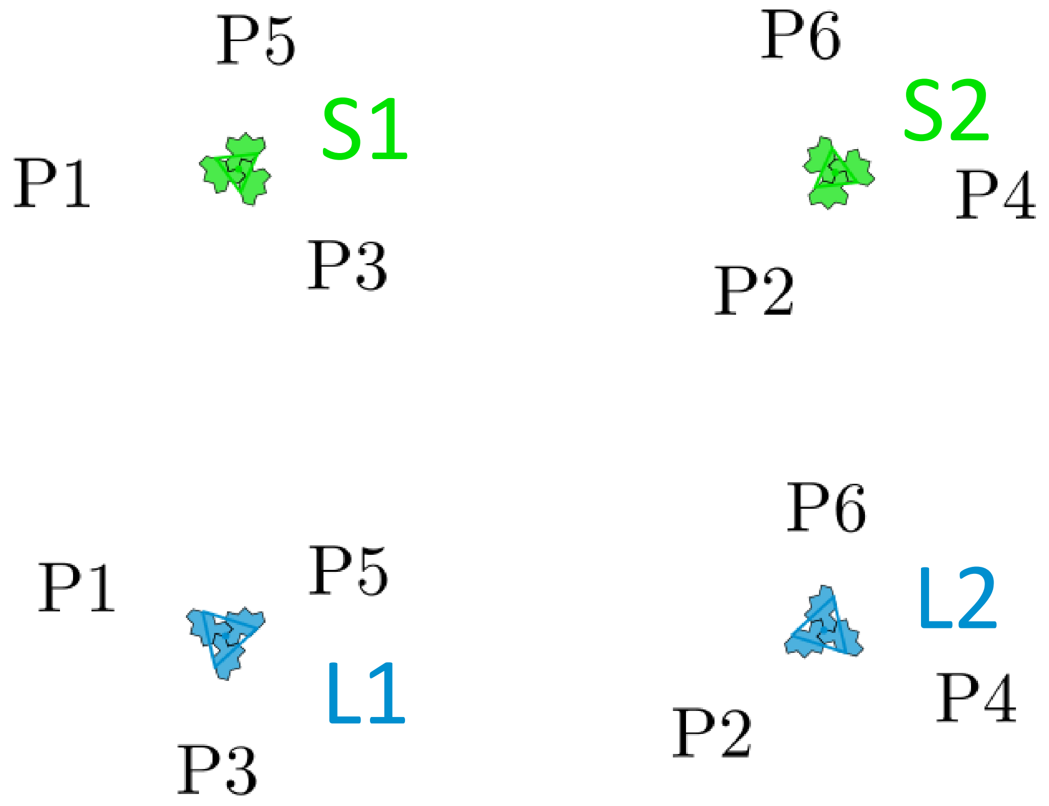


P6

FOUR possible seeds with 3-fold symmetry

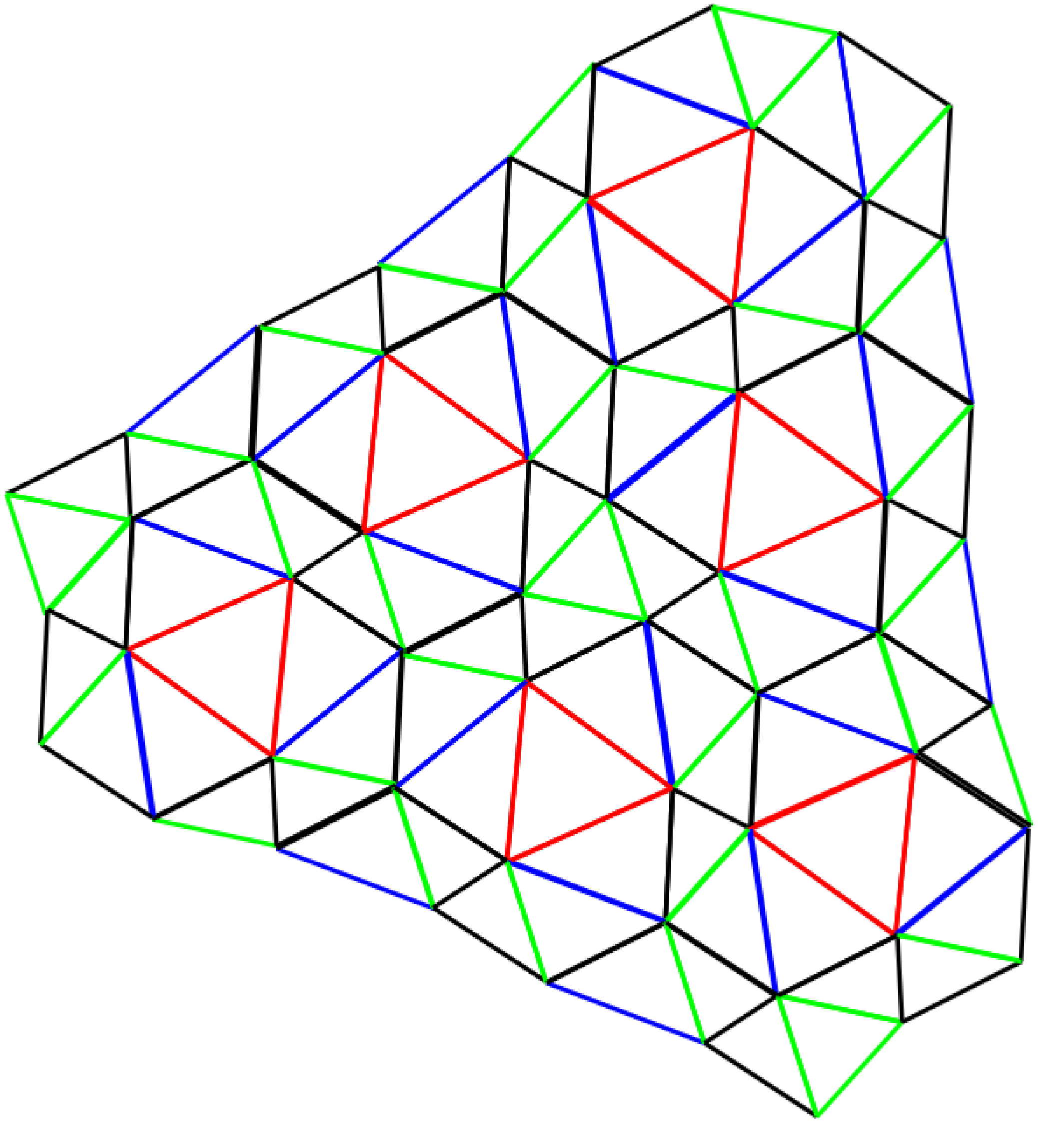
- COMBINE Three 2-arms rotated from each other by 2pi/3

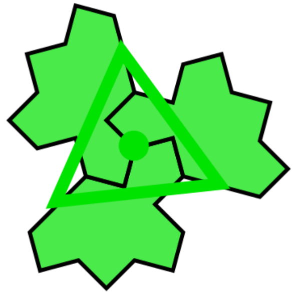


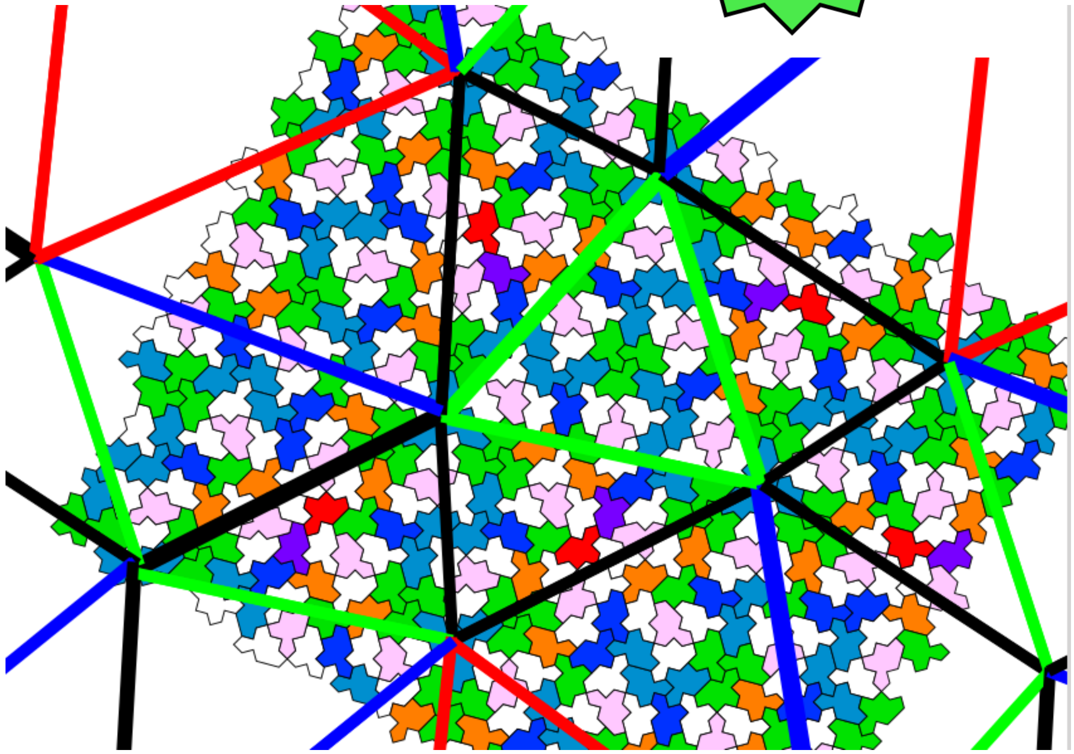


- Question: are all of them present in the infinite tiling ???

Using set of triangles

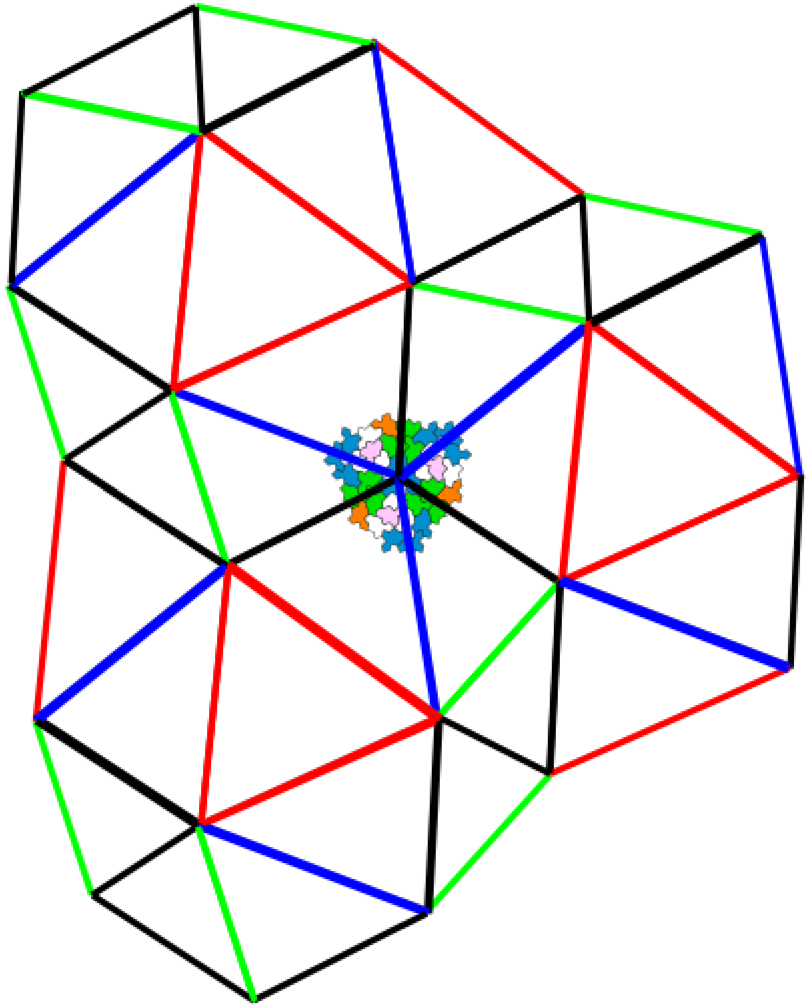




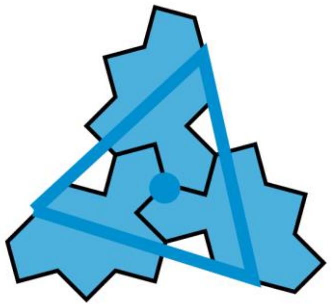


3-fold-S2

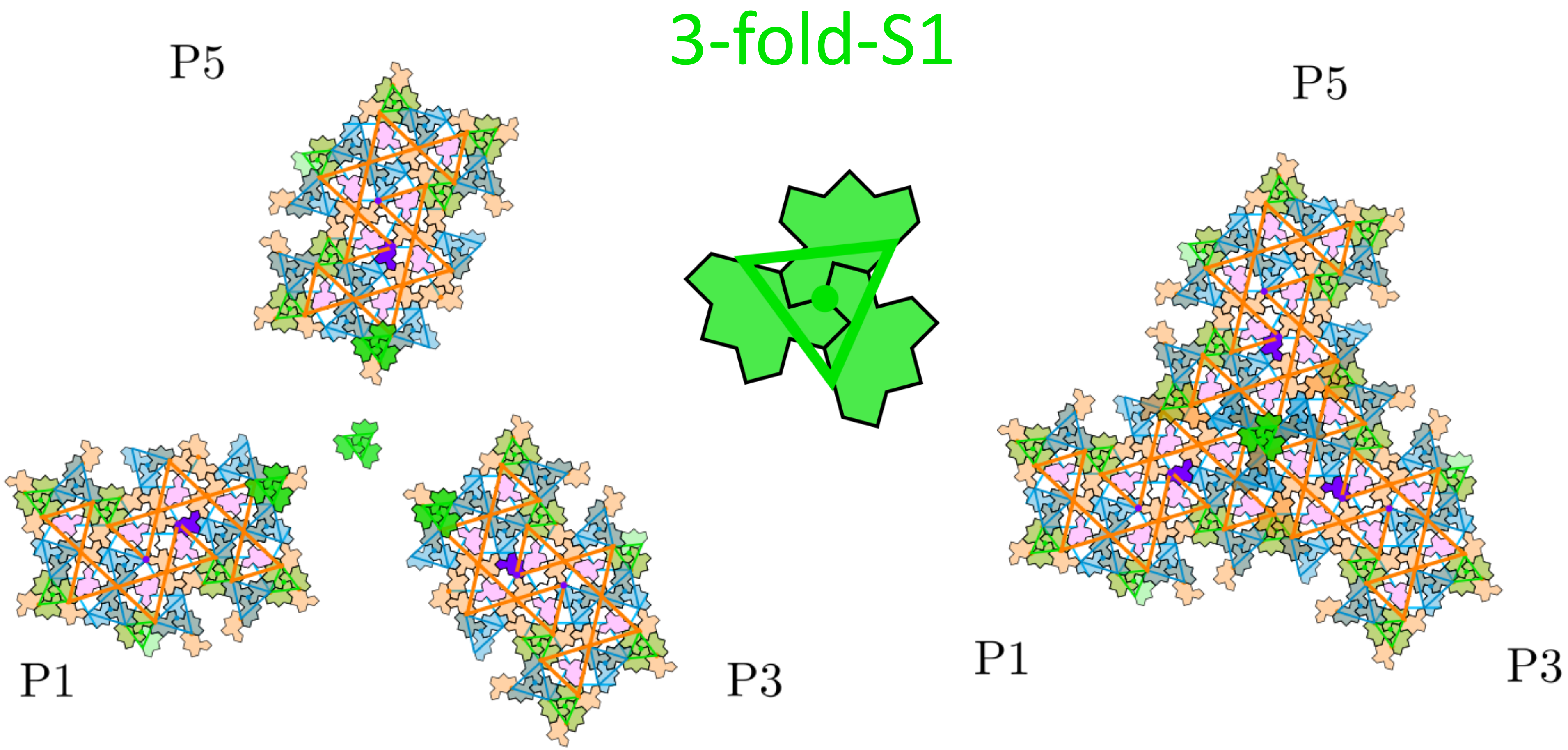
Using set of triangles



3-fold-L2

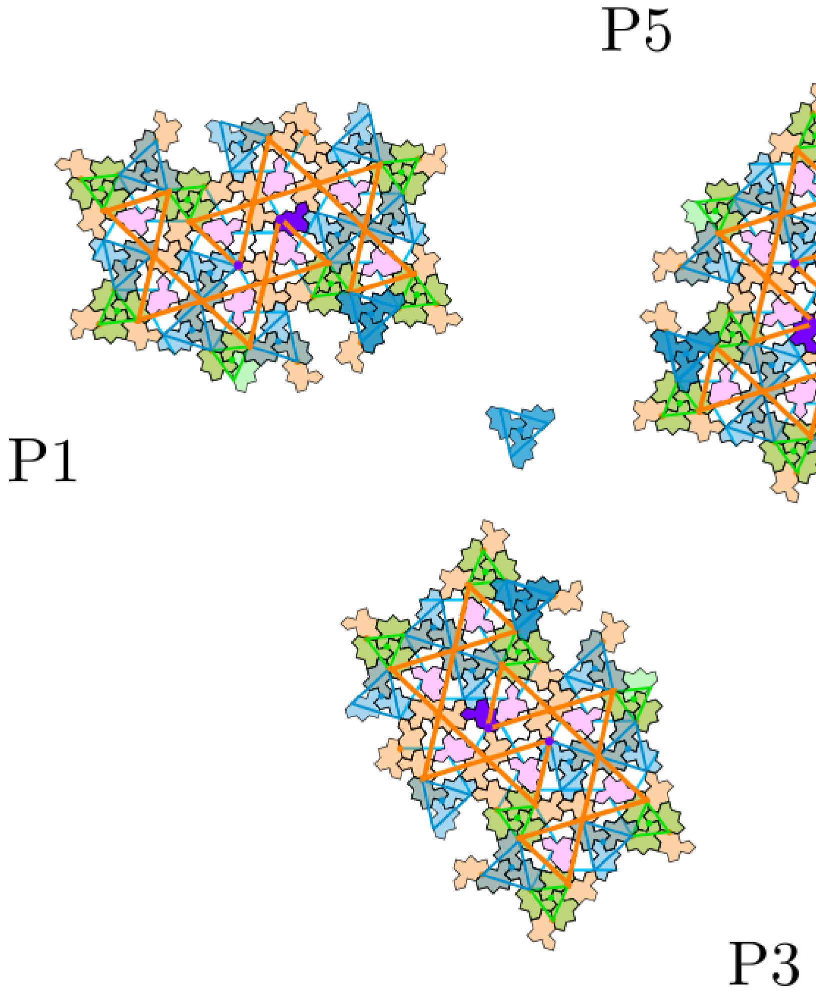


Seed 1 with 3-fold symmetry

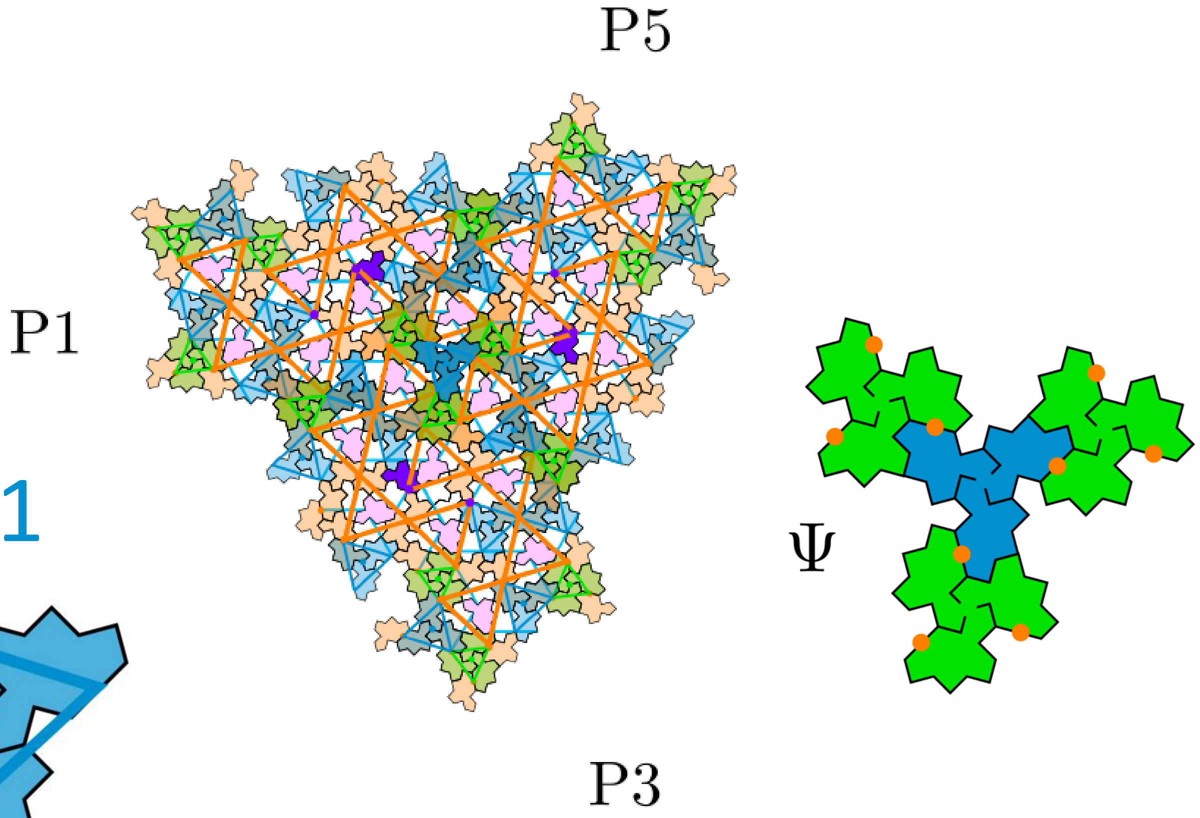


- Green star S1 at the center

Seed 2 with 3-fold symmetry

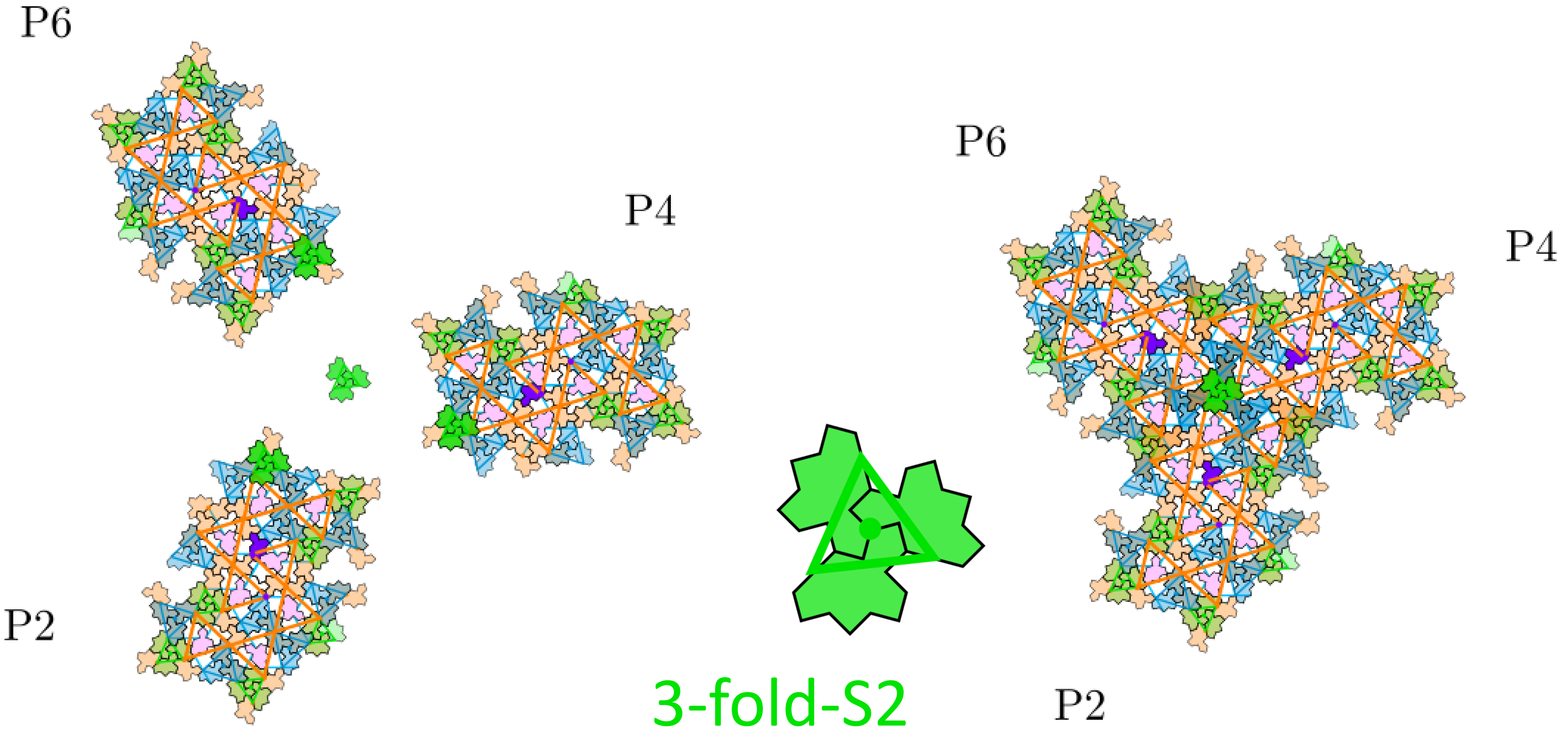


3-fold-L1



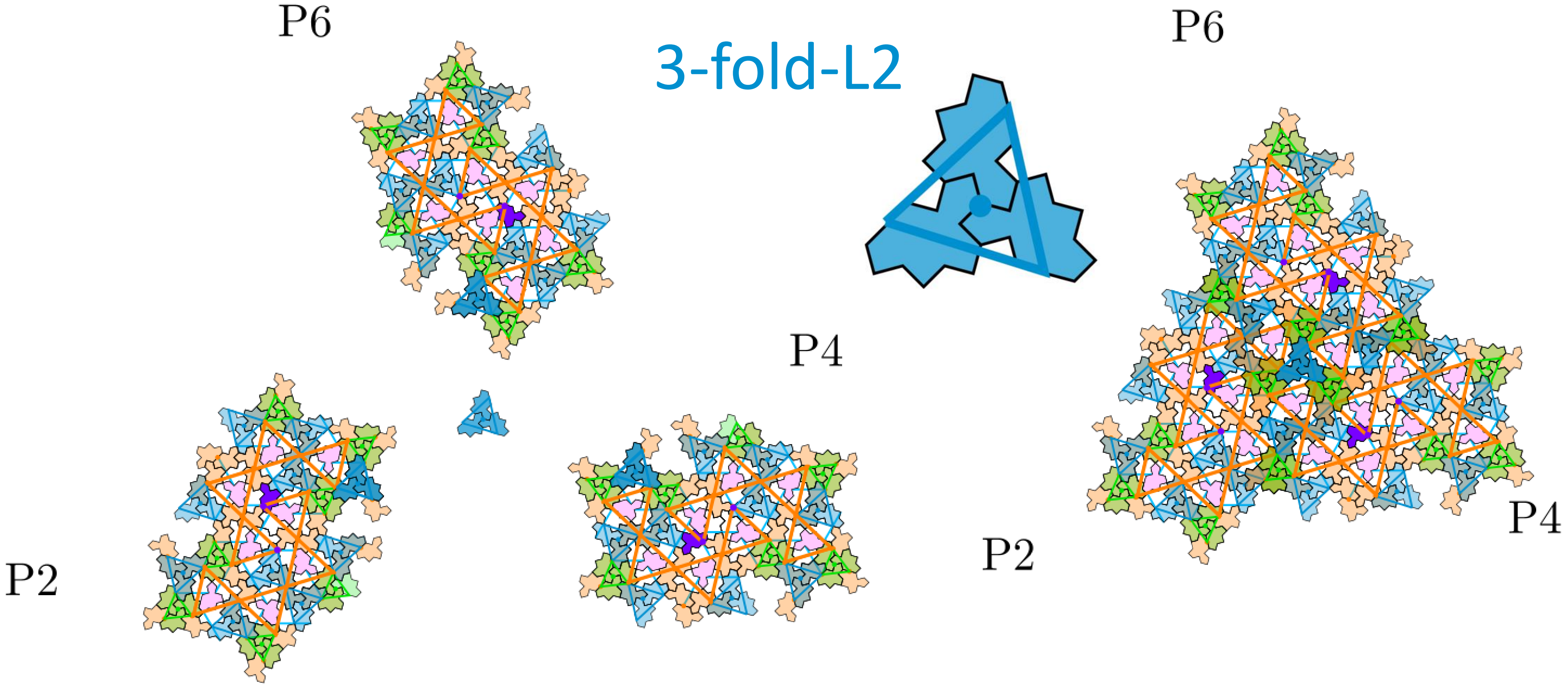
- blue star L1 at the center
- Lone blue star surronded by three S2 green stars

Seed 3 with 3-fold symmetry



- Green star S2 at the center

Seed 4 with 3-fold symmetry



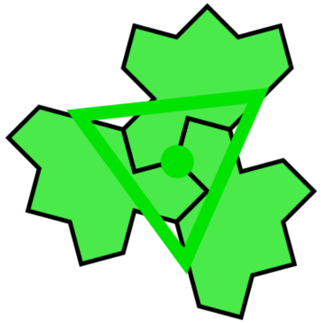
- blue star L2 at the center

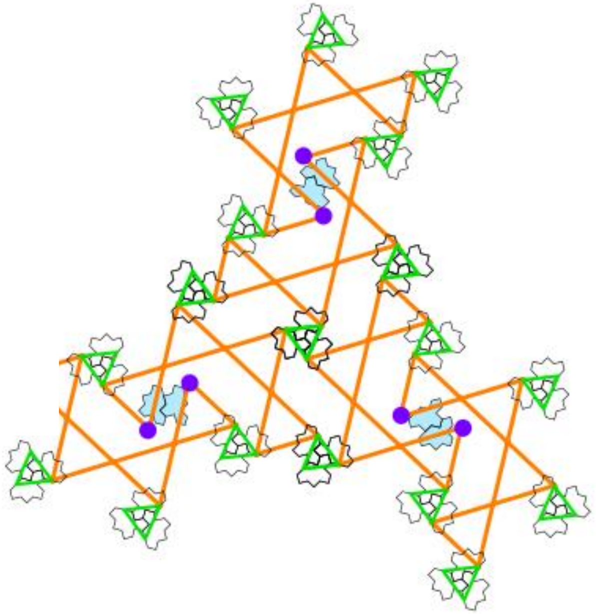
Two possible seeds with global 3-fold symmetry

seed1

- Green star at the center

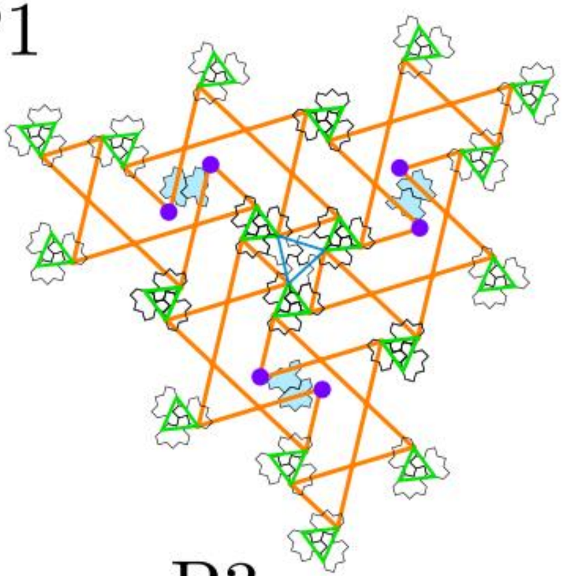
3-fold-S1





seed2

- blue star at the center

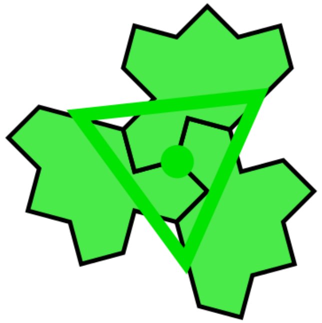


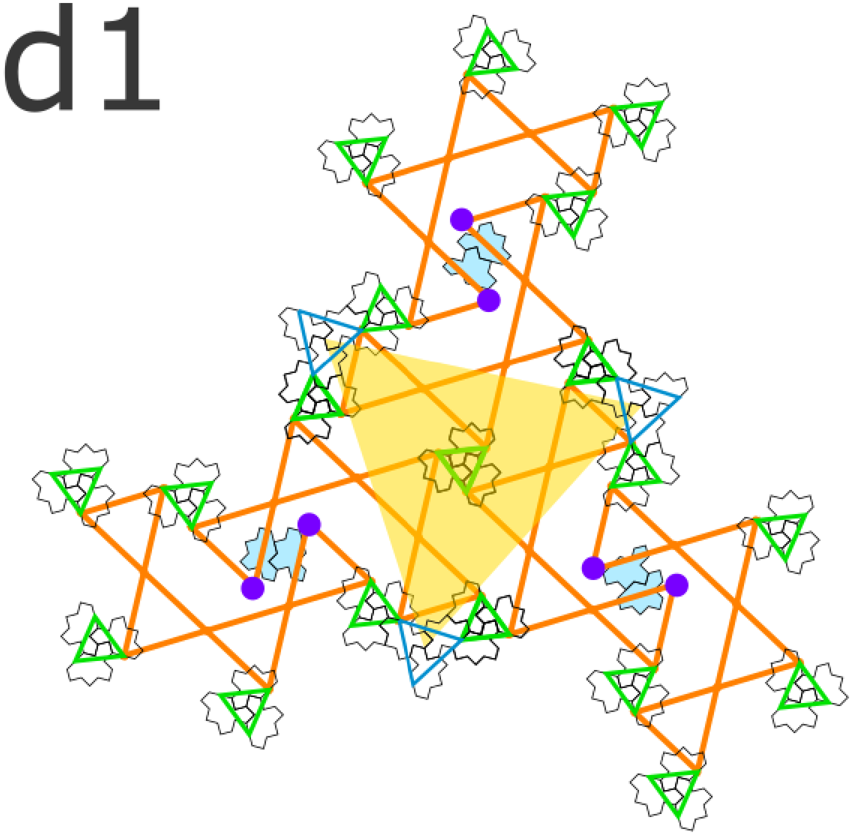
3-fold-L1

- Three 2-arms rotated from each other by 2pi/3

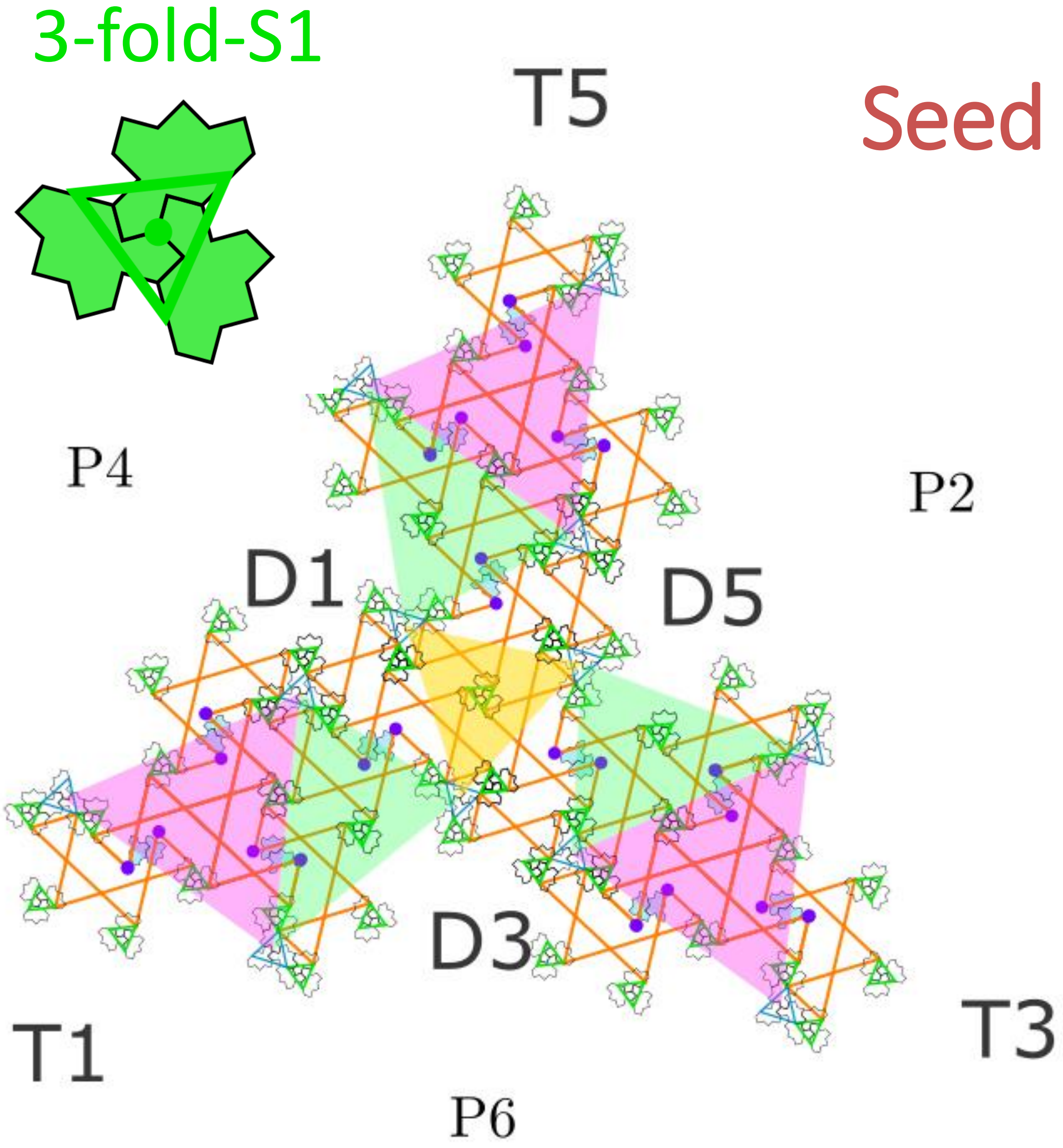
Seed 1

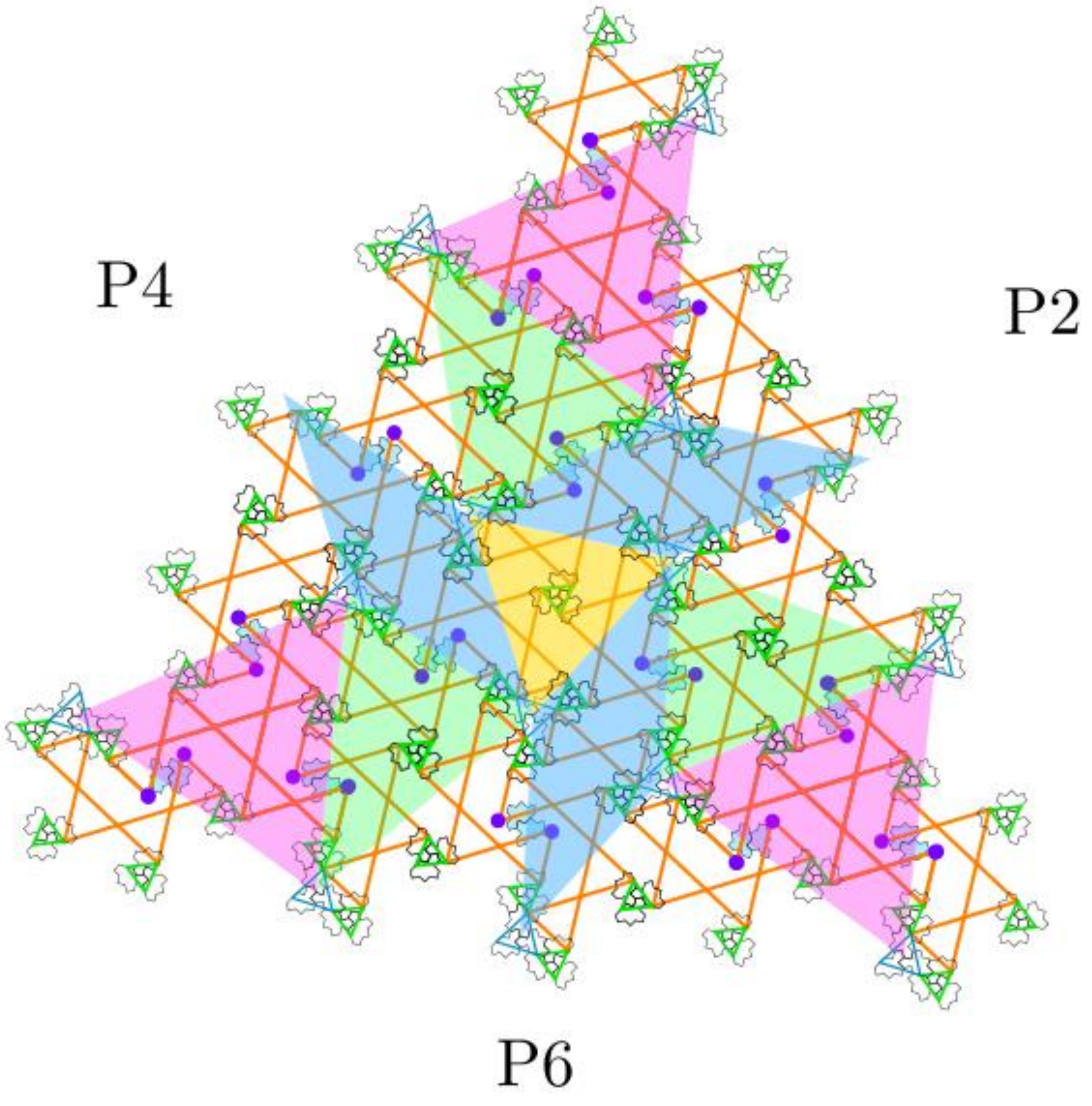
3-fold-S1



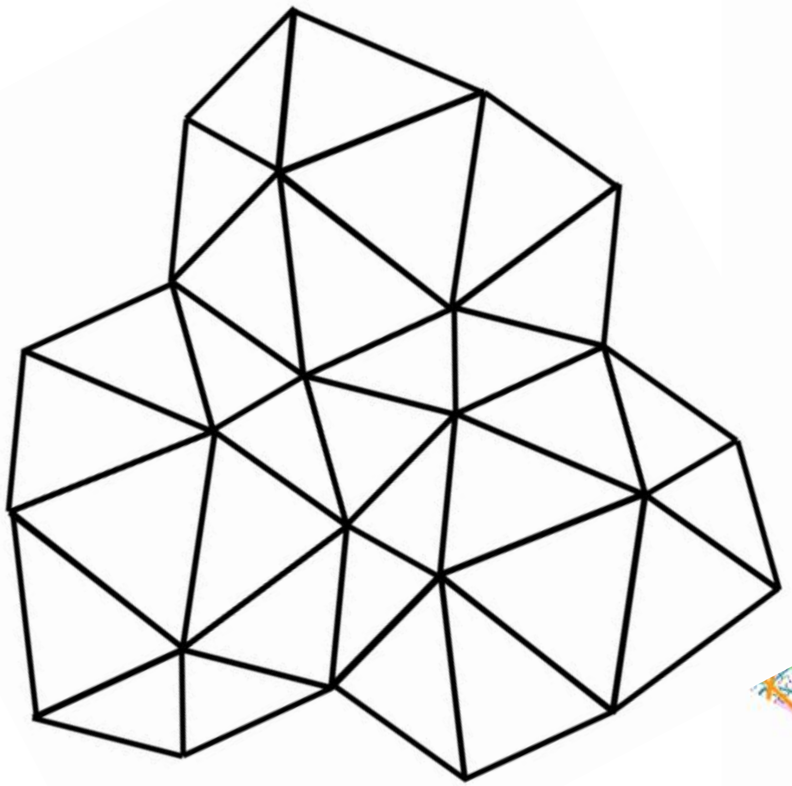


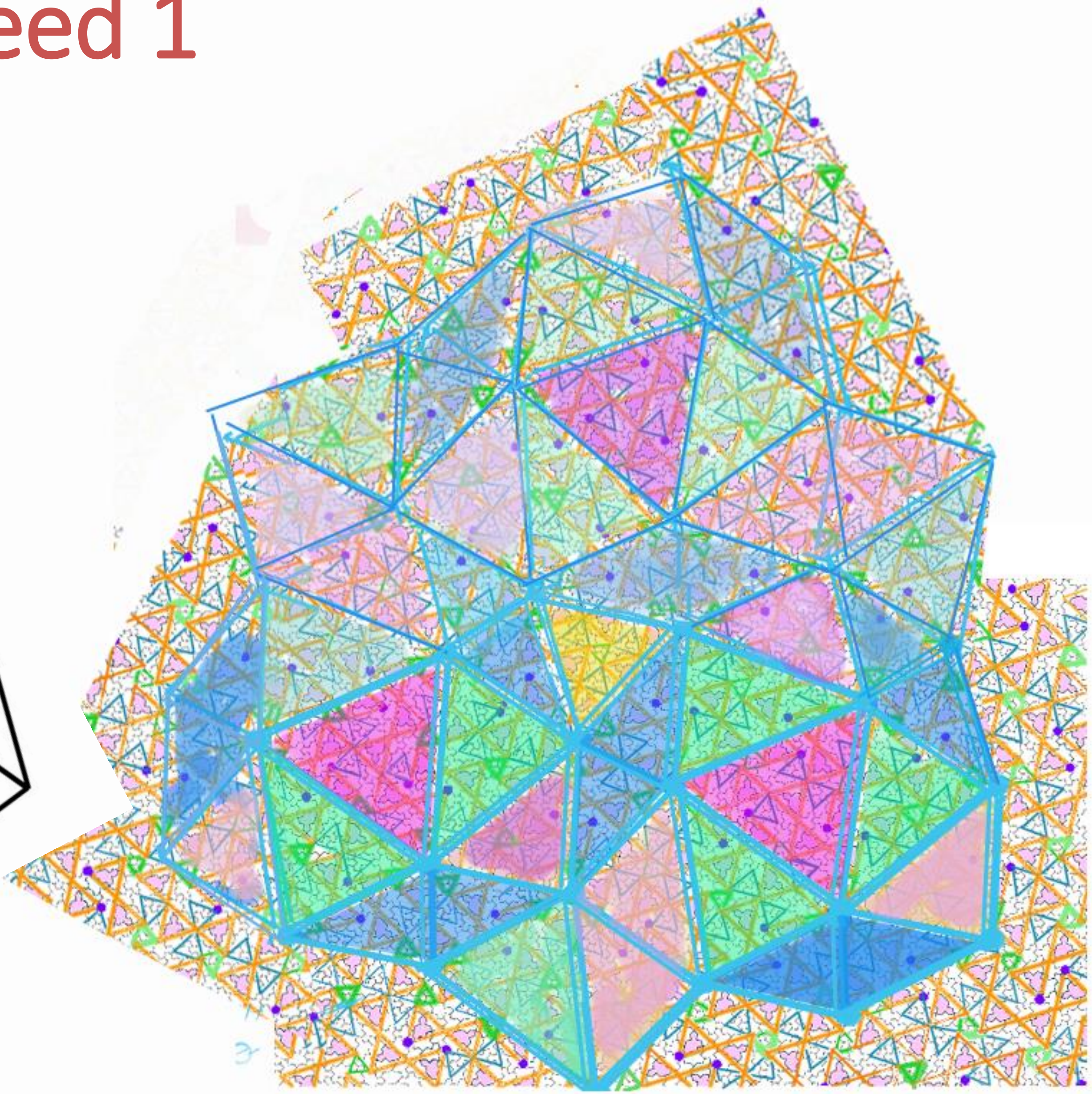
Seed 1

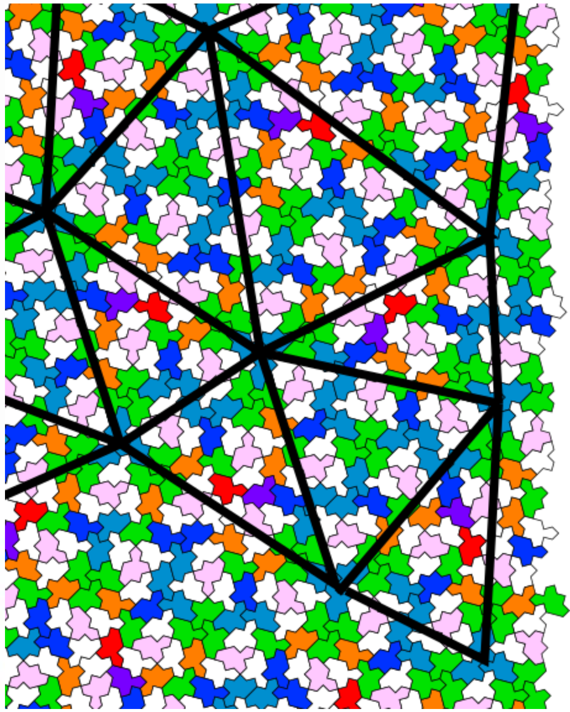




Seed 1

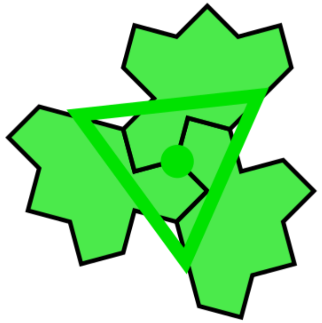


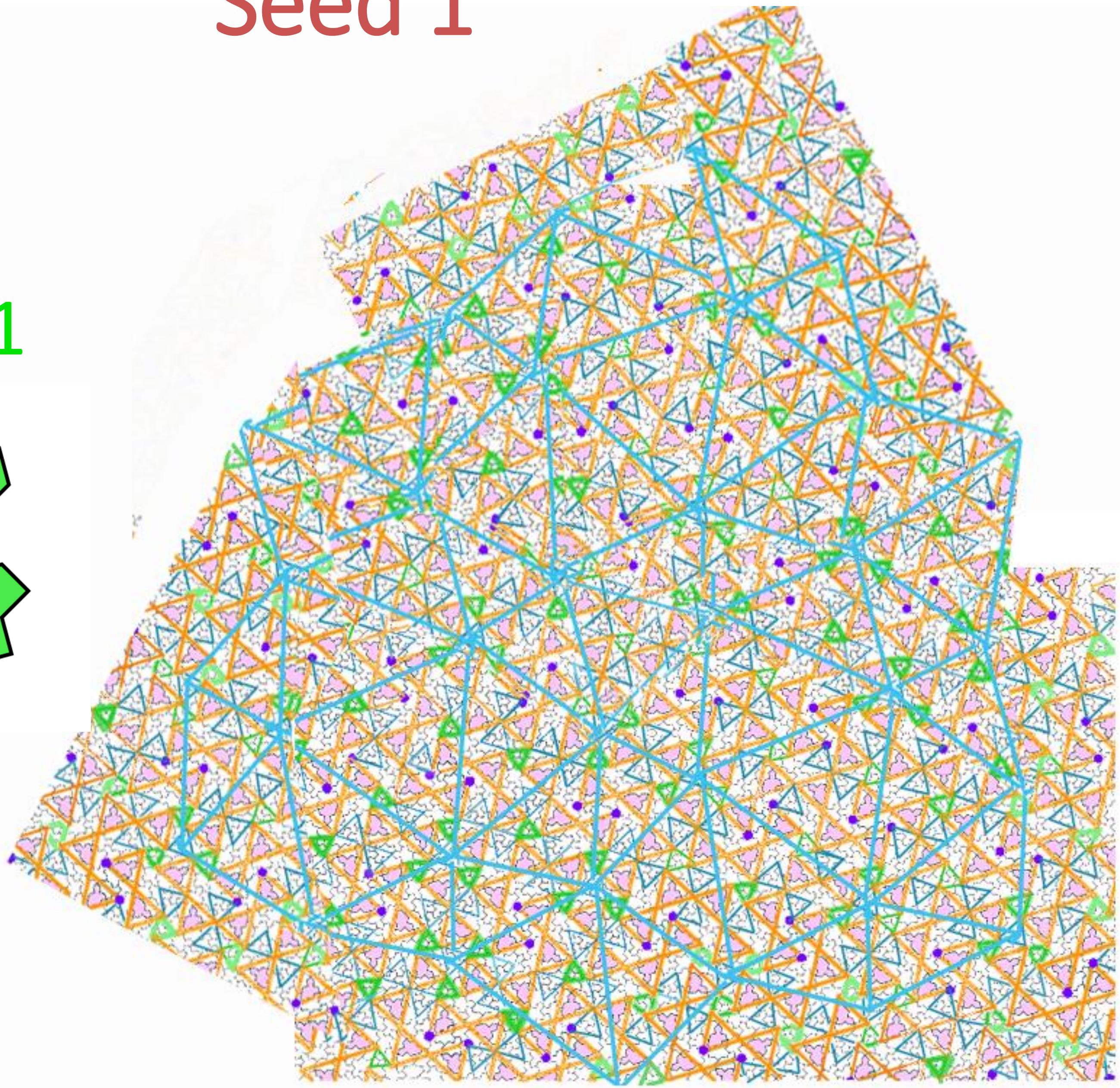


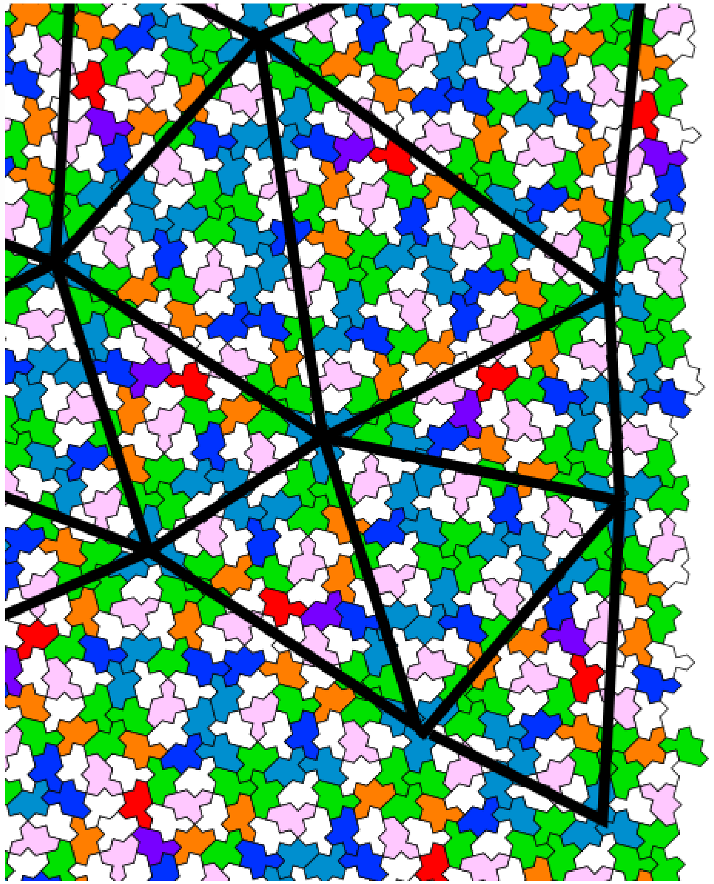


Seed 1

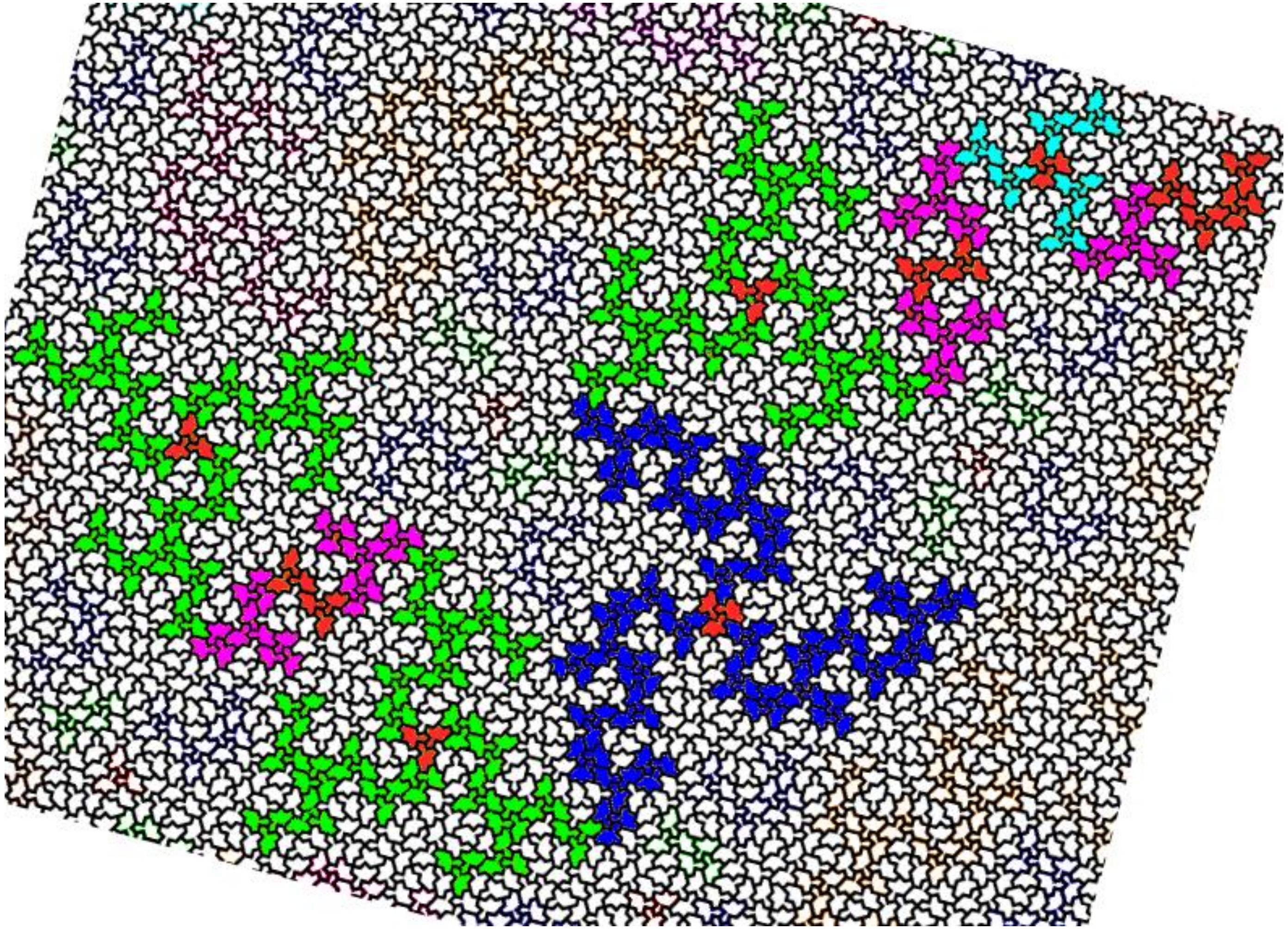
3-fold-S1





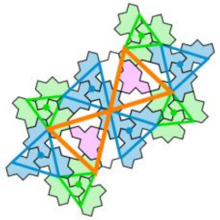


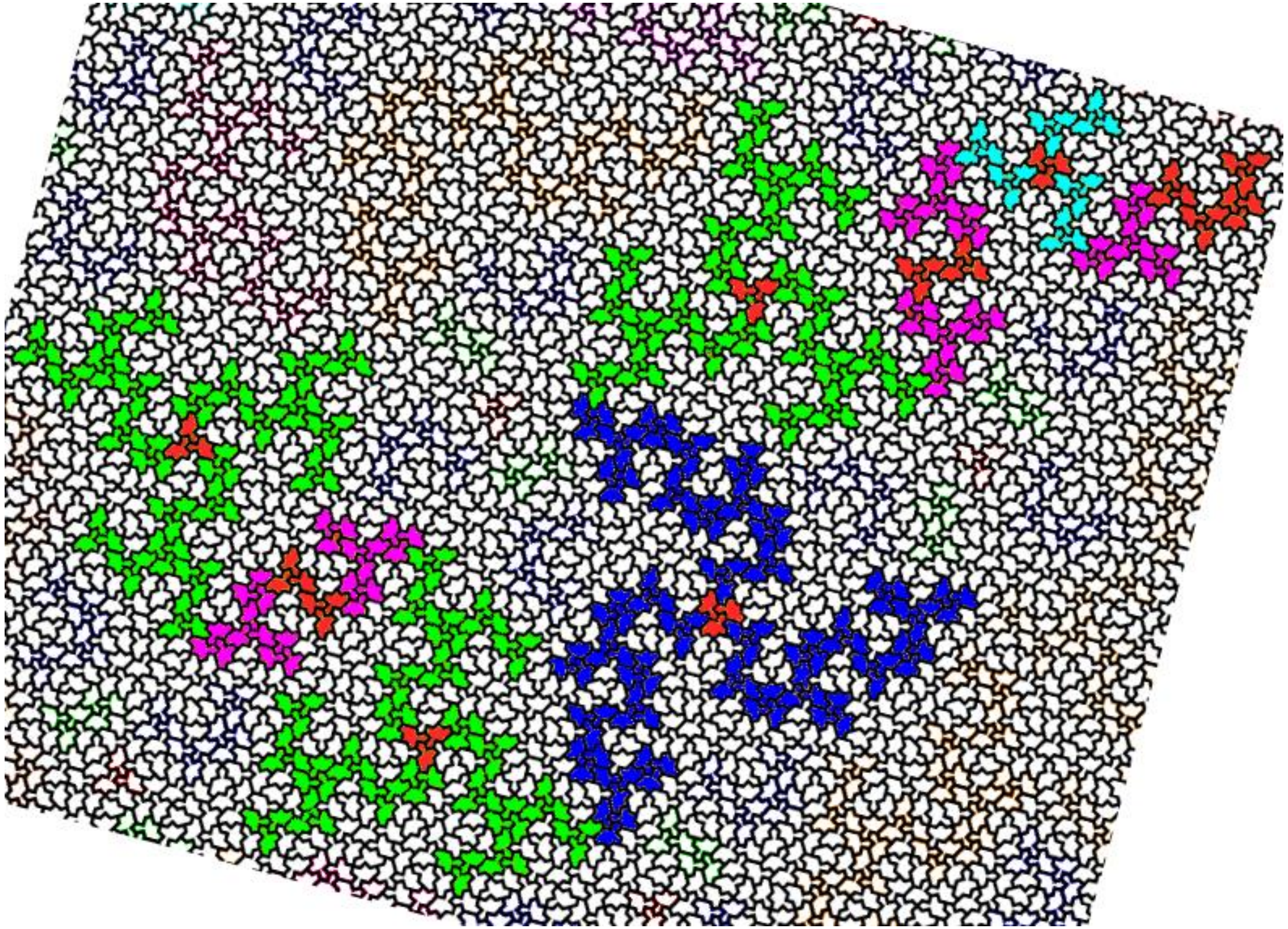
Star clusters by Boris Horvat: example of cluster 147

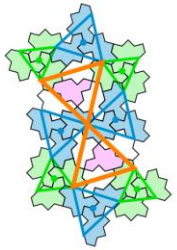


- Mirror + rotation of pi/12 for comparision

cluster 147: two diabolos







- -> to finish … Identify the type of the local 3-fold symmetry centers

Results so far

- Local 3-fold and 2-fold symmetry centers are identified
- Violet points -> different way to identify (chimera …)
- Analysis of the size/amount of stars clusters depending on the level (Boris Horvat)
- Different patterns can be used to decompose the whole tiling:
- - 2-arms plus 3-arms+diabolo
  - full clusters
  - versions of ‘anatomy’ etc …

Helpful to identify the global symmetry centers !

- Identifcation of 2-fold global symmetry center (center of diabolo figure)
- two possible 3-fold global symmetry centers: center of green triangle and center of blue triangle. They correspond to three 2-arms (P1/P3/P5) rotated by 2pi/3 around the center)
- triangular array of blue stars surrounded by three green stars

TO DO: inflation schemes around symmetry centers

Different possible methods should give the same result

- H7/H8 (by S. Tathan) applied to a symmetric subset of clusters

-> might by the most straightforward method for implementation as code is already running (but in RUST language) ?

- Full clusters with translations (Marianne …)

-> will need more work to do a complete python implementation

- Substitution rules for large colored triangles (JFS)

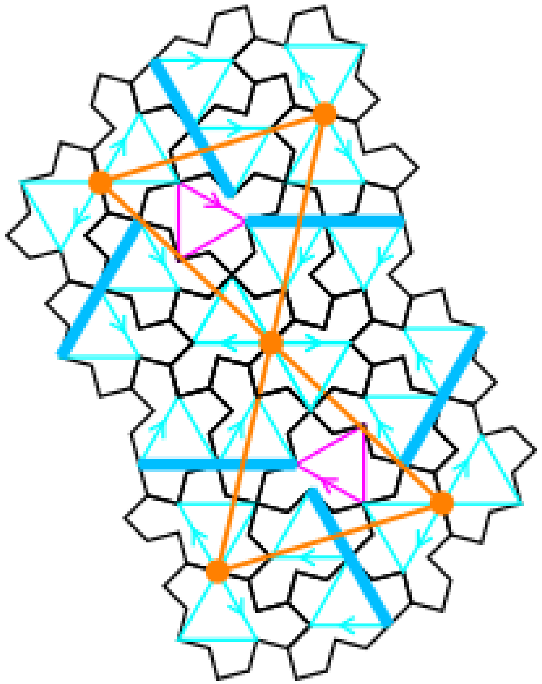
-> will need more work to figure out the substitution rules

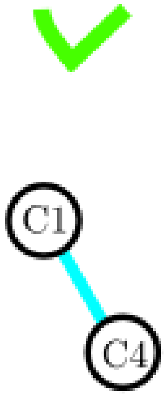
- generate only the star clusters based on 2-fold and 3-fold local rotations and some additionnal rules ???

-> to test the idea first …

Homochiral inflation: use H7/H8 by Simon Tathan ?

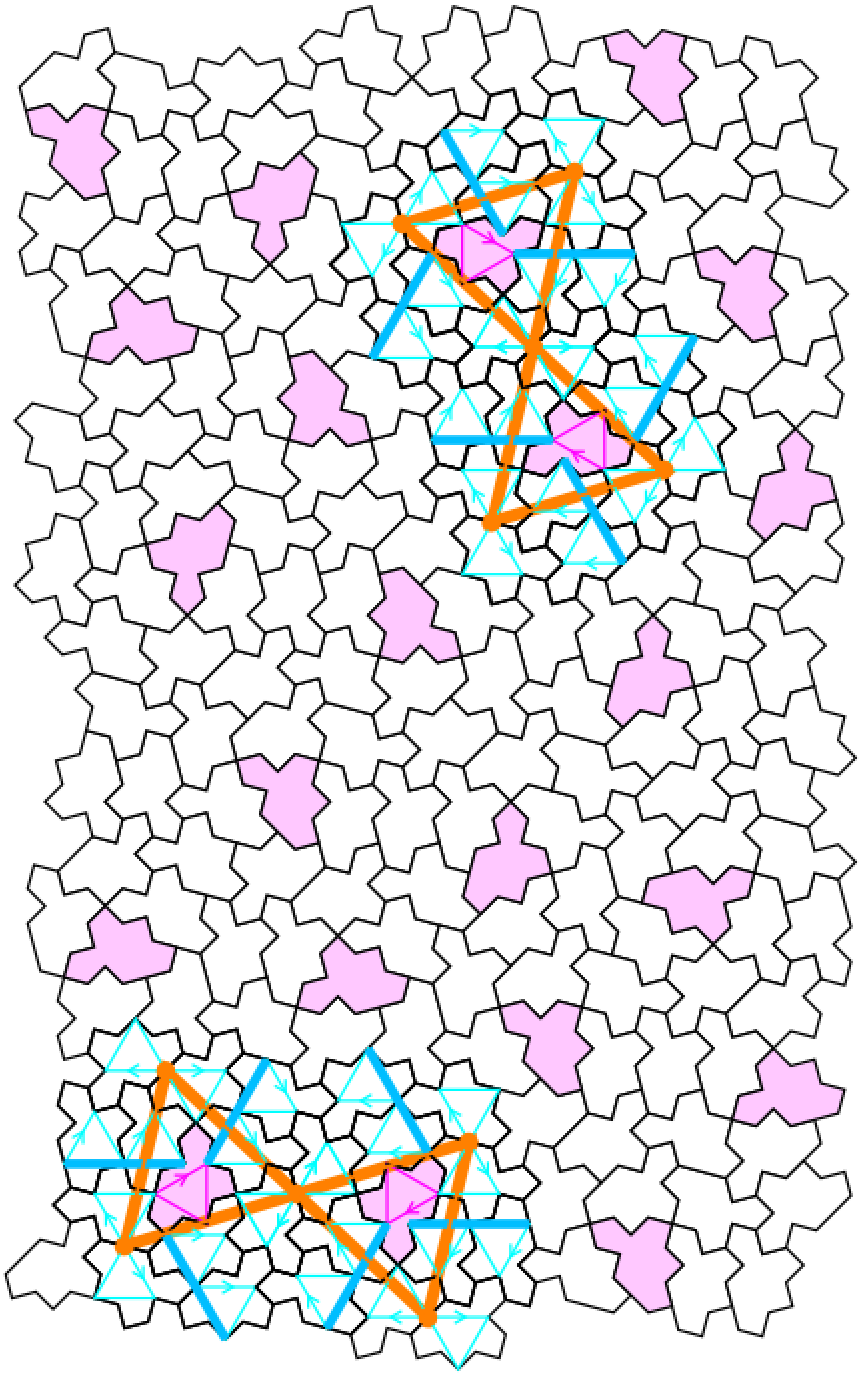
- For 2-fold symmetry, apply inflation to two clusters in C1-C4 relative orientations :

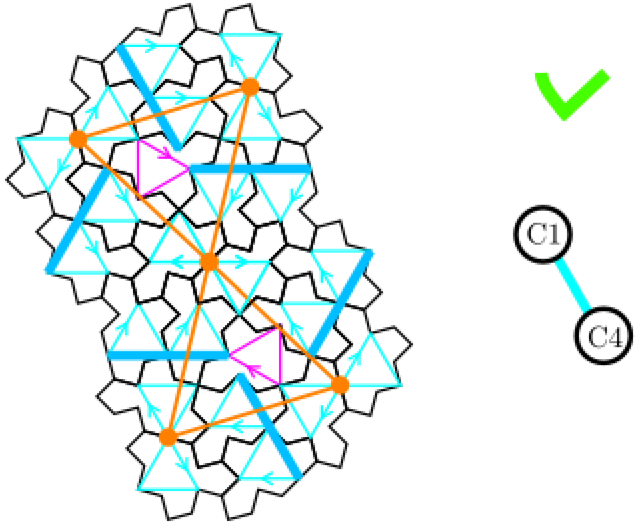


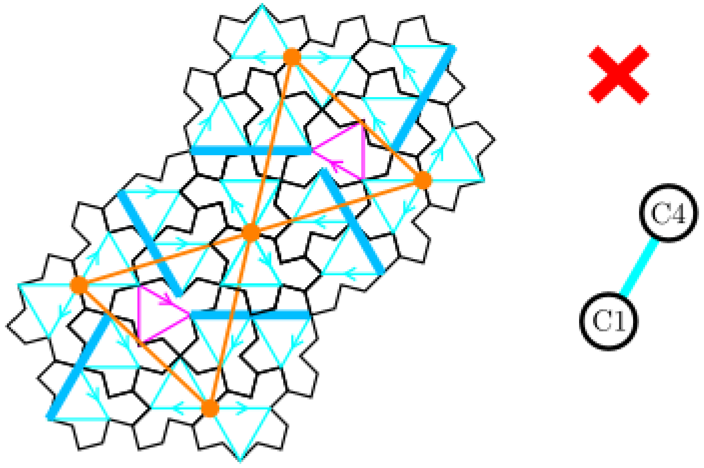


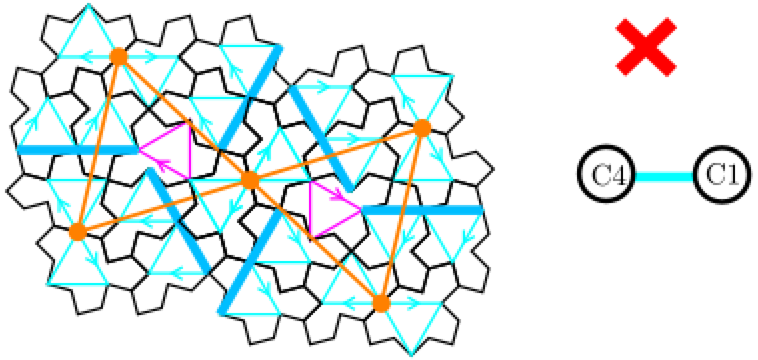
- Should keep the 2-fols symmetry at all inflation steps
- TO DO: test it manually and contact S. Tathan afterwards

Diabolo pattern





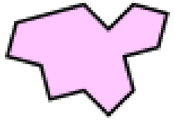




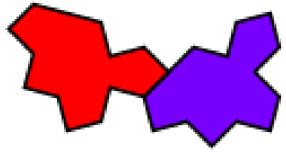
Identification of the greek letters for tiles

- ODD tiles
Pink color

Γ Γ_2



Θ

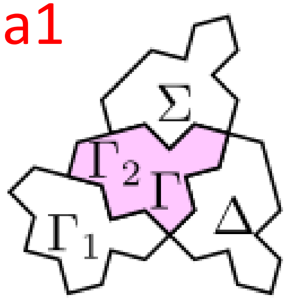


Λ

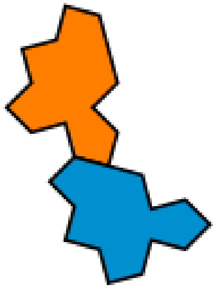
- \theta and \lambda always in couple

- Small cluster \sigma, \delta, \gamma1
white color

Γ_1 Σ Δ

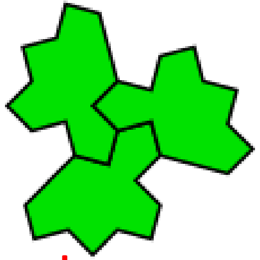


Ξ



All \chi tiles are linked to a \pi tile and consequently to a large star.

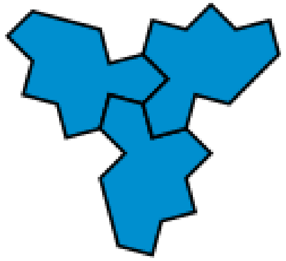
Φ



- \phi: Small green stars + some isolated EVEN tiles

Π

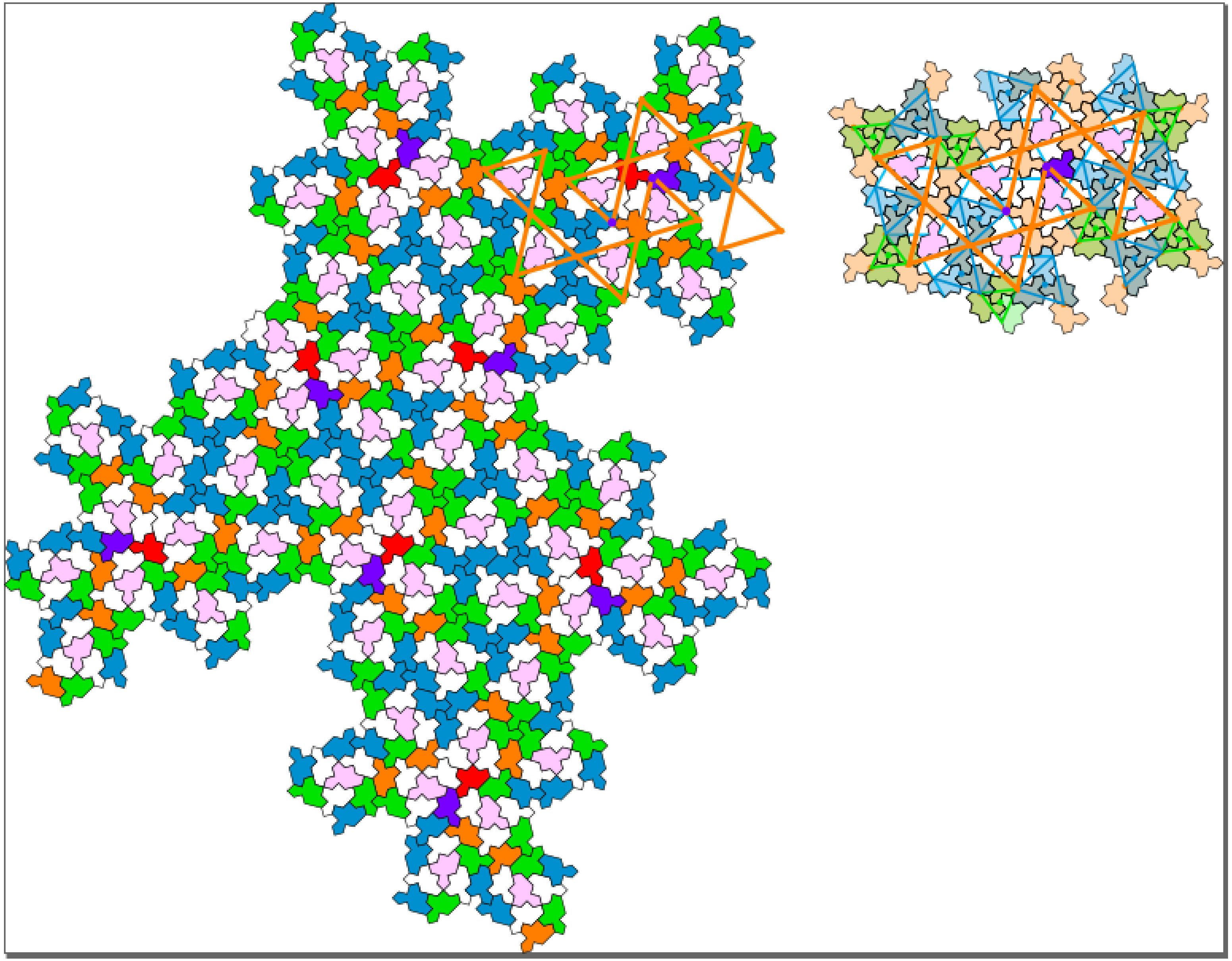
Ψ



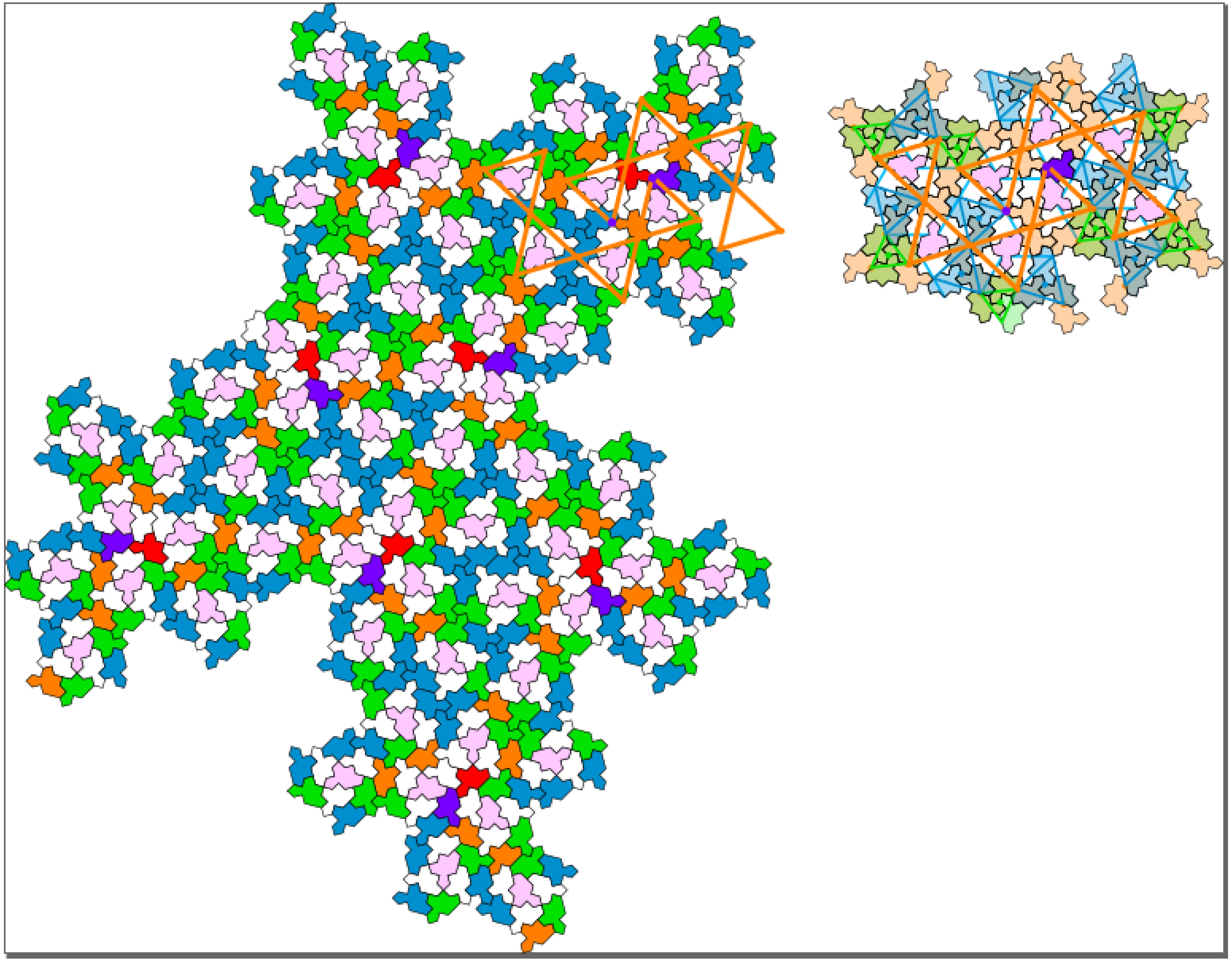
Π

- Blue large stars correspond to all \Pi and \Psi tiles with different combinations

Inflation using python code



Inflation using python code

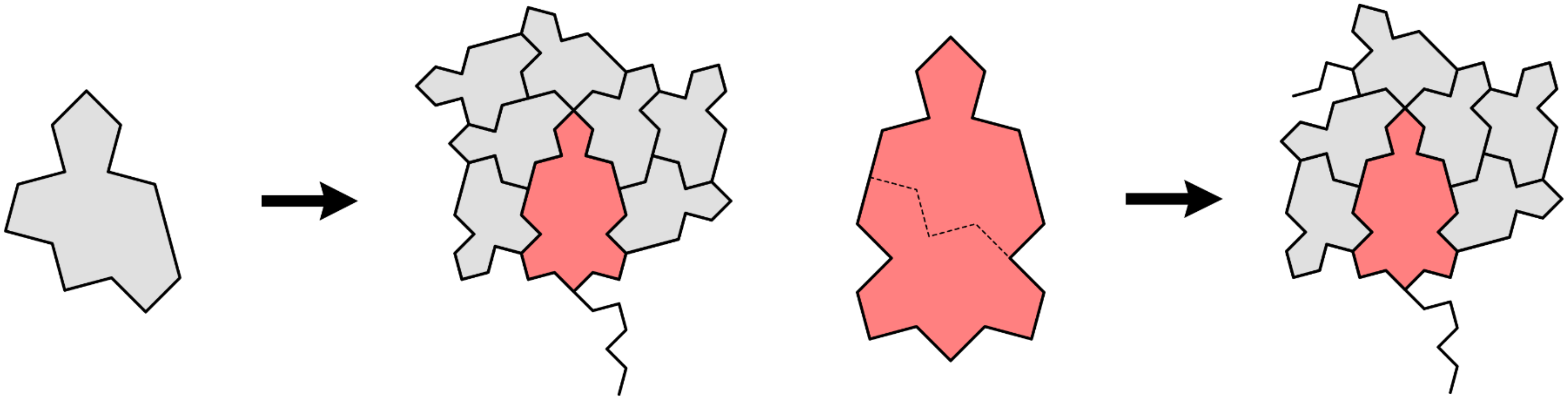


H7/H8 by Simon Tathan

https://www.chiark.greenend.org.uk/~sgtatham/quasiblog/aperiodic-refine/

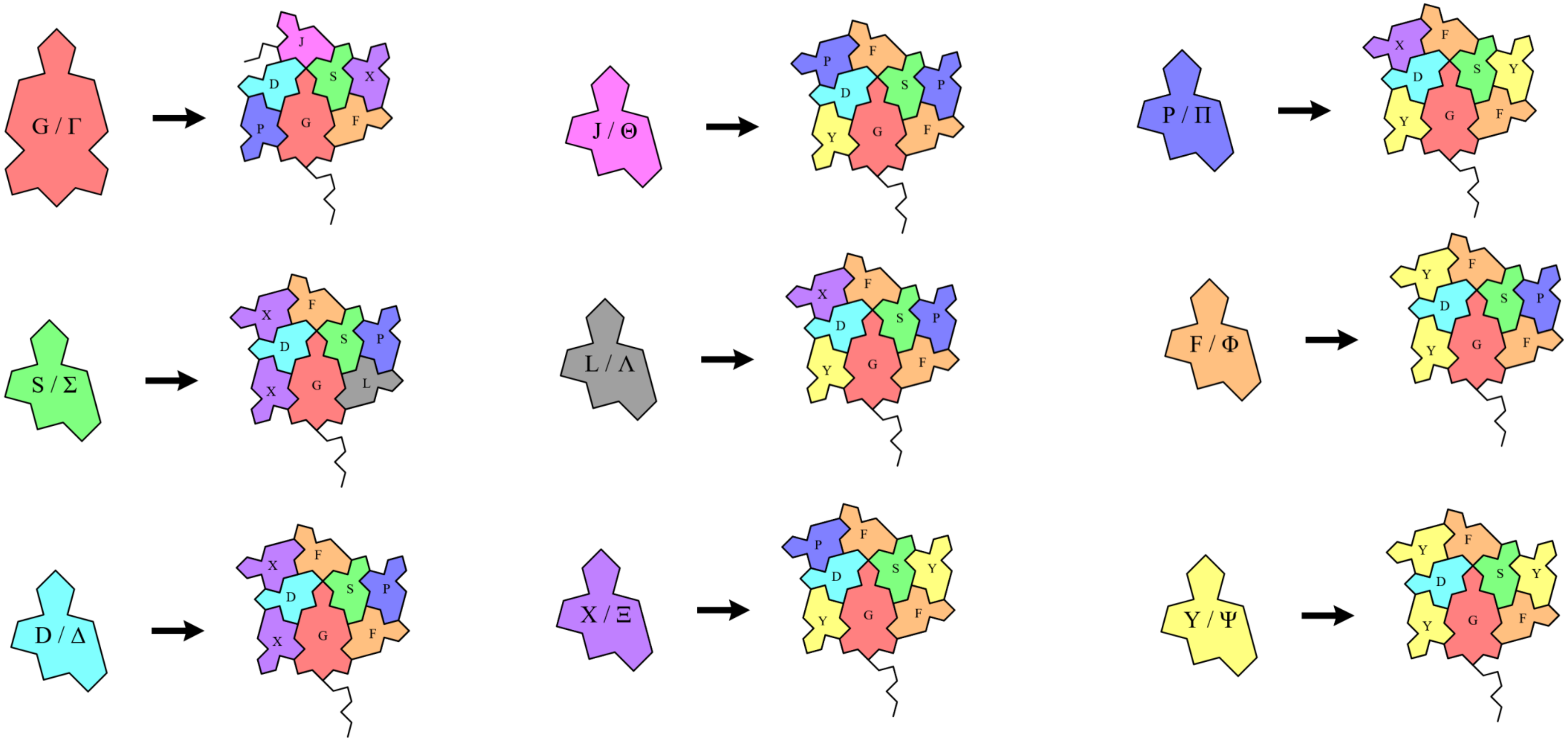
https://arxiv.org/abs/2512.16595

Code in RUST language:
https://crates.io/crates/substitution-tiling-transducerhttps://crates.io/crates/substitution-tiling-transducer



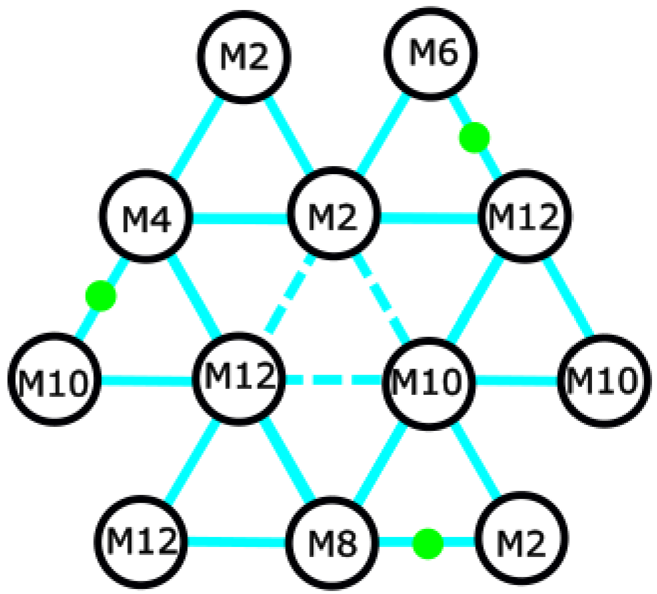
The nine homochiral substitutions by Simon Tathan

https://www.chiark.greenend.org.uk/~sgtatham/quasiblog/aperiodic-refine/

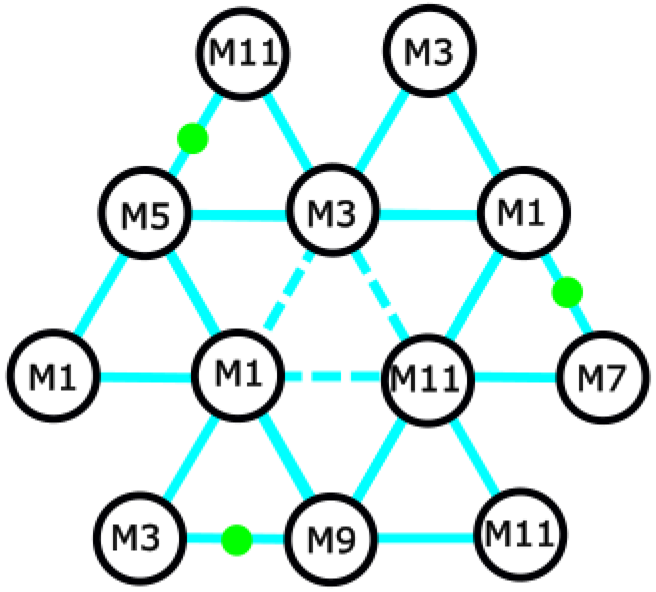


Inflation step 1: substitution rule for C1

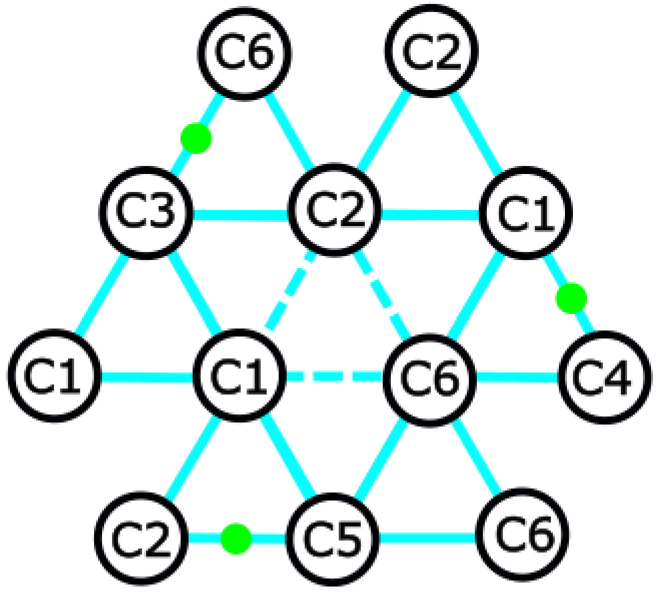
EVEN tiles in C1



ODD tiles for step 1

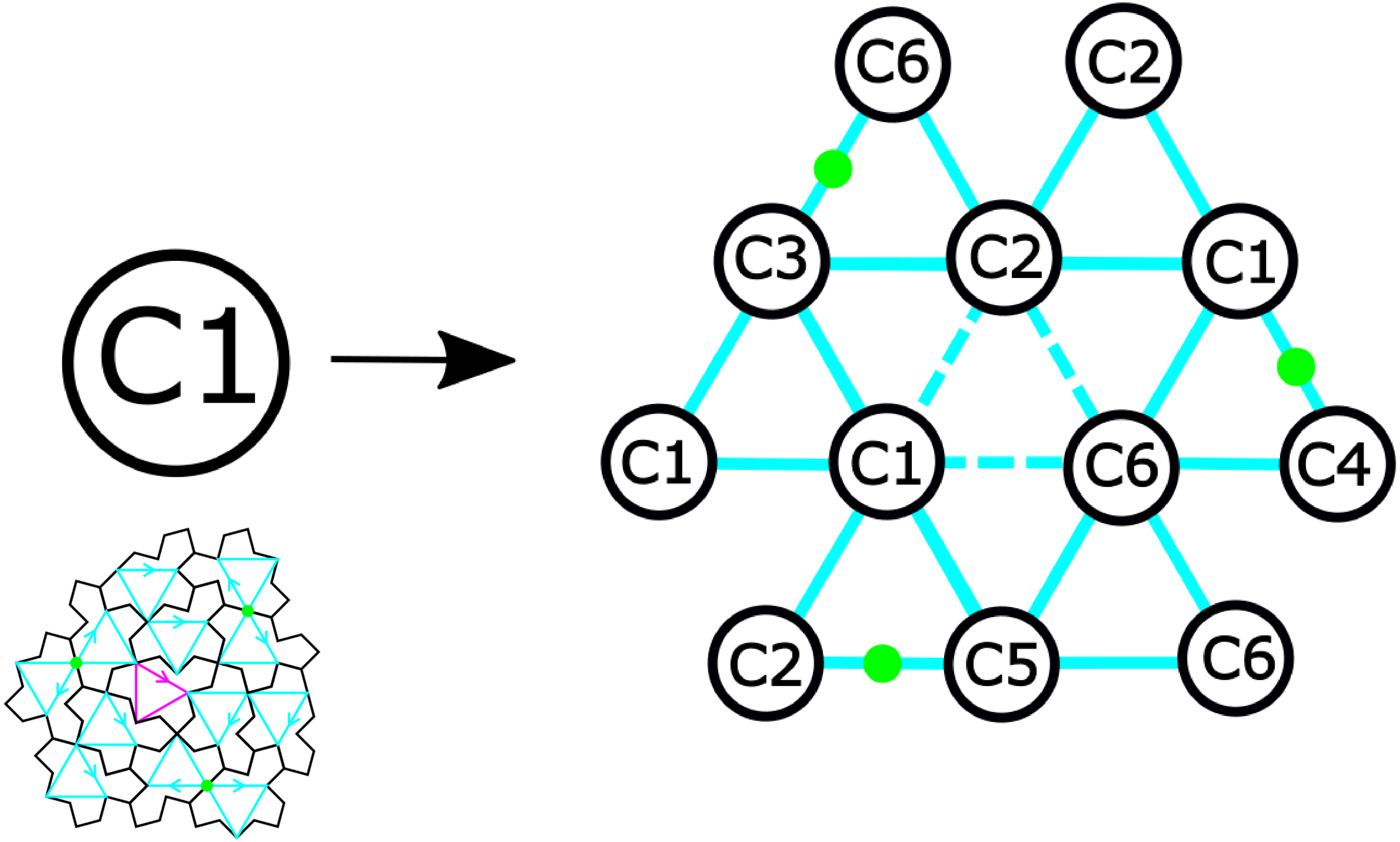


same as full clusters



- ODD tiles for step 1 are obtained by adding 1 (mod 12) to all EVEN tiles in the C cluster (6 tiles)
- Directions of translated and half-turned tiles are exchanged (**see the change of position for green points**). Then three tiles are translated (M1, M3, M11) and three are half-turned (M5->M11, M1 -> M7, M9 -> M3)
- ODD tile index is equivalent to the orientation of a full C cluster ☺

Inflation step 1



Homochiral python code on github

- https://github.com/marimperorclerc/homochiral

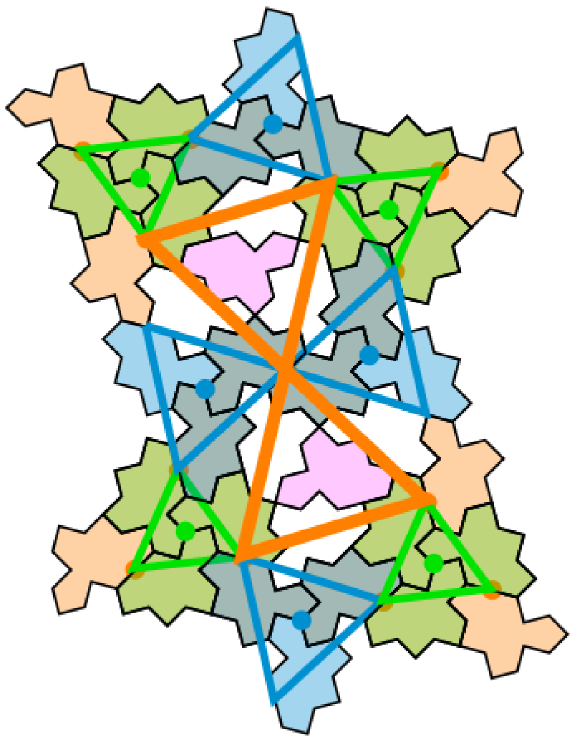
python code for drawing clusters with Tile(a,b) in svg graphics

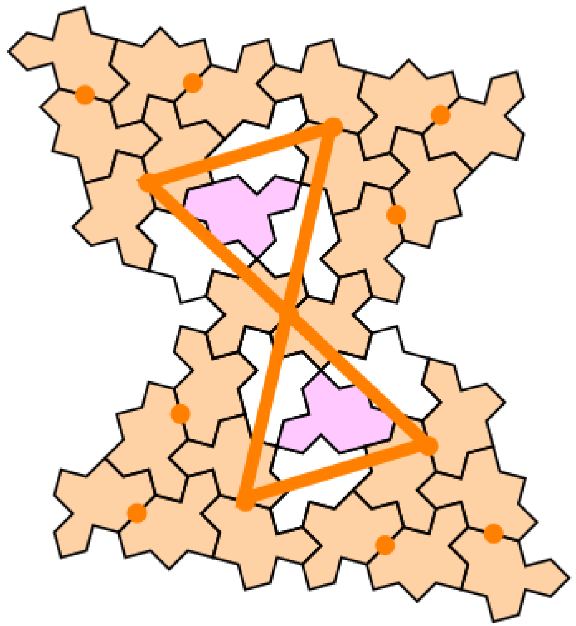
References

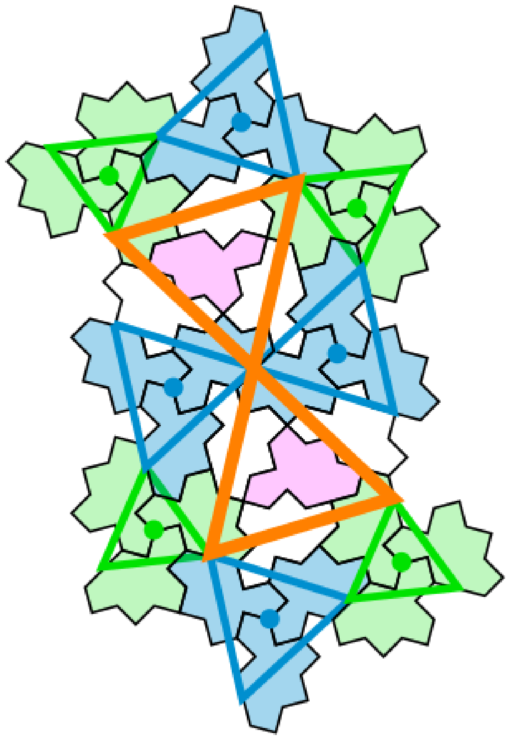
Code is based on the drawsvg package:
https://pypi.org/project/drawsvg/
And the python code with alternate chirality for inflation for spectre tiling:
https://github.com/shrx/spectre/blob/master/spectre.py

- Adapted here (spectre-colored-tiles.ipynb) with different color choices
- See some .svg files in \svg-spectre folder

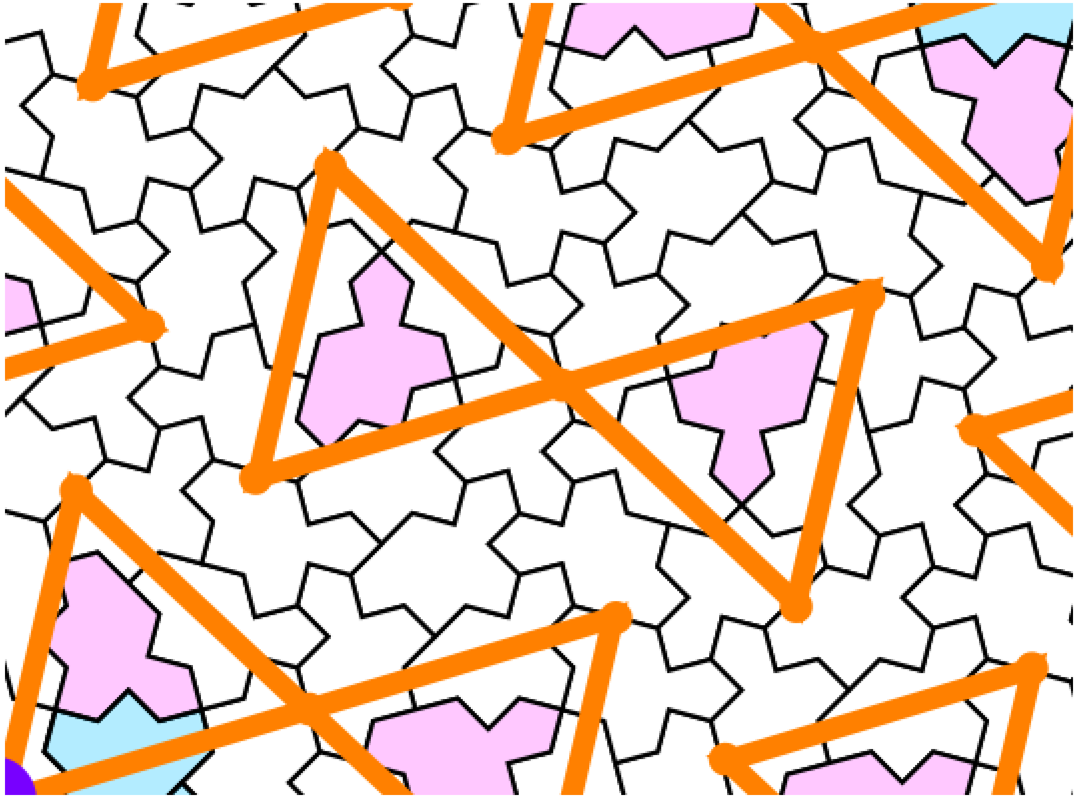
DIABOLO





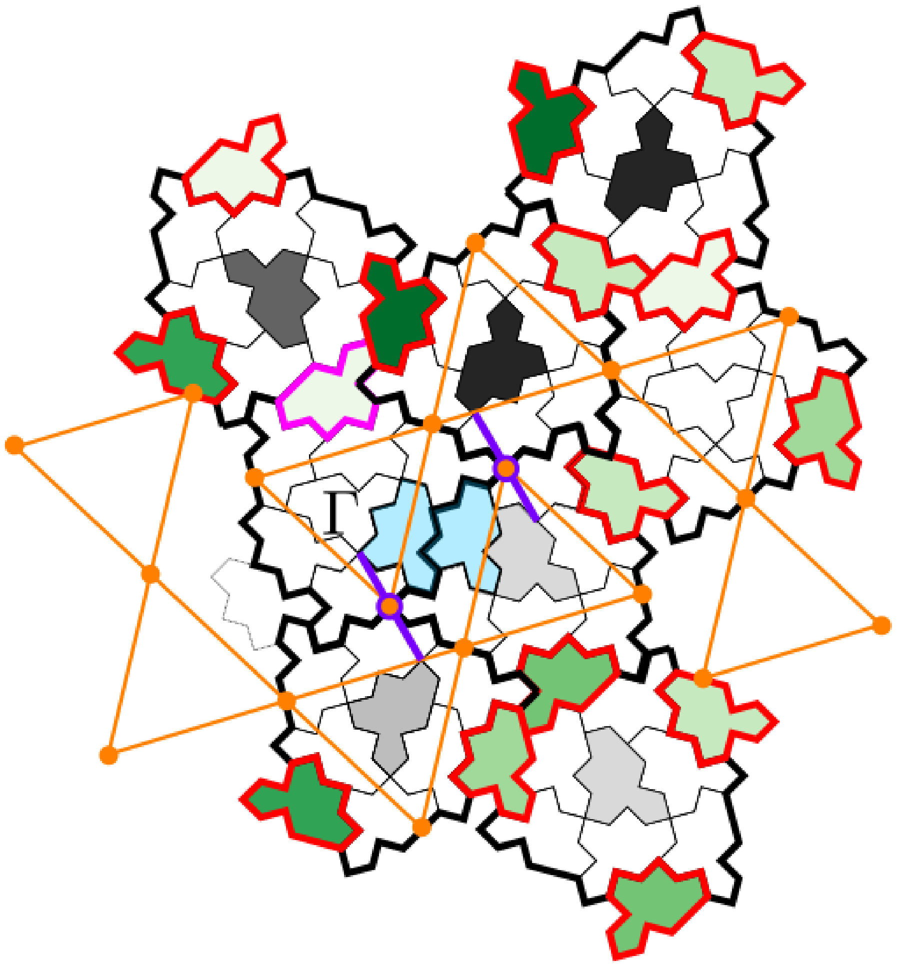


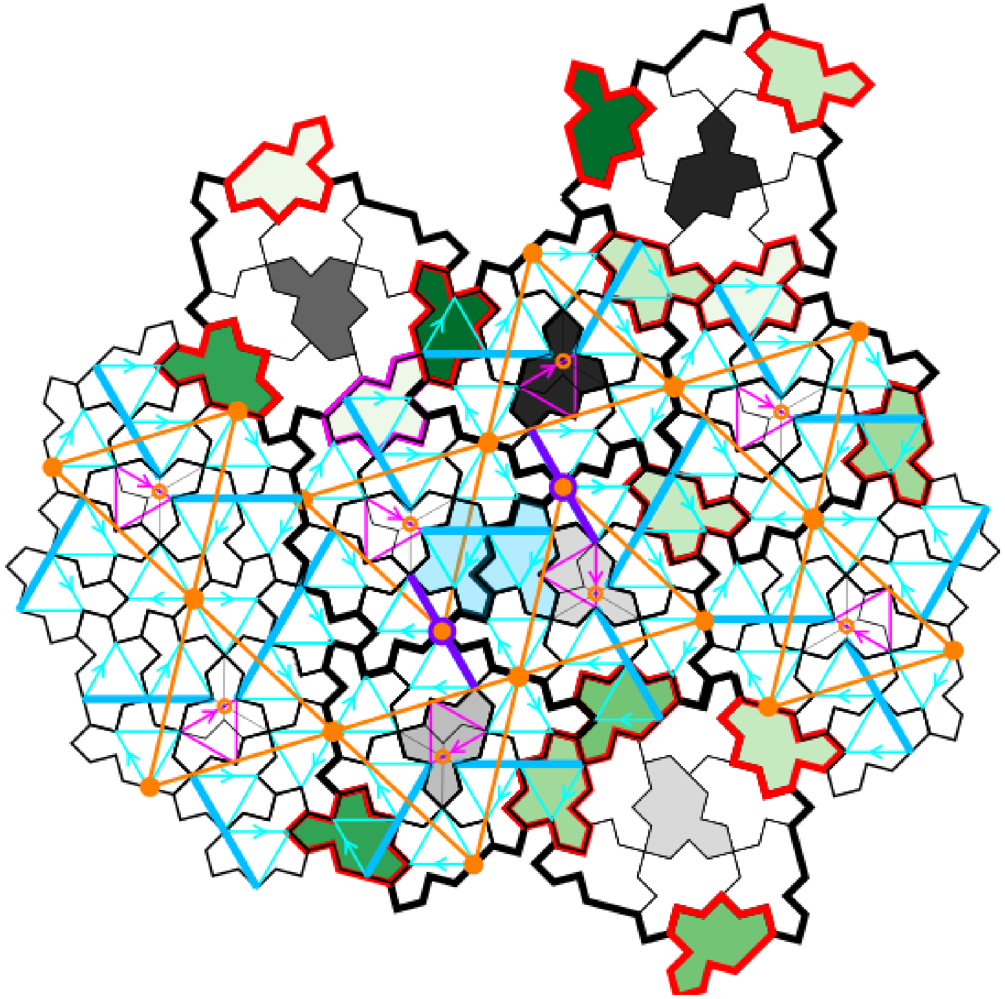
Diabolo pattern



- Local 2-fold axis

Link with triangular pattern

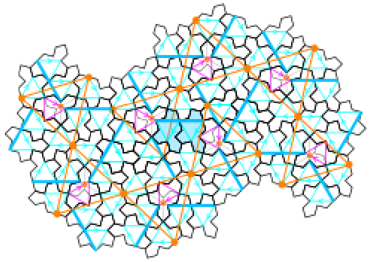




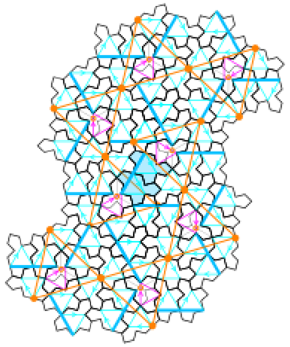
- The two other environnements with a violet lines are always coupled together

Six orientations of the 2-arms

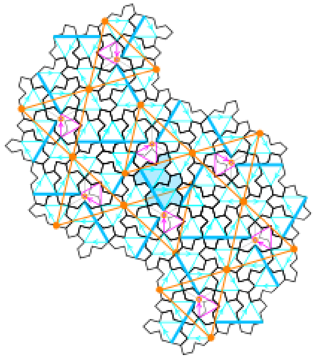
P1

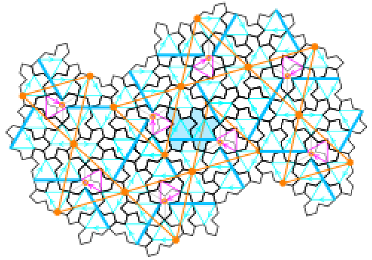


P2

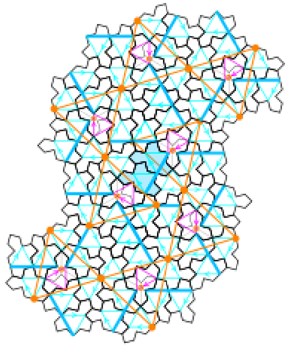


P3

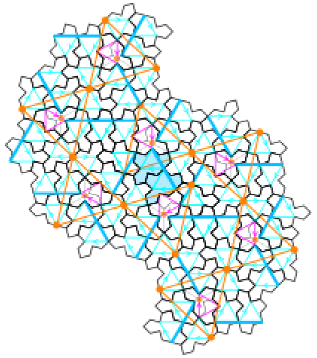




P4

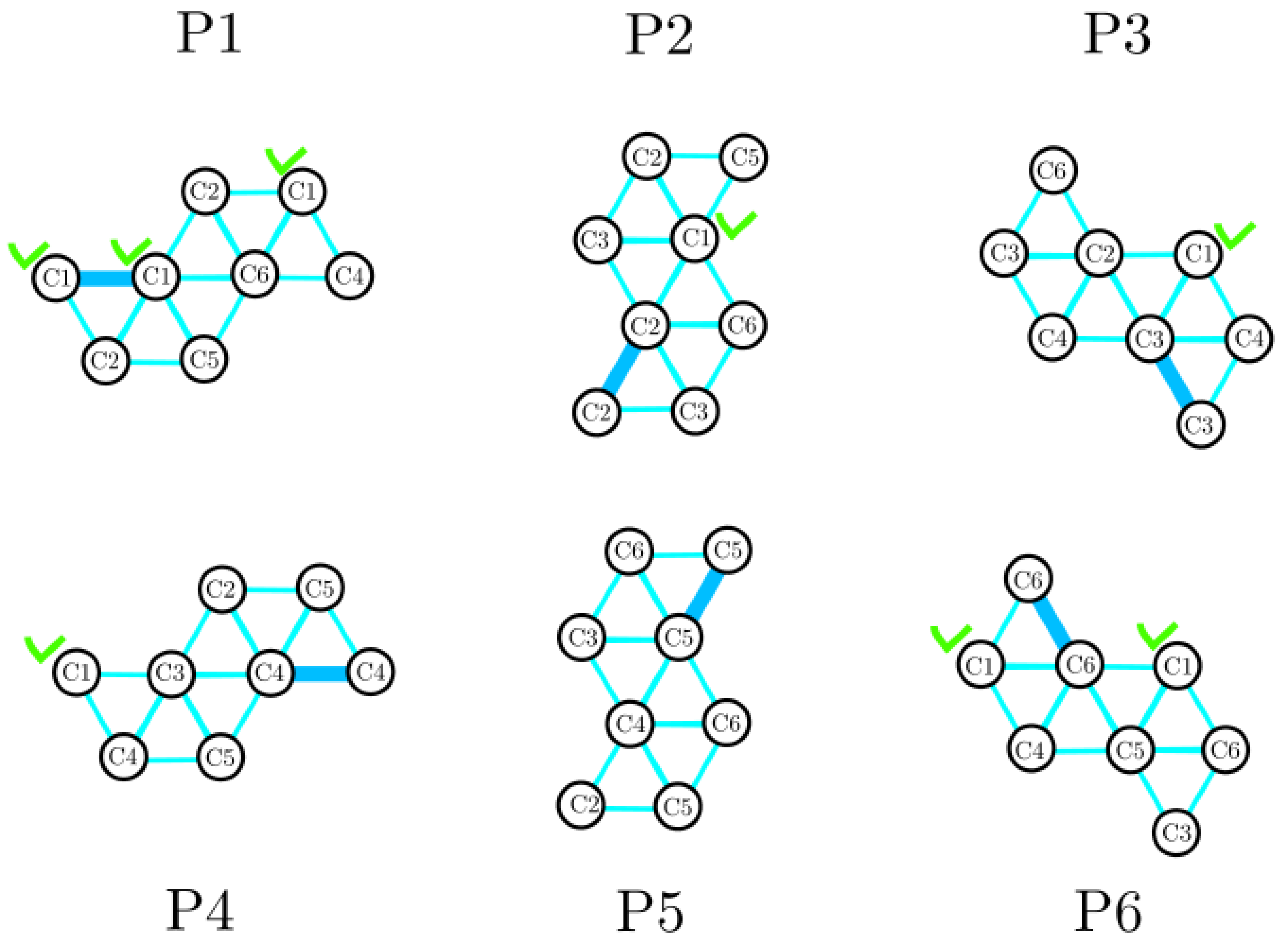


P5



P6

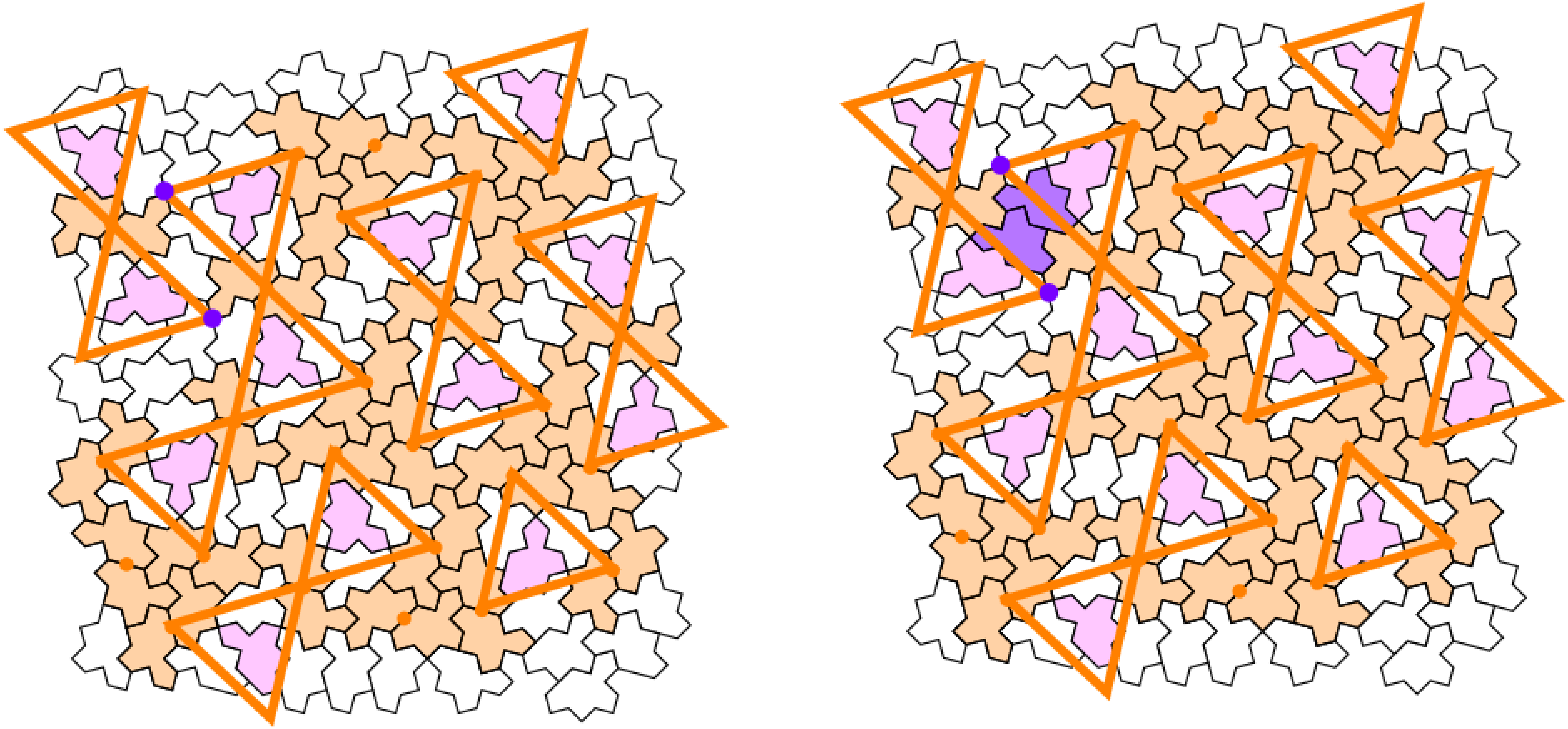
Six orientations of the 2-arms



- 8 combinations for a given cluster orientation (green crosses)

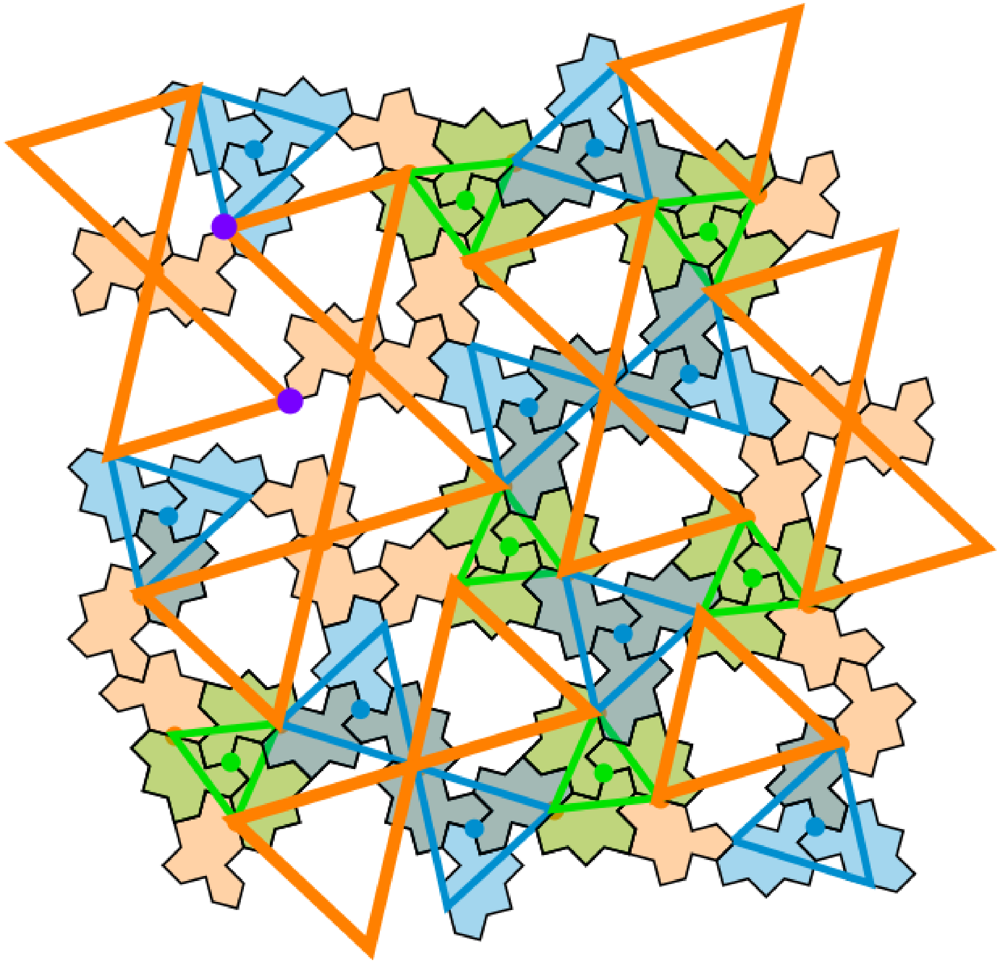
Example: Set of all local 2-fold symmetry centers

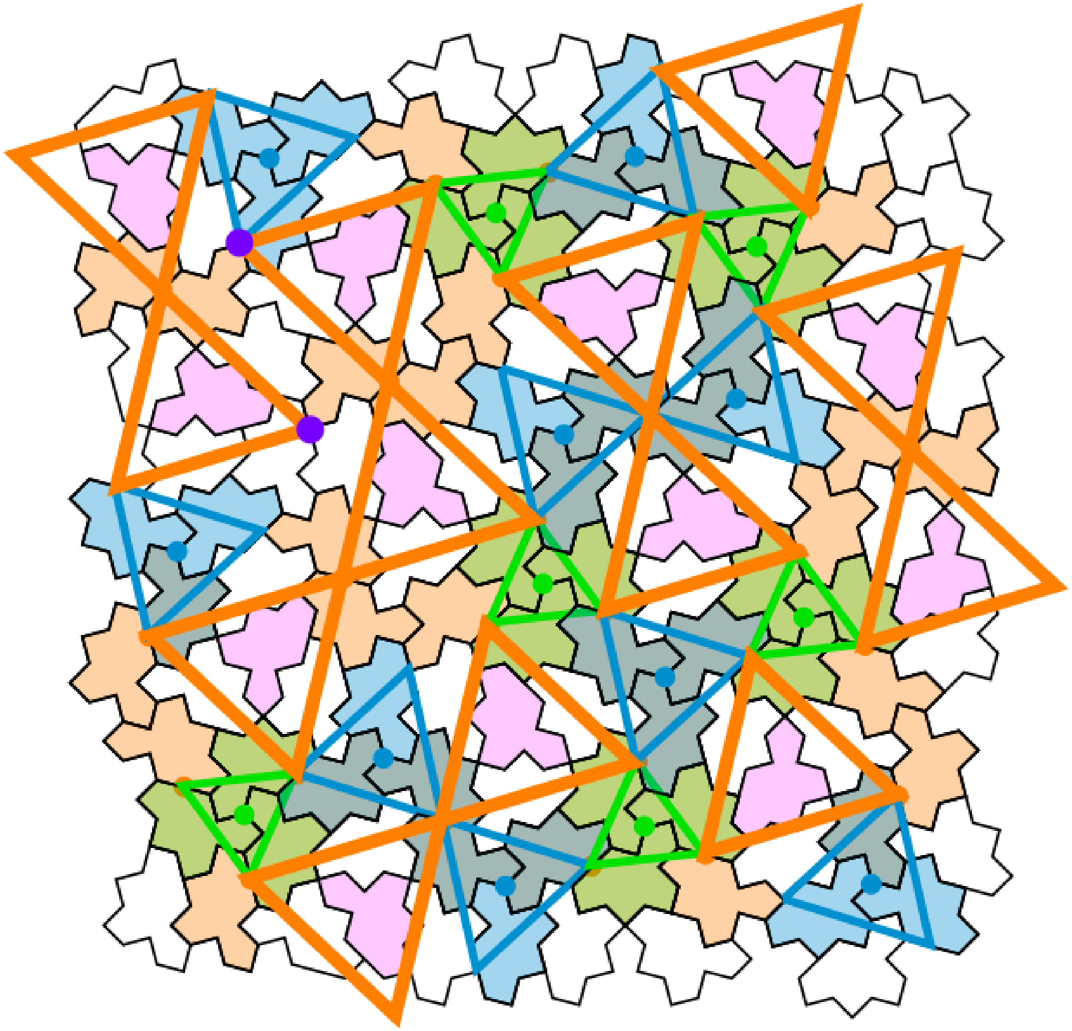
2-fold



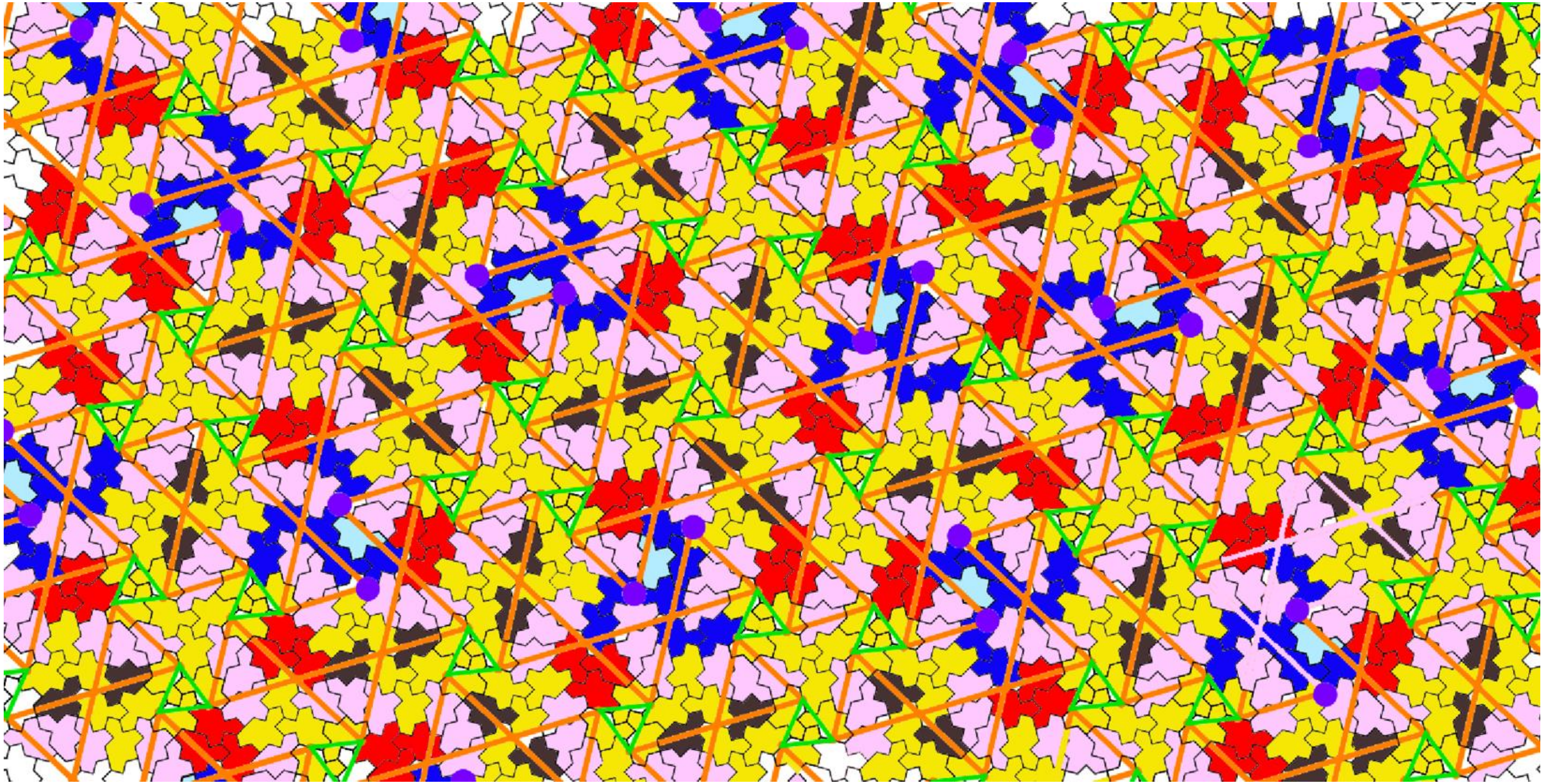
- around each ODD monotile (pink), three points (orange triangle)
- Violet points are not local 2-fold centers, but all other points are (orange)

superposition

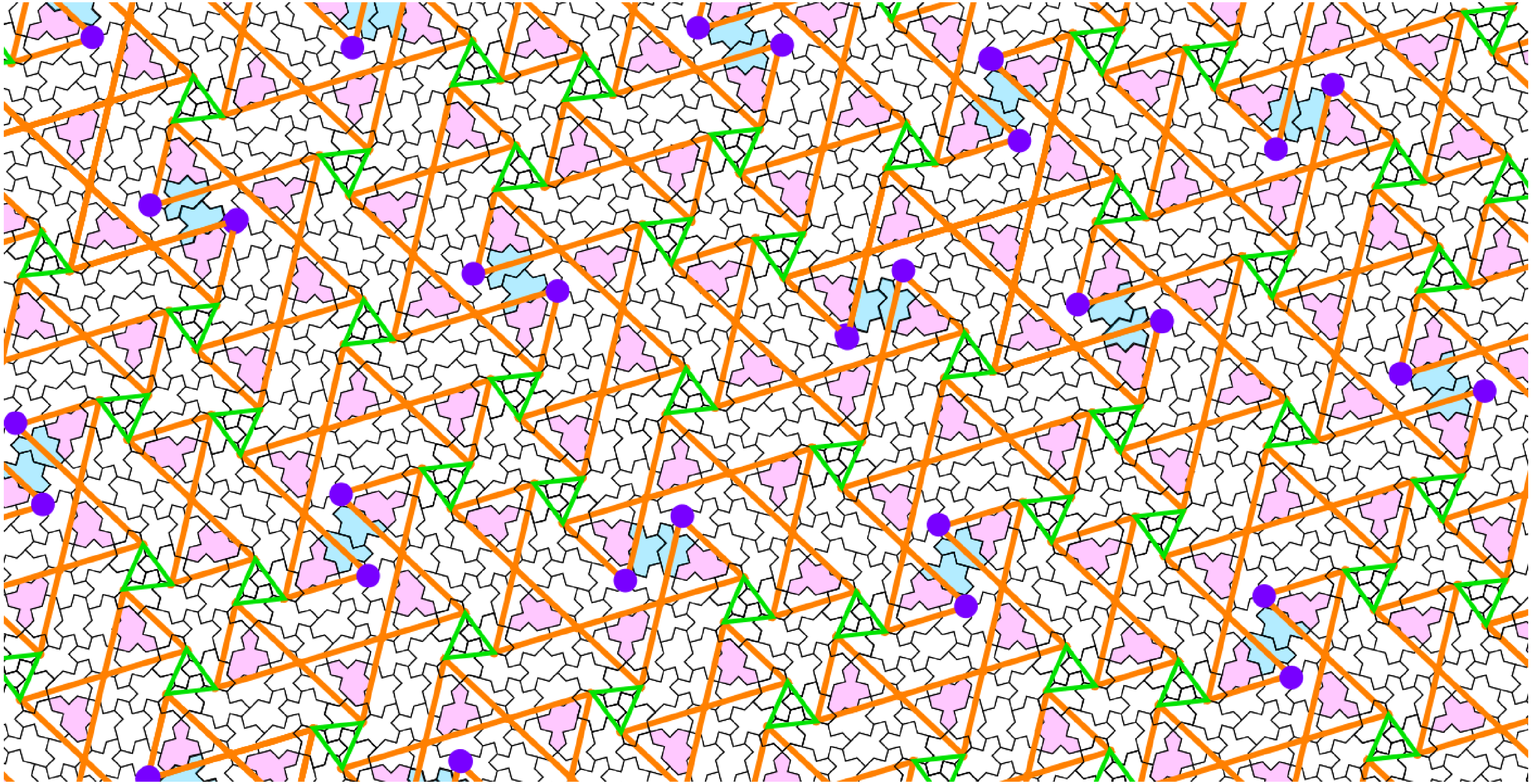




Comparision

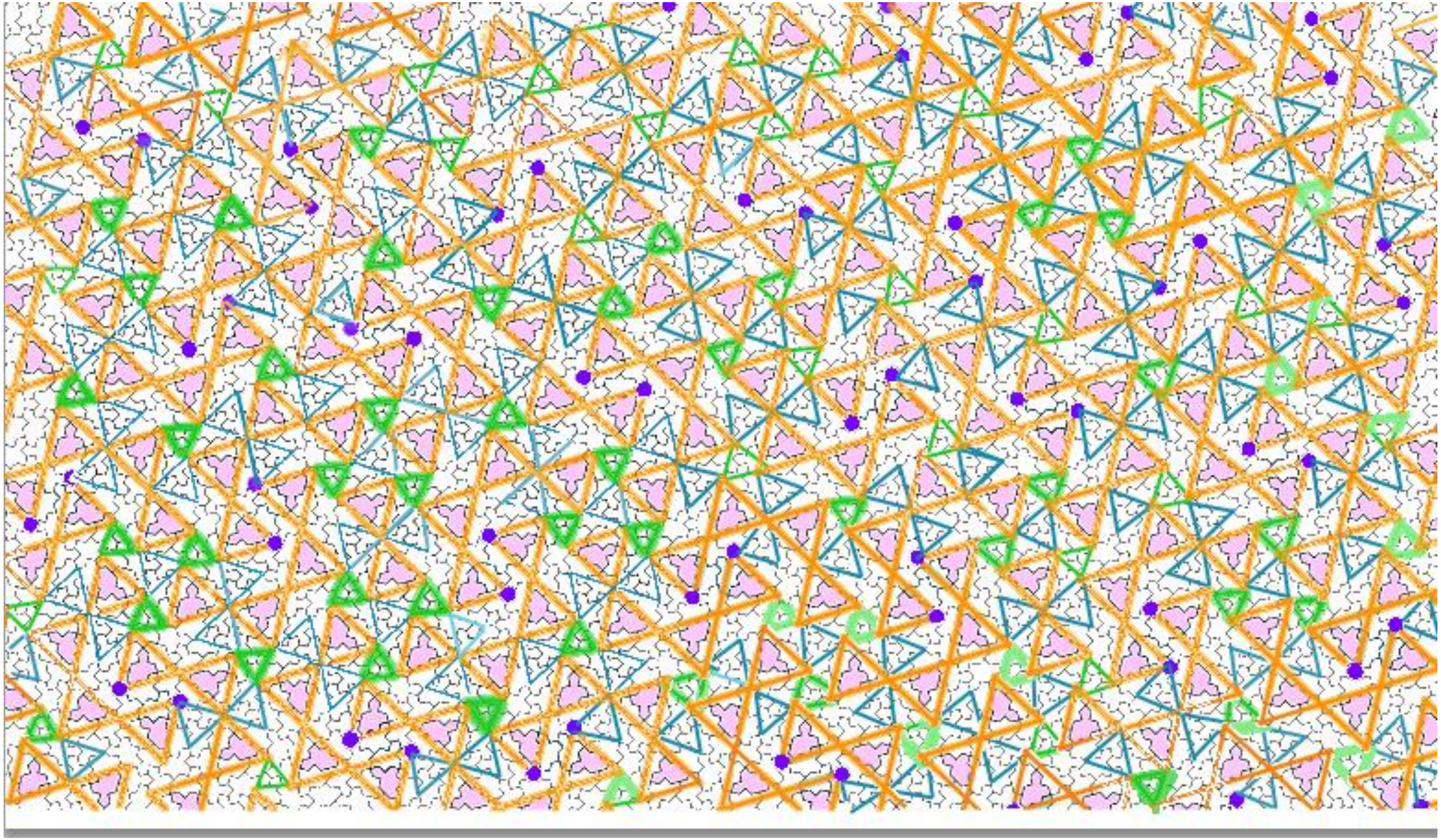


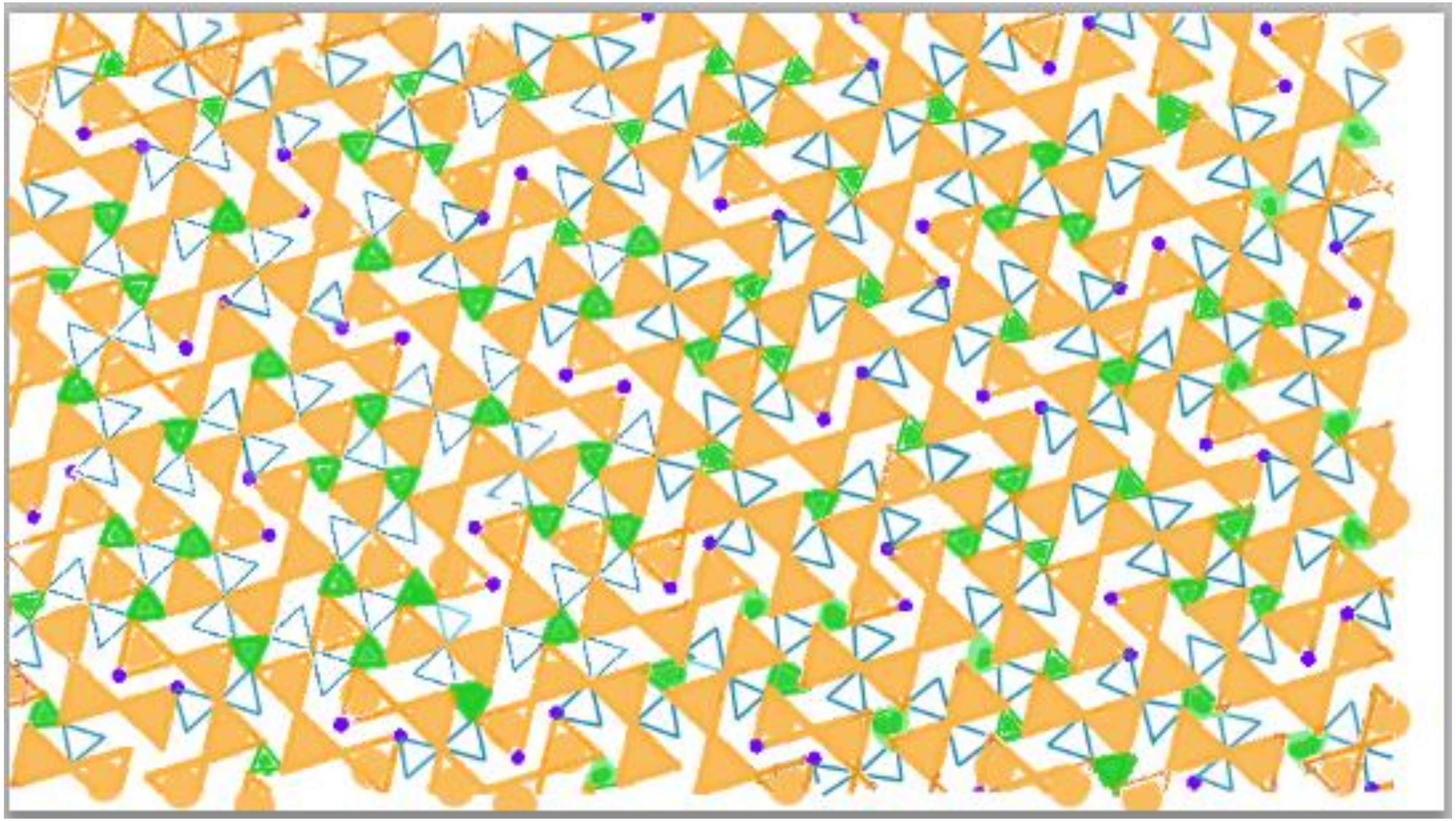
Small stars connect the orange triangles



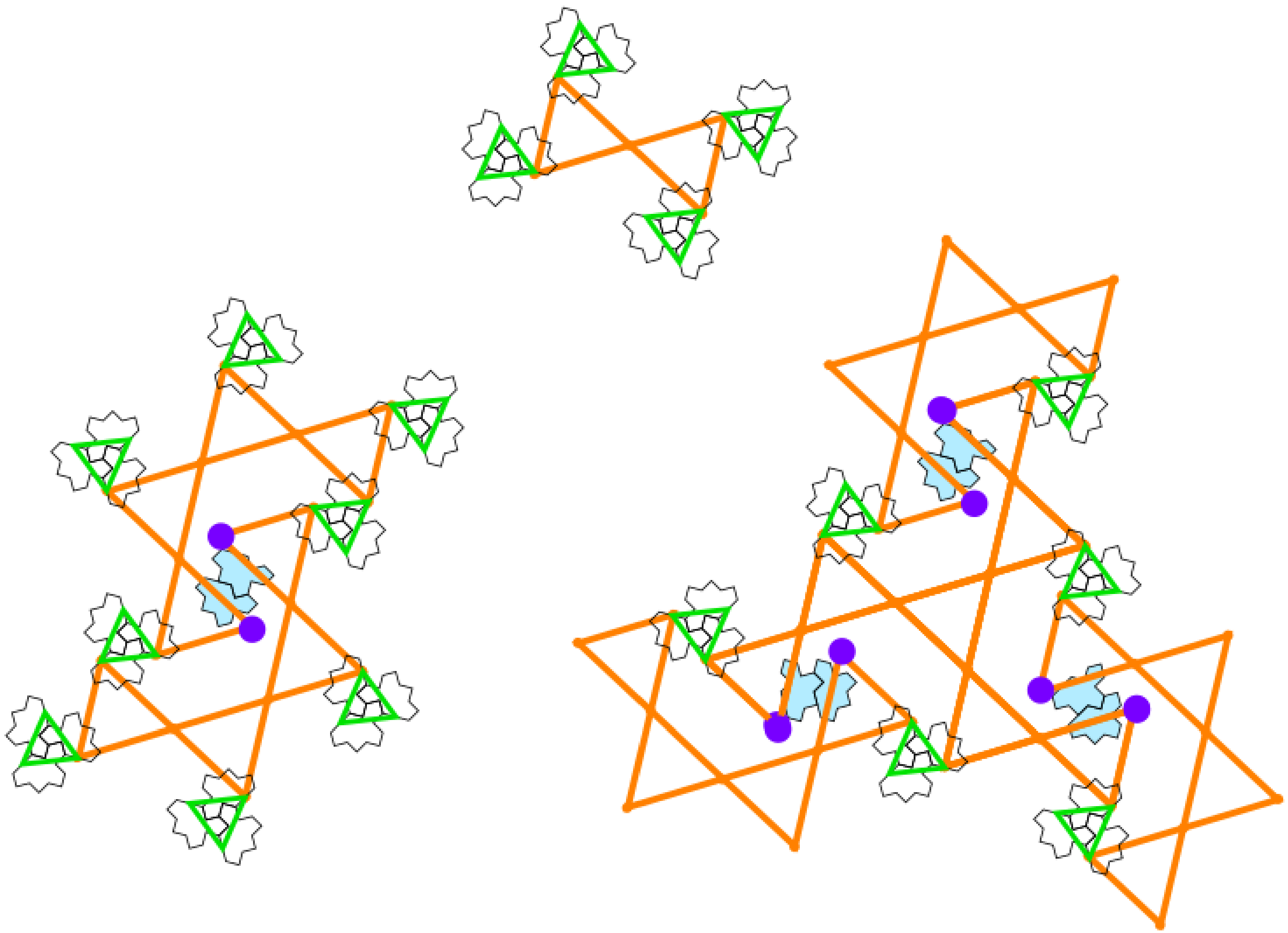
- Possibility to obtain the relative positions between diabolo, 2-arms and 3-arms

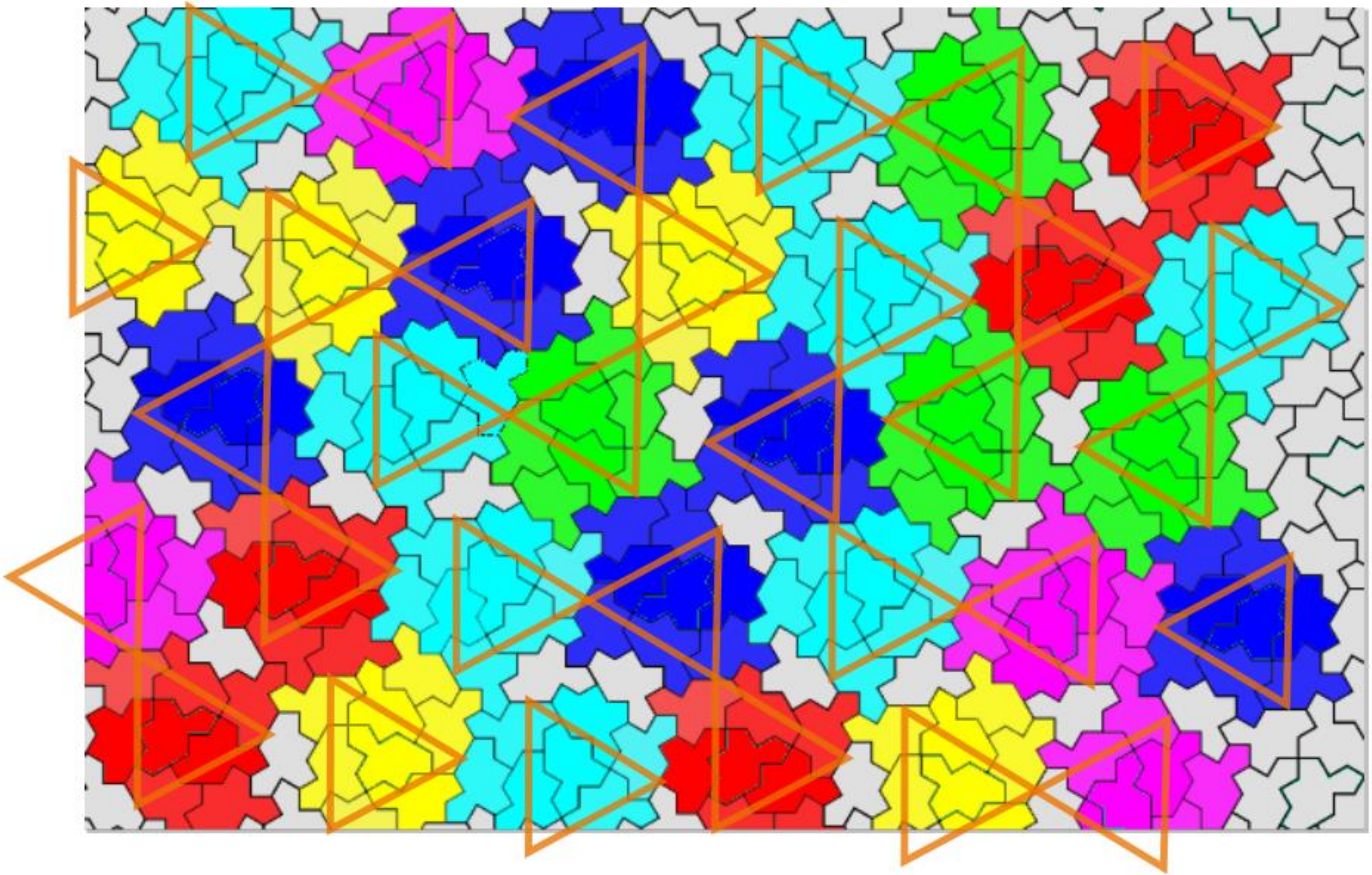
SMALL AND LARGE STARS





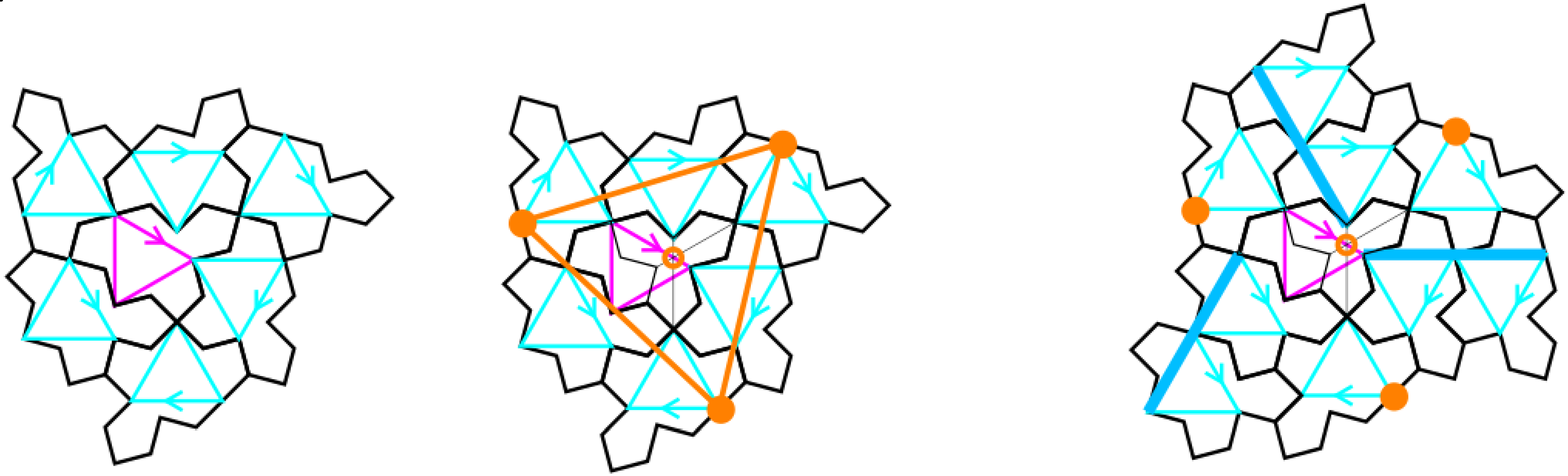
New puzzle





- Tile(1,1)

full cluster

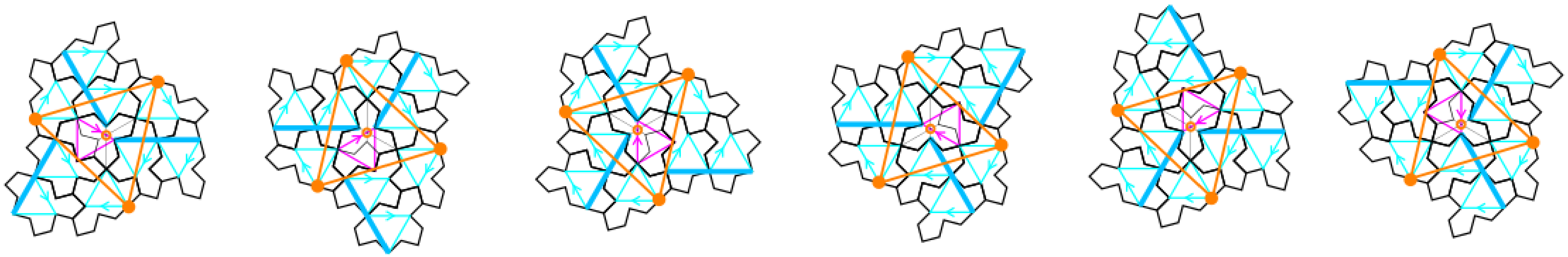


- C cluster: one ODD tile at the center + 6 EVEN tiles around
- Three glue points (orange color): regular triangle centered on the trifle shape of the ODD tile
- Full cluster: 3 more EVEN tiles obtained by translations (three dark blue lines)

full cluster with its six rotations: n(2pi/6)

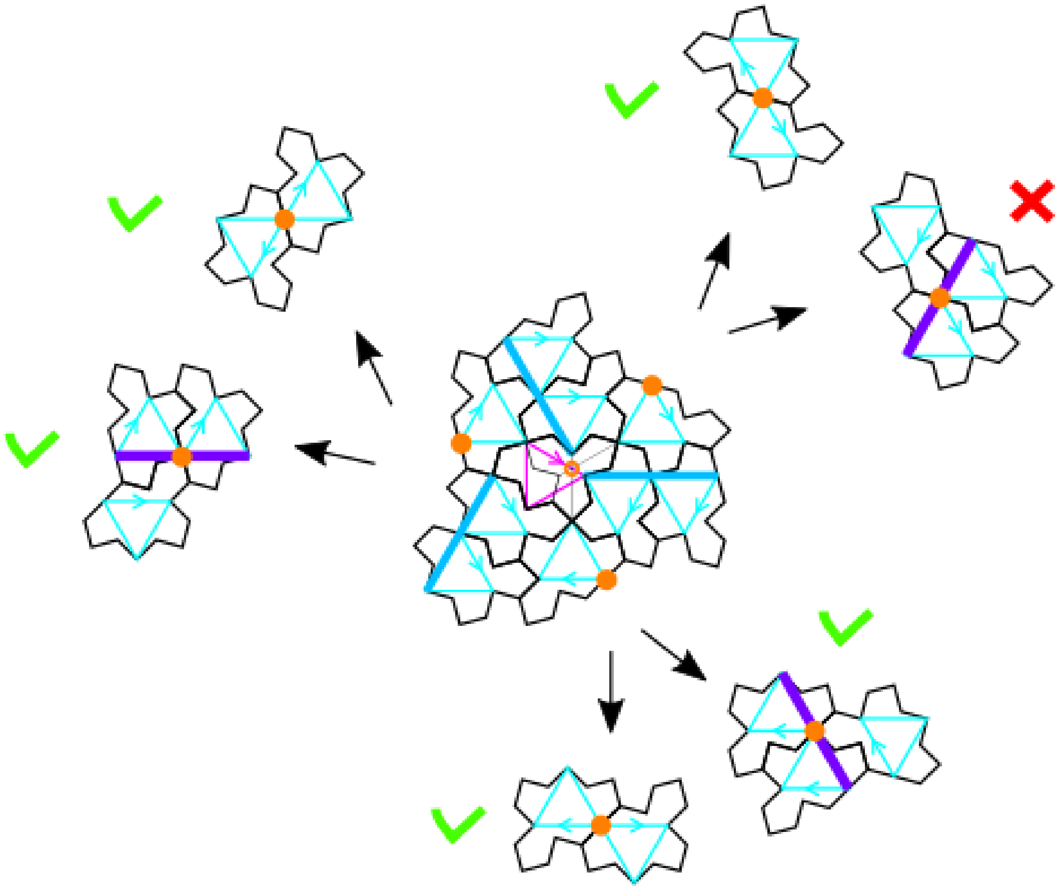
- Tile(1,1)

C1 C2 C3 C4 C5 C6



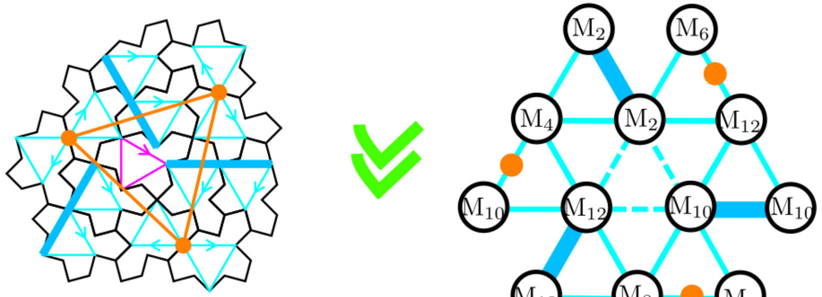
- 13 tiles in a full cluster: one ODD tile at the center + 9 EVEN tiles around
- Blue triangles (EVEN) have one horizontal edge when pink (ODD) have a vertical one

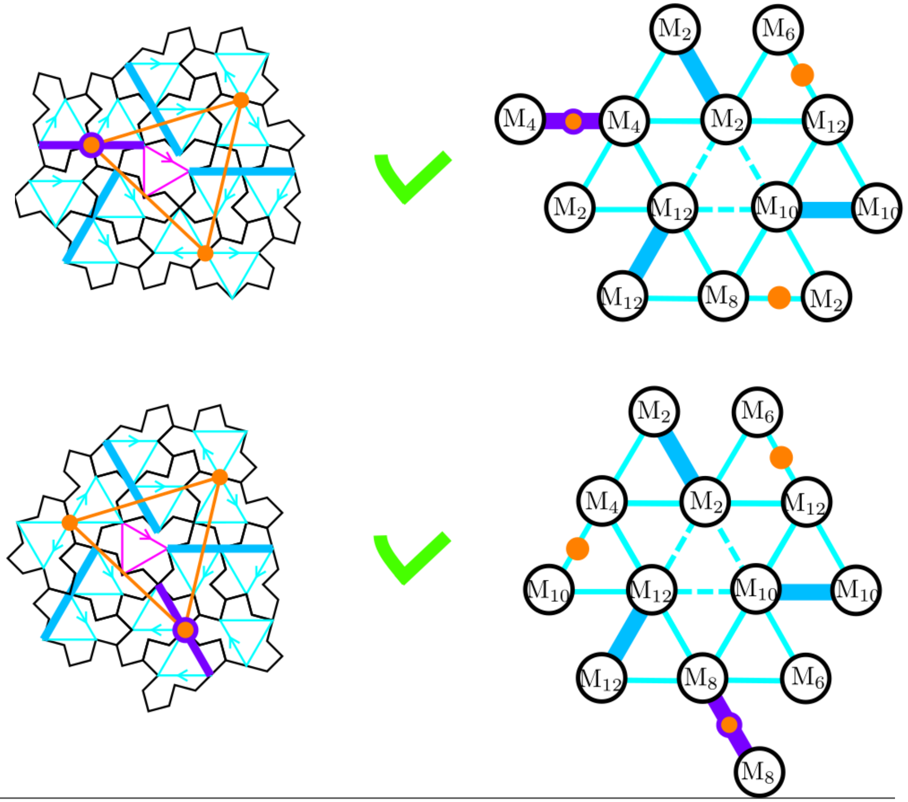
Possible configurations at the glue points ?



- Half-turned or translated monotile (violet line)

Three possible configurations at the glue points

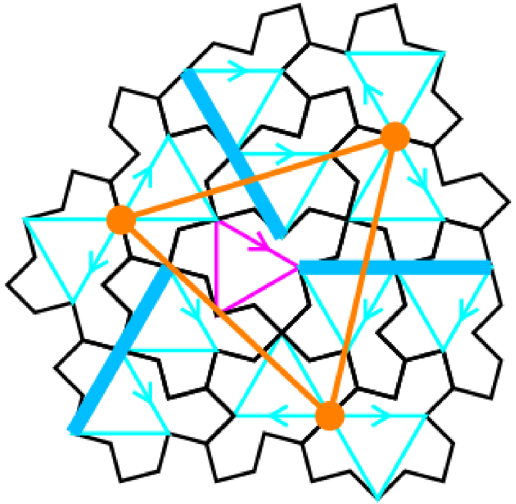


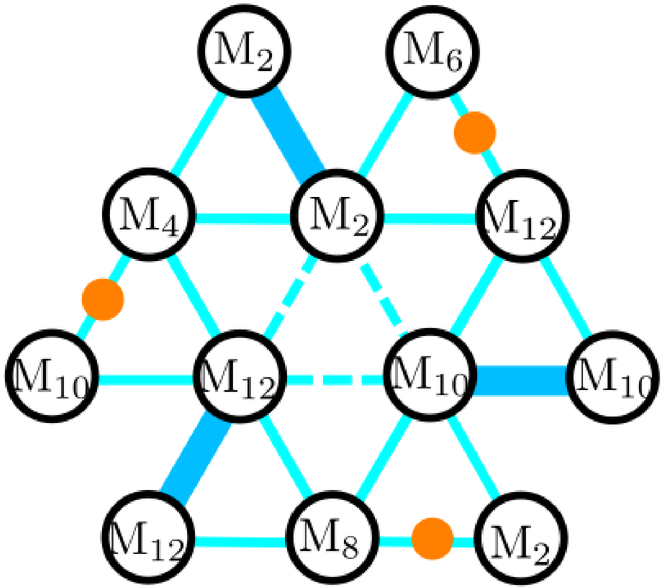


- configuration with three half-turned tiles is the most common
- The other two are always coupled together in the tiling:
- One violet line in a 2-arms and three violet lines in a 3-arms

Orientation of the EVEN tiles in the most frequent configuration

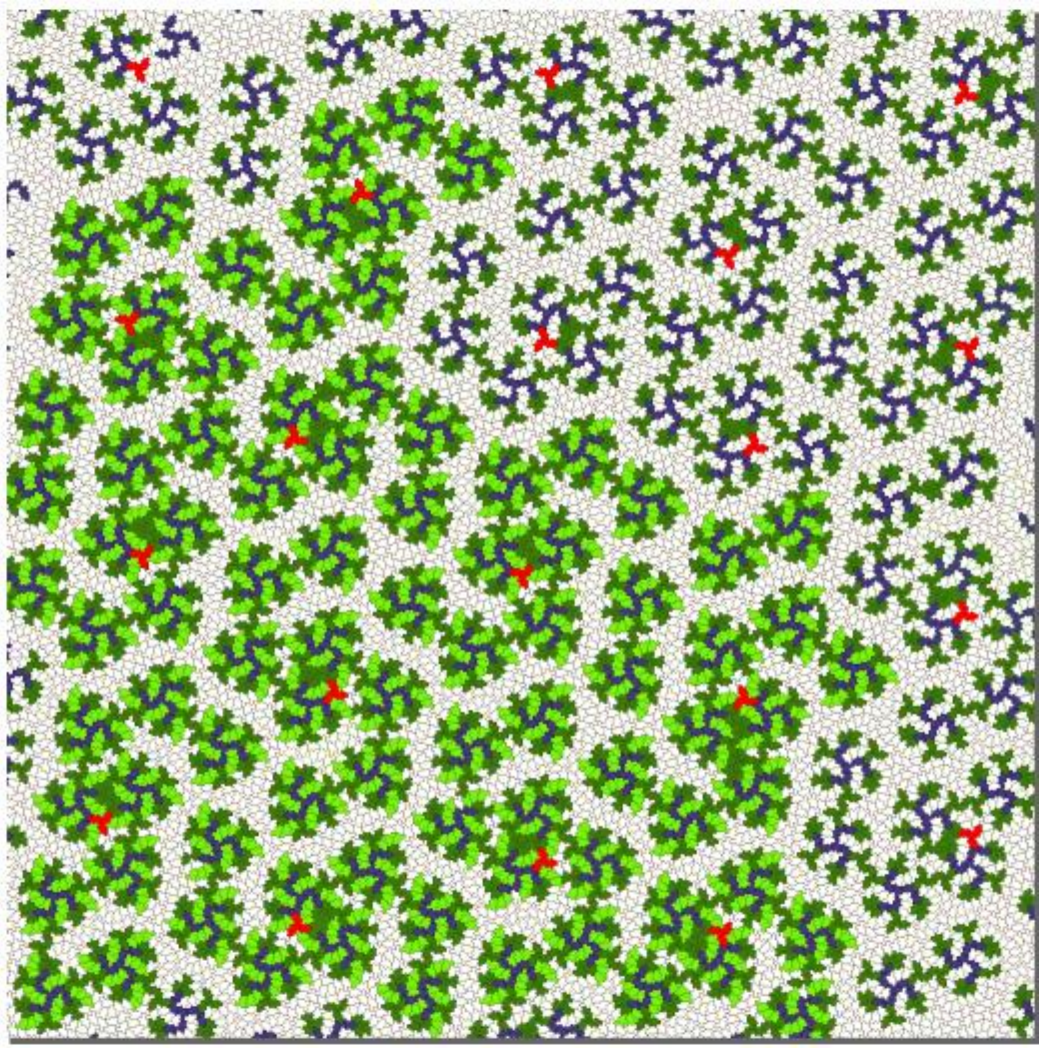
- Tile(1,1)

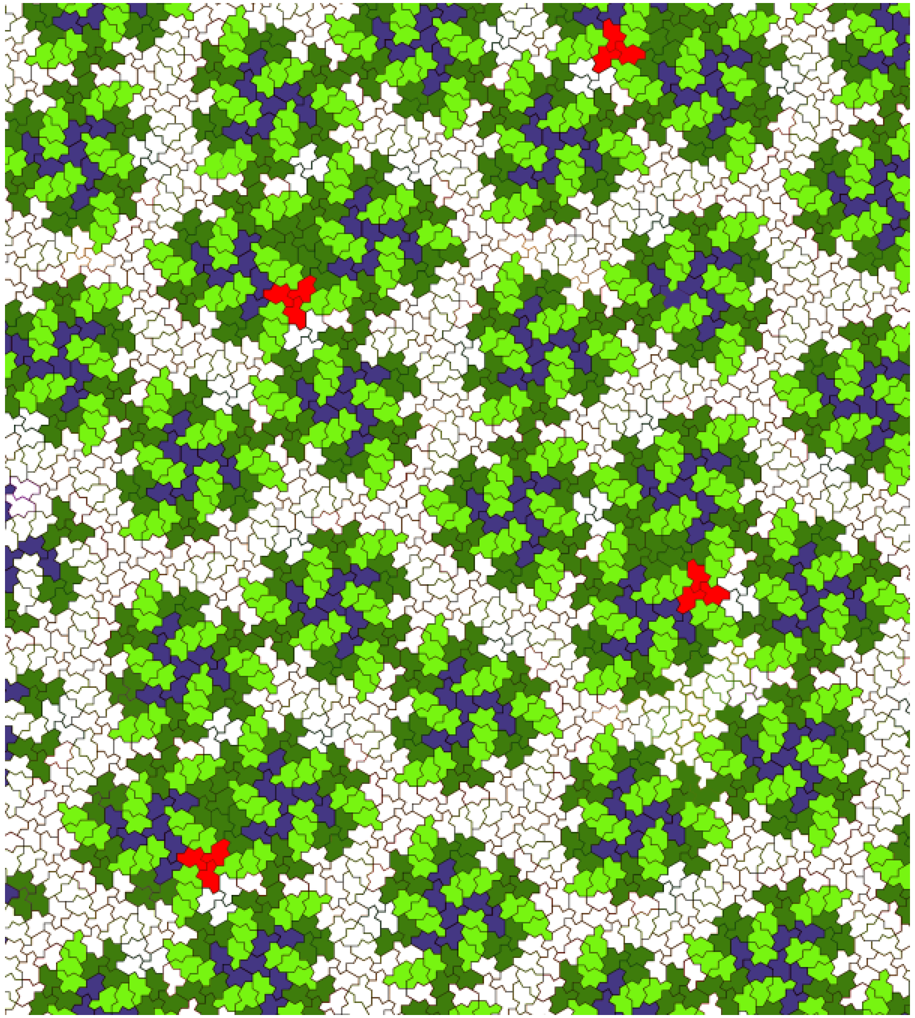




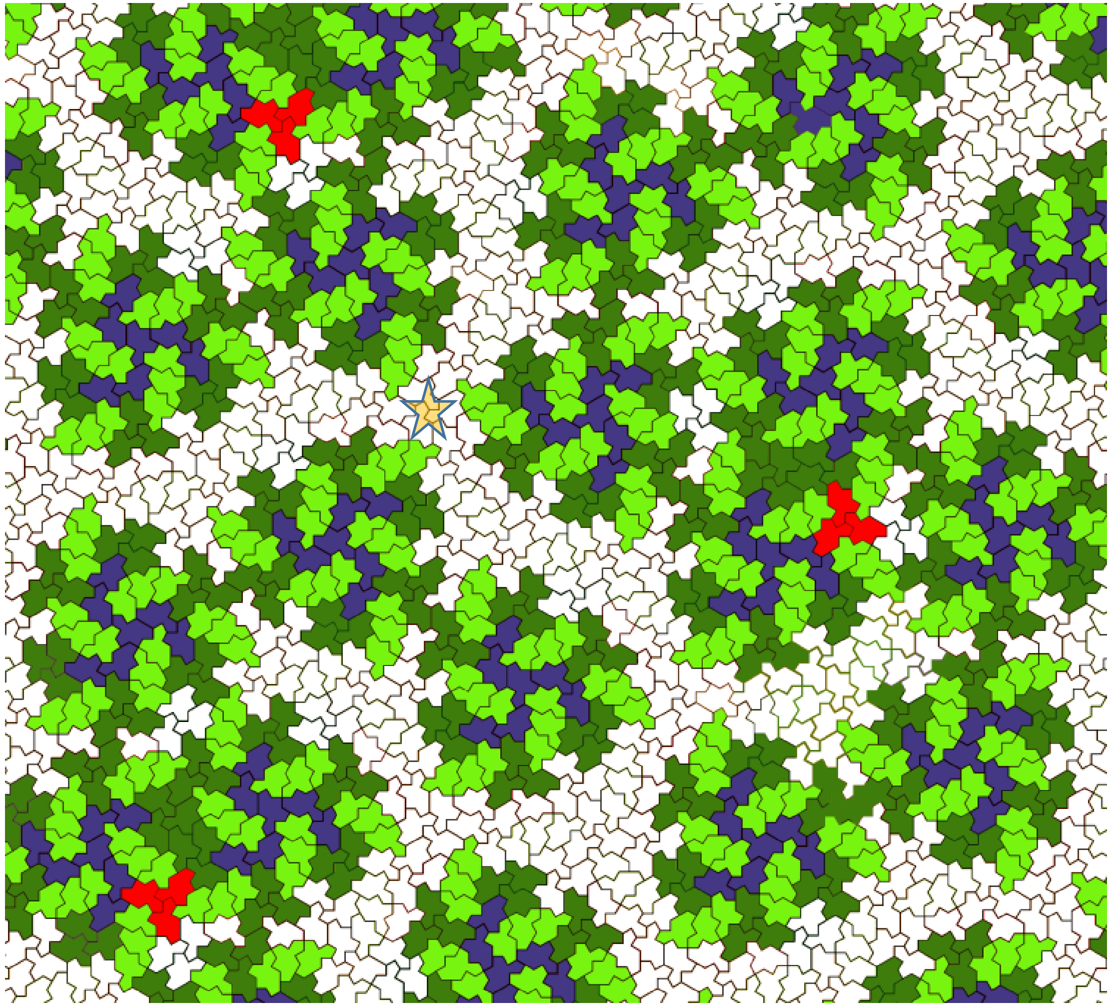
- 6 tiles in the C-cluster (corona made of M12, M4, M2, M12, M10, M8)
- 3 translated (M2, M10, M12) (along blue lines, the edges of blue triangles)
- 3 half-turned (M4->M10, M12 -> M6, M8 -> M2)

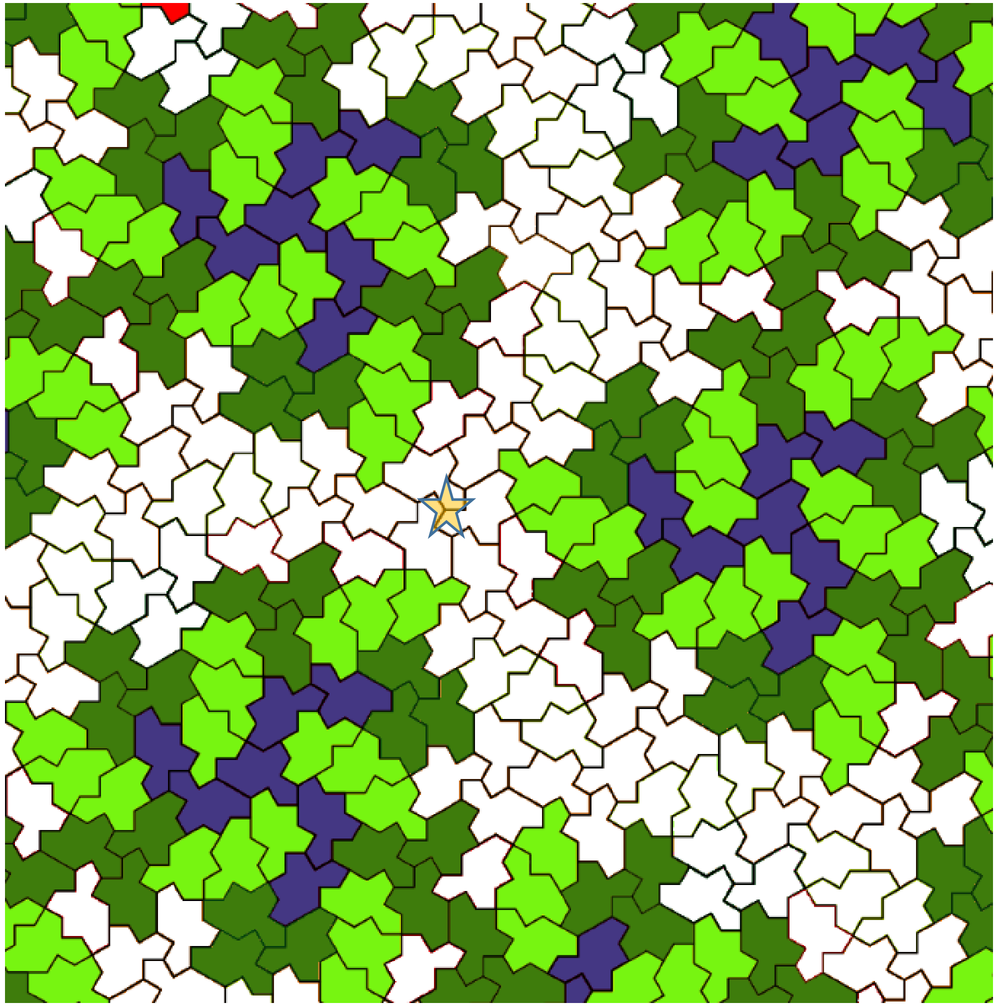
Diabolo/2-arms/3-arms at level 2 by Boris Horvat



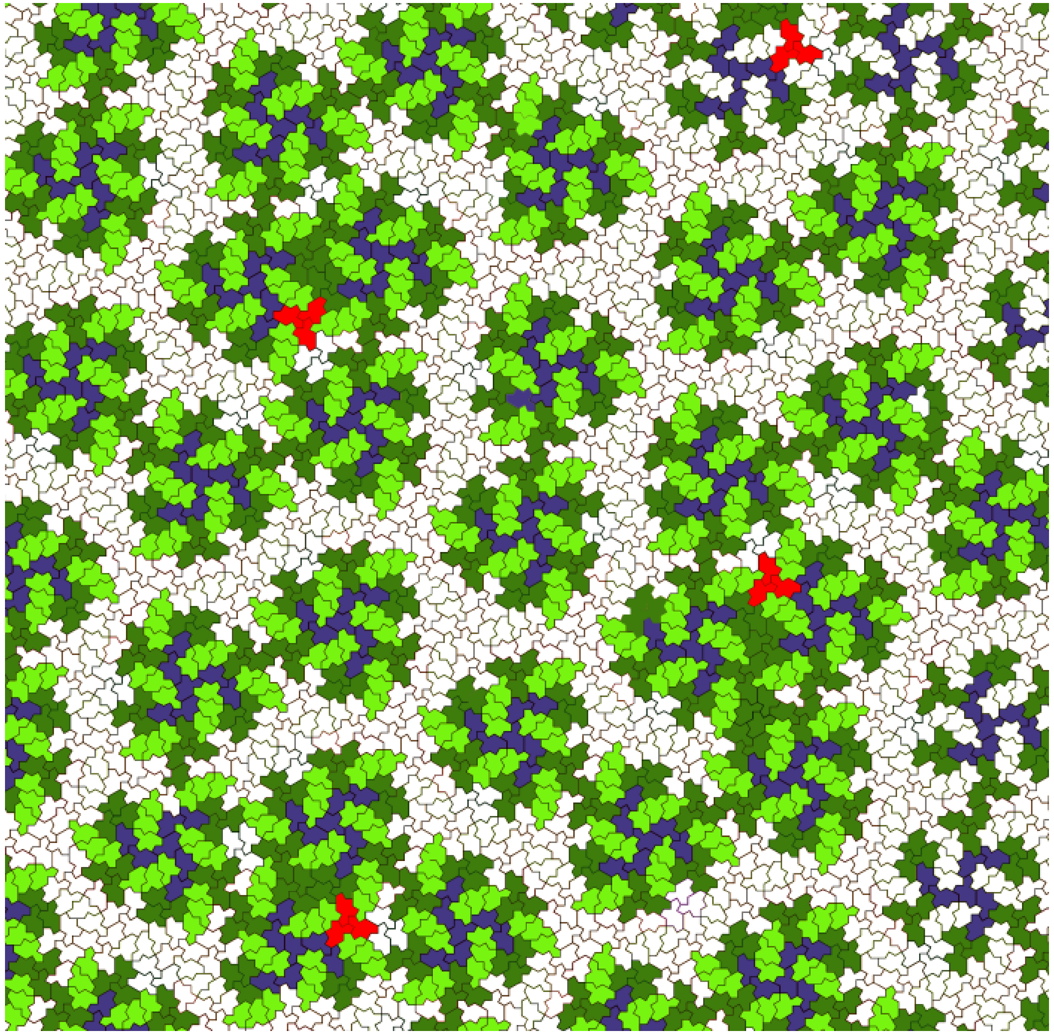


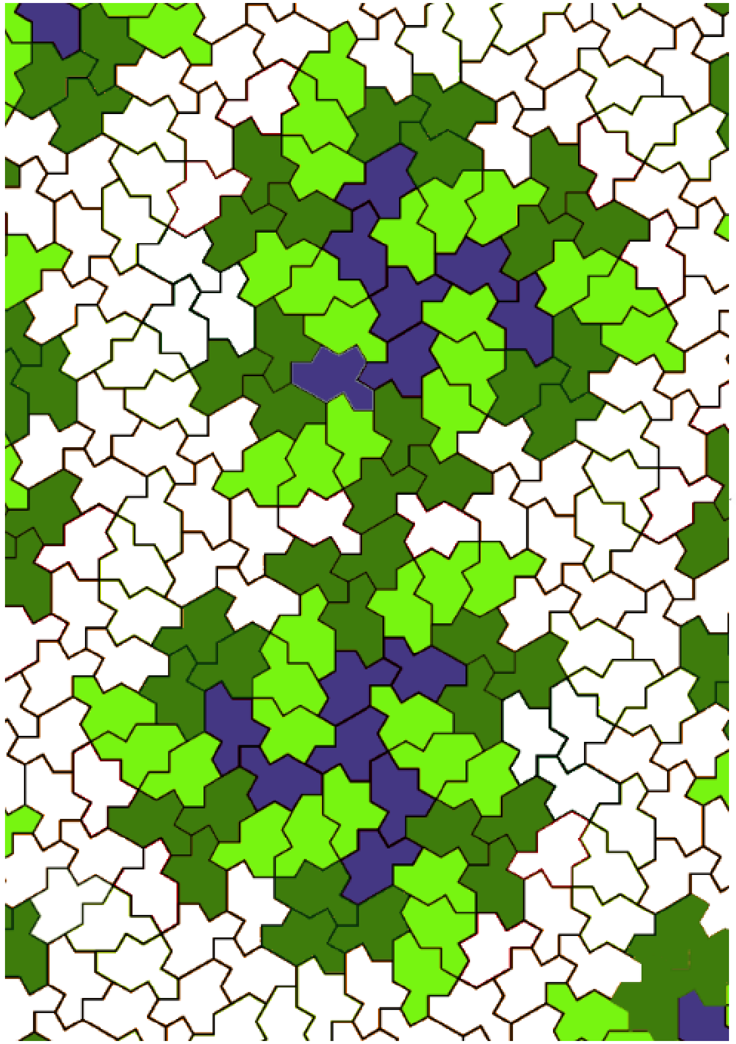
Diabolo/2-arms/3-arms at level 2 by Boris Horvat

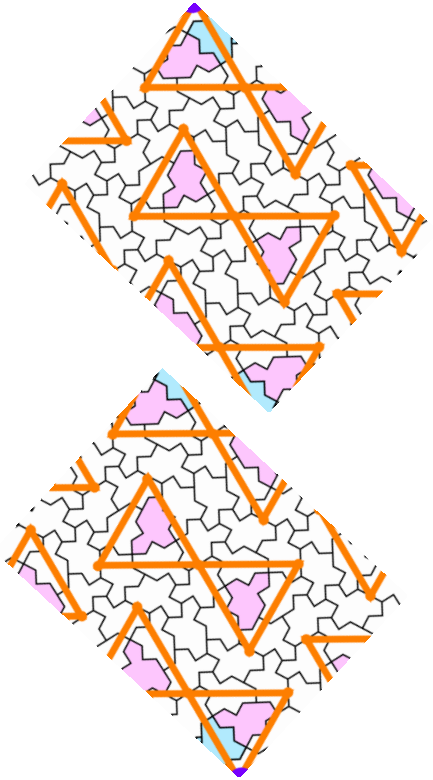




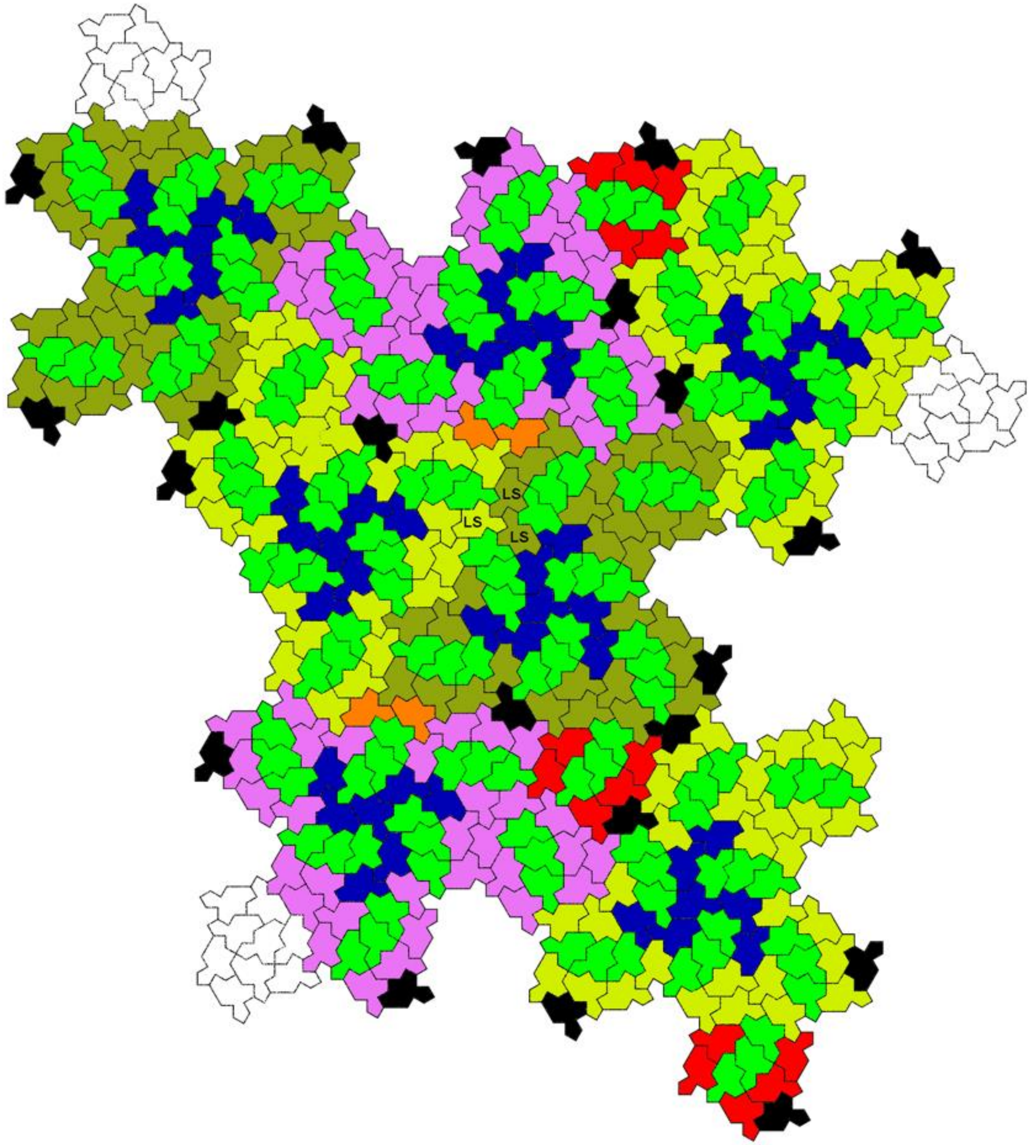
Diabolo/2-arms/3-arms at level 2 by Boris Horvat



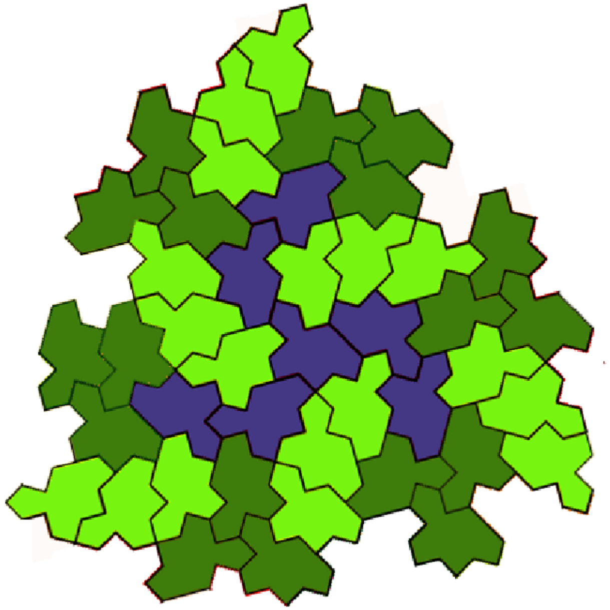


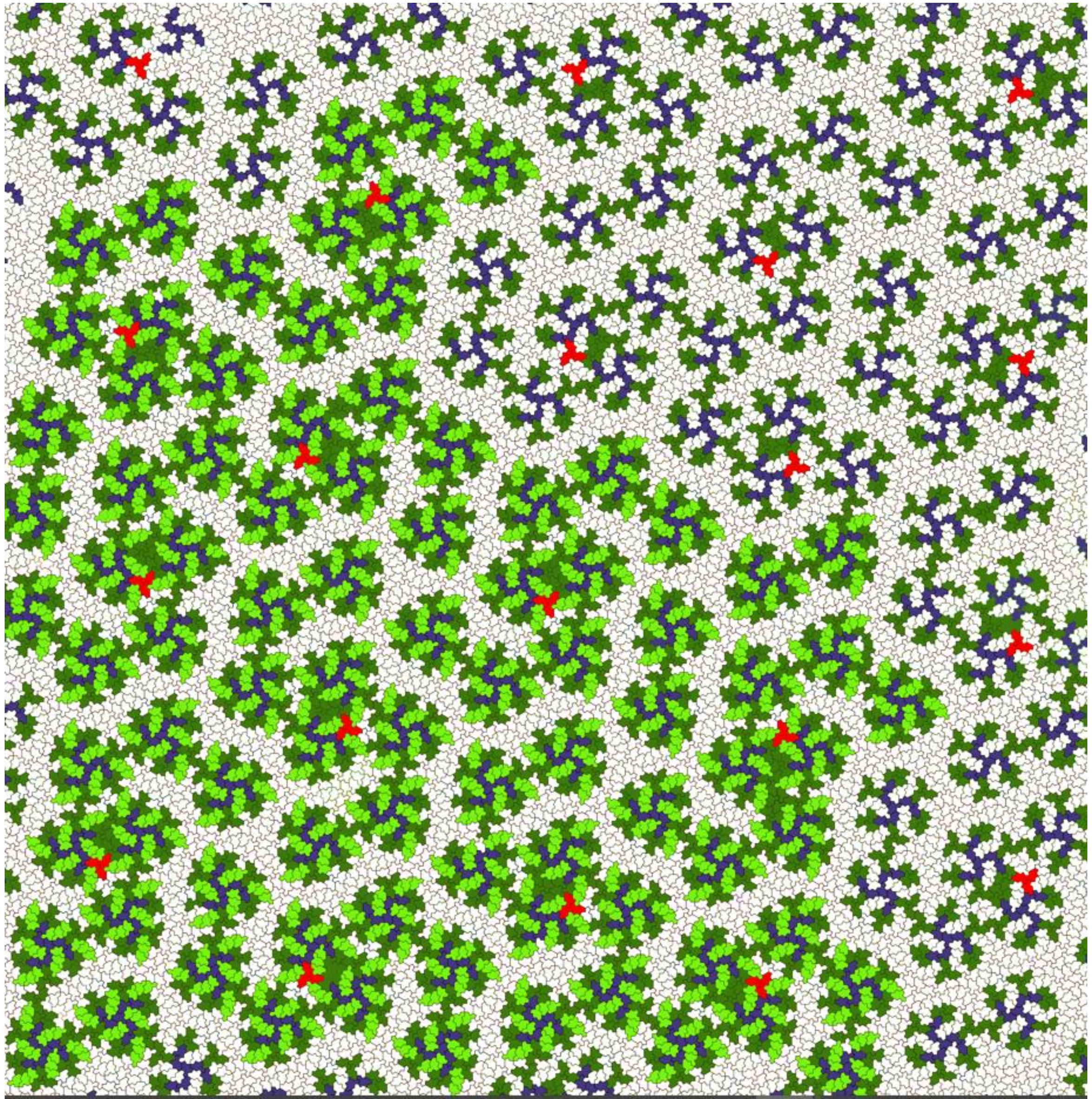


Diabolo/2-arms/3-arms at level 2 by Boris Horvat



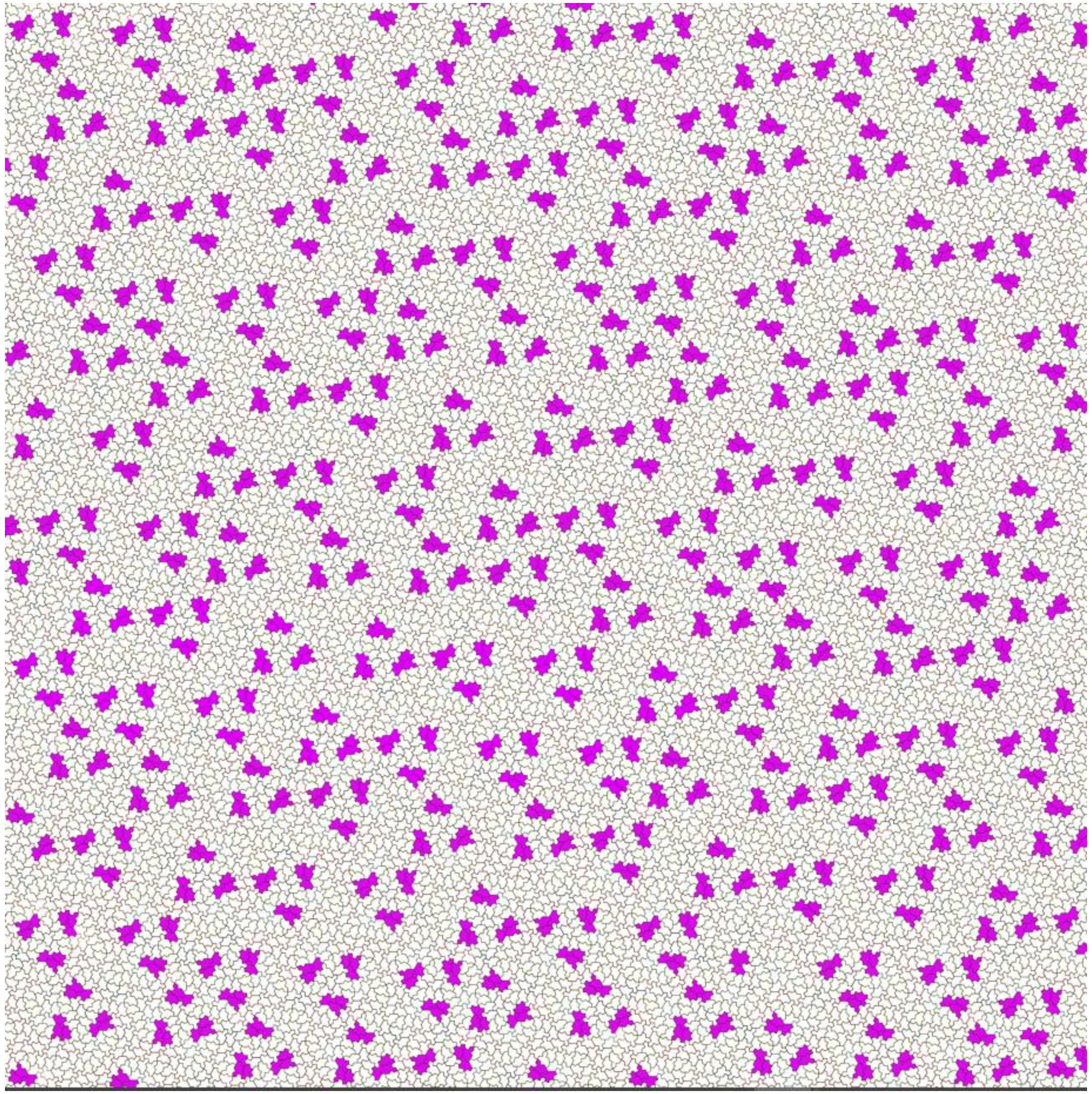
Chimera level 2 by Boris Horvat





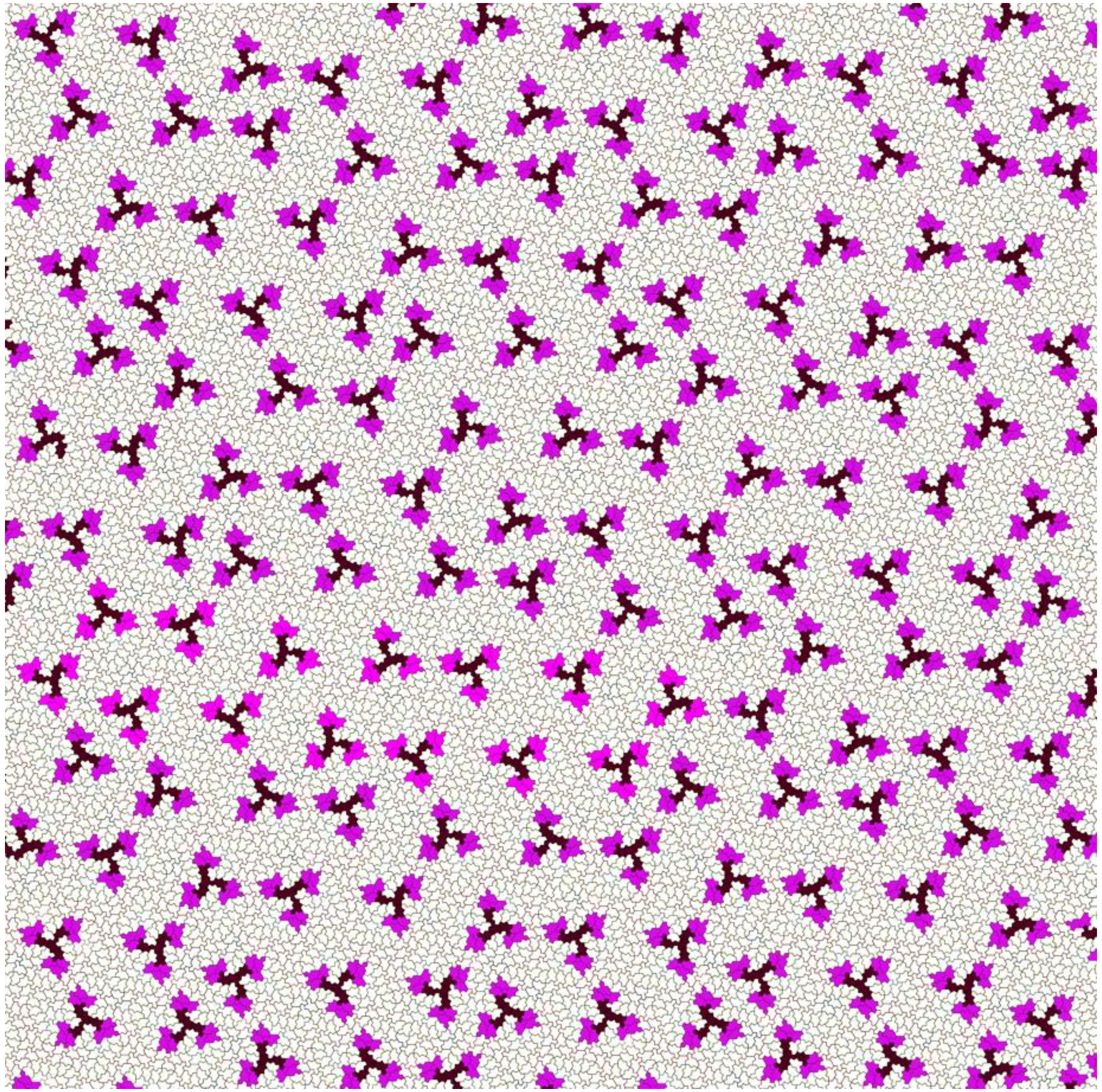
Triplets Marianne by Boris Horvat

Diabolo
2-arms
3-arms
are visible !

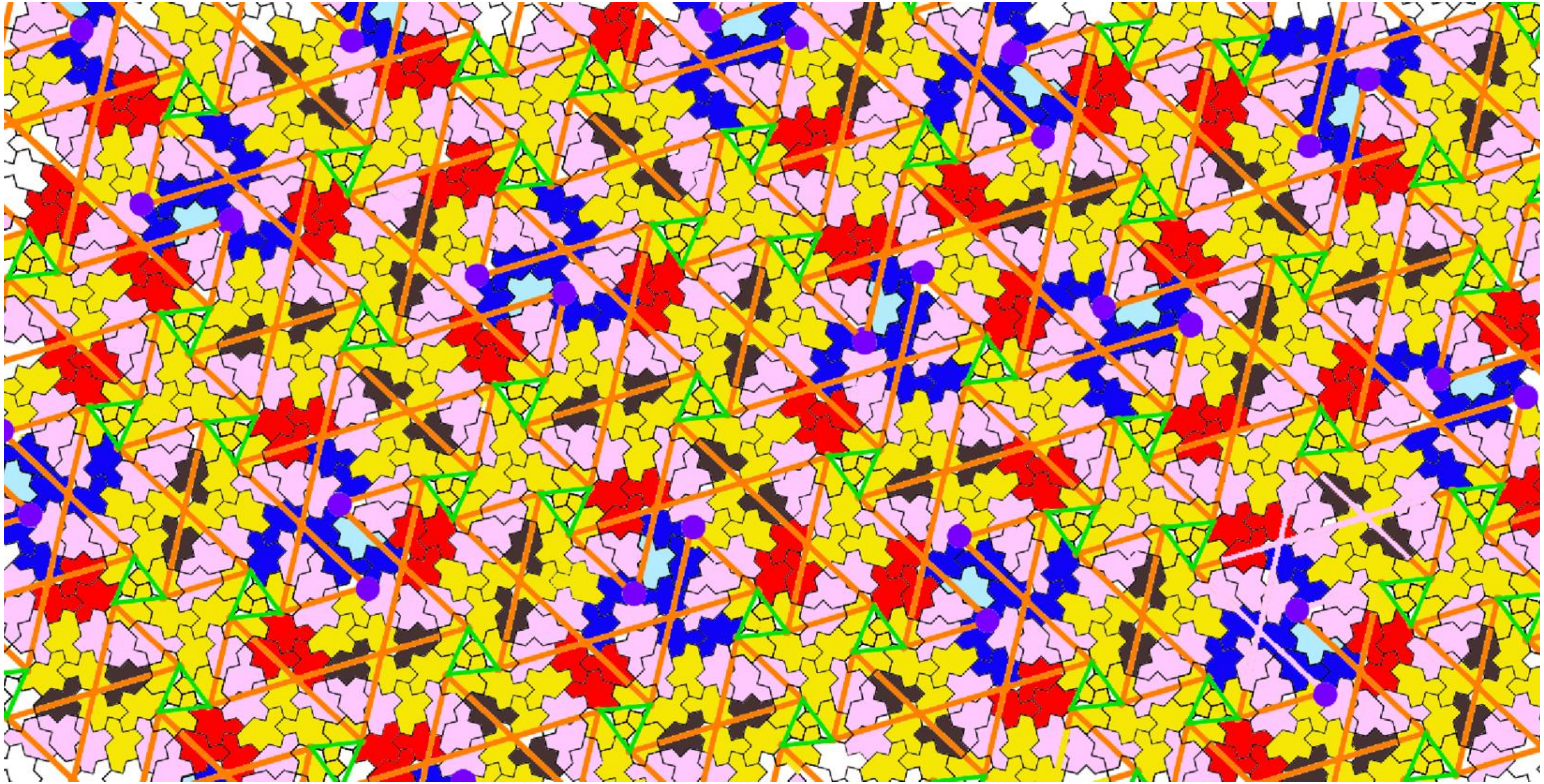


Triplets and propellers by Boris Horvat

Diabolo
2-arms
3-arms
are visible !



Comparision



Espaces entre triangles oranges